

Substrate-Mediated Coordination Across the Inter-Self Perimeter: Extending the Coordination Knowledge Substrate Pattern to Cross-Organizational Coordination and Population Scope under Unified Human Governance

Wenxin Li *Independent Researcher* ORCID: 0009-0004-8065-3235

May 2026

Abstract

Paper 1 (Li, 2026a) defended the Coordination Knowledge Substrate (CKS) as a design pattern at the coordination and decision layer, with the substrate handling coordination and governance, the LLM handling high-dimensional reasoning, and the division drawn at the governance boundary at cell scope. Paper 2 (Li, 2026b) extended CKS to AI Self scope and toward the enterprise brain Self with the foundational separation of fast-pattern instinct from deliberate reasoning into independently-evolving layers under unified human governance. This paper extends substrate-mediated coordination one further step — across the inter-Self perimeter and at population scope — defending six architectural commitments.

The first establishes the **shared substrate** as the architectural object through which two or more CKS-governed Selves coordinate — temporarily constructed for an interaction, perimeter-spanning across distinct Selves’ home governance perimeters, with all six Paper 1 commitments holding within scope. The second commits **Full Aspect Integration (FAI)** as the canonical operation over the shared substrate — n-ary aspect-level merge with full-merge as architectural default, exchange bounded to reasoning-layer content per Paper 2’s instinct/reasoning separation extended at the inter-Self boundary. The third commits **three-tier conflict handling** within the shared substrate — preserve as substrate-level state by default; resolve via orchestration with rules as inspectable substrate content under explicit human authority; escalate to humans across joint authority of participating Selves’ governance. The fourth commits a **four-locus evolution-feed mechanism** connecting inter-Self events to home-perimeter Paper 2 evolution machinery — hand-off at FAI dissolution, per-mechanism feed structure, a three-case layer-routing rule sending DNA-layer content to DNA evolution and action-layer content to action-feedback evolution while instinct evolution takes no FAI input by architectural commitment, and asymmetric ingestion sourced from per-perimeter governance. The fifth commits **configuration as substrate content with recursive applicability** — every dimension of the FAI mechanism is itself substrate content under all six Paper 1 commitments, with the recursion bottoming at human-authored governance authority. The sixth offers the extension claim that **population-scale collective evolution** is the architectural object Paper 3’s five preceding claims compose into at population register when accumulated FAI events across many CKS-governed Selves operate under joint authority across population-level governance perimeters. Population-scale CKS-governed Self deployments do not yet exist; the architectural commitment is to what the inheritance produces at population scope through composition, with specific population-scale dynamics awaiting empirical observation — the architectural commitment rendered confidently, the empirical-verification status of the population-scale extension rendered with calibrated humility.

The contribution is a design pattern, not an empirical evaluation. The unifying architectural move across all six claims is the extension of substrate-mediated coordination across the inter-Self perimeter under unified human governance, composed at population scope through accumulated FAI events under joint authority across population-level governance perimeters — Paper 1’s *outside-the-model vs. inside-the-model* spine extended through the inter-Self perimeter and into population-scale collective evolution. Paper 3 inherits Papers 1 and 2’s architectural commitments and extends them one further step at trilogy completion, sitting at the intersection of multi-agent systems and inter-agent coordination, federated learning and cross-organizational ML, governance-as-code and policy-as-substrate work, multi-coordinator and joint-authority architectures, substrate-as-content and recursive-governance literature, and trilogy continuity with Papers 1 and 2 — articulating a design pattern available to be adopted, varied, composed with adjacent patterns, or argued against.

Empirical validation through real-world cross-organizational deployment at scale is downstream work.

Table of Contents

1. Introduction

- 1.1 Problem framing — the coordination gap at the inter-Self perimeter
- 1.2 Contribution statement — extending substrate-mediated coordination across the inter-Self perimeter and population scope
- 1.3 Roadmap

2. Framing and Scope

- 2.1 The cross-claim spine
- 2.2 Biology vocabulary positioning at Paper 3's scope
- 2.3 What Paper 3 is not

3. Related Work

- 3.1 Multi-agent systems and inter-agent coordination
- 3.2 Federated learning and cross-organizational ML
- 3.3 Governance-as-code and policy-as-substrate work
- 3.4 Multi-coordinator and joint-authority architectures
- 3.5 Substrate-as-content and recursive-governance literature
- 3.6 Trilogy continuity with Papers 1 and 2
- 3.7 The four-family closest-neighbor structure at trilogy-completion register
- 3.8 Where Paper 3 sits

4. Claim 1 — The shared substrate as the architectural object of inter-Self coordination

- 4.1 Claim statement and the substrate-mediator extension across the inter-Self perimeter
- 4.2 The named foil — opaque agent-to-agent communication and its structural limits
- 4.3 Closest-neighbor engagement at tied-top across axes
- 4.4 Three differentiation axes
- 4.5 Reviewer-objection responses
- 4.6 Papers 1 and 2 inheritance and downstream placement

5. Claim 2 — Full Aspect Integration as the canonical operation over the shared substrate

- 5.1 Claim statement and FAI as the canonical operation over the shared substrate
- 5.2 The named foil — opaque inter-agent merge and the merge-mechanism gap
- 5.3 Closest-neighbor engagement at three tied-top clusters
- 5.4 Three differentiation axes
- 5.5 Reviewer-objection responses
- 5.6 Inheritance accounting and downstream placement

6. Claim 3 — The three-tier inter-Self conflict-handling mechanism

- 6.1 Claim statement and the three-tier mechanism over the shared substrate
- 6.2 The named foil — opaque conflict resolution and the conflict-handling gap
- 6.3 Closest-neighbor engagement organized by tier
- 6.4 Three differentiation axes with asymmetric inheritance shape
- 6.5 The inheritance-active finding and reviewer-objection responses
- 6.6 Inheritance accounting and downstream placement

7. Claim 4 — The four-locus evolution-feed mechanism at the FAI hand-off boundary

- 7.1 Claim statement and the four-locus mechanism at the FAI hand-off boundary
- 7.2 The named foil — opaque post-deployment update and the evolution-feed gap

- 7.3 Closest-neighbor engagement organized by locus
- 7.4 Three differentiation axes with asymmetric inheritance shape
- 7.5 The inheritance-active finding and reviewer-objection responses
- 7.6 Inheritance accounting and downstream placement

8. Claim 5 — Configuration as substrate content with recursive applicability

- 8.1 Claim statement and the enumeration-and-elaboration register at the configurable-dimensions and recursive-governance angle
- 8.2 The named foil — opaque coordination-mechanism configuration and the configuration gap
- 8.3 Closest-neighbor engagement organized by commitment
- 8.4 Three differentiation axes with asymmetric inheritance shape
- 8.5 The third inheritance-active finding and reviewer-objection responses
- 8.6 Inheritance accounting and downstream placement

9. Claim 6 — Population-scale collective evolution as the trilogy-completion architectural object

- 9.1 Claim statement and population-scale collective evolution as the trilogy-completion architectural object
- 9.2 The named foil — opaque population-scale coordination and the population-scale-dynamics gap
- 9.3 Closest-neighbor engagement organized by family
- 9.4 Three differentiation axes with asymmetric inheritance shape
- 9.5 The fourth inheritance-active finding and reviewer-objection responses
- 9.6 Inheritance accounting and trilogy closing

10. Thought Experiment

- 10.1 The opening FAI event
- 10.2 Population scope
- 10.3 Alignment with prior claims
- 10.4 What the thought experiment is not

11. Discussion and Implications

- 11.1 The substrate-mediator ladder — six rungs at paper-level scopes
- 11.2 Six-organizational-pattern coherence — territory through family
- 11.3 Six-FL-boundary pattern — joint-authority-mechanics through population-scale-dynamics
- 11.4 The four-inheritance-active-findings compounding pattern
- 11.5 The six-foil-parallel pattern across §§4.2–§9.2
- 11.6 Calibration registers across the trilogy — varied landscape, not closed taxonomy
- 11.7 Restatement objections — paper-level response across the six claims
- 11.8 Capability-motivated objection — foundation labs solve this anyway
- 11.9 Scope-and-framing objections — why these specific scopes
- 11.10 Trilogy positioning, future work, and the calibrated-humility voice anchor

12. Conclusion

References

1. Introduction

1.1 Problem framing — the coordination gap at the inter-Self perimeter

Paper 1 (Li, 2026a) defended the Coordination Knowledge Substrate (CKS) as a design pattern at the coordination and decision layer of an AI system, with substrate handling coordination and governance, the LLM handling high-dimensional reasoning, and the division drawn at the governance boundary rather than

at the capability boundary. Paper 1 stood at cell scope — a unit of coordinated work, demonstrated through one proof-of-concept in regulated-industry reporting in commodity tools. Paper 2 (Li, 2026b) took Paper 1’s modular cell as starting unit and extended the pattern to the AI Self — cells composing into aspects, aspects into Selves, all under unified human governance, with the foundational separation of fast-pattern instinct from deliberate reasoning into independently-evolving layers. Paper 2 closed at the enterprise brain Self as architecturally coherent design pattern, rendered concretely through one thought experiment continuing Paper 1’s POC lineage at enterprise scope, with empirical validation in real-world enterprise deployment as downstream work.

What Papers 1 and 2 did not address is what happens when the coordinated whole stops being the largest architectural object the design pattern reasons about. Paper 2 explicitly bracketed inter-Self autonomous dynamics as out of scope — how multiple enterprise brain Selves interact across organizational boundaries was left as a class of extension theories that would each stand on its own claim set rather than on arguments Paper 2 develops. Two further scope-shift directions sit outside Papers 1 and 2 jointly and motivate Paper 3. First, Selves owned by distinct organizations under distinct authority structures encounter each other and have coordination work to do that crosses the boundary between them — work that involves substantive content, conflicts, decisions, and rationale that no single Self holds authority over. Second, many such inter-Self interactions accumulate across a network of CKS-governed Selves over time, producing population-scale dynamics whose architectural-pattern shape — what propagates, what diverges, what governance configurations enable or constrain — is itself a question.

The architectural question is real and is sharpening rather than fading as multi-agent AI deployment matures. Contemporary architectures address inter-AI coordination at scale, but commit to it at registers that leave the substrate-content commitment Paper 1 establishes outside the architectural pattern. Foundation lab multi-agent systems at scale (OpenAI AgentKit, Anthropic Claude multi-agent with MCP, Google Gemini Enterprise Agent Platform) deliver functional coordination through capability scaling and platform-plus-protocol selection, with governance pushed out to deployment-shape and contract/institutional registers. Agent-protocol ecosystems (MCP, A2A, ANP, IBM ACP, IETF and W3C drafts at protocol-spec register) standardize communication mechanics with governance left to implementers. Federated-learning consortia at hundreds-of-organizations scale (FLARE, FATE, OpenFL across healthcare, pharma, government labs, financial sector) deliver cross-organizational coordination through coordinator-mediated aggregation under consortium-governance contracts. Cross-organizational governance frameworks (Gaia-X federation services, IDS-RAM, data trusts and cooperatives, JAMS AI Disputes Clauses and Rules — JAMS being a US-based arbitration provider supplying AI-specific dispute-resolution mechanisms) commit joint authority at contract, institutional, or protocol-parameter layer rather than as substrate-content architectural commitment. Each of these supplies functional inter-AI coordination at progressively larger scopes; none commits at architectural-pattern register to the configuration in which the exchange medium is itself first-class human-governed substrate content carrying Paper 1’s six commitments within scope.

Paper 3’s response is not that these contemporary architectures should be displaced — they deliver functional coordination at scope honestly. The response is that some commitments — substrate-content commitment for the medium of coordination, joint authority across distinct governance perimeters as architectural property, conflict preservation as first-class state, recursive applicability of configuration as substrate content — must remain architecturally present in the inter-AI exchange itself, for the same reasons Paper 1 named at cell scope and Paper 2 extended at Self scope, but now at a third scope: the inter-Self perimeter and, by composition, the population register. The unifying architectural move is the extension of substrate-mediated coordination across the inter-Self perimeter under unified human governance, with a temporary shared substrate as the medium of coordination, Full Aspect Integration as the canonical operation over it, three-tier conflict handling within, an evolution-feed connecting events to home-perimeter Paper 2 evolution machinery, configuration recursively applicable as substrate content, all composing at population scope as collective evolution under joint authority across population-level governance perimeters. Paper 1’s *outside-the-model* vs. *inside-the-model* spine extends through Paper 3 unchanged: inter-Self coordination, governance, and conflict-handling live in the shared substrate outside any single LLM; instinct-layer content remains within each Self’s LLM and does not cross the inter-Self perimeter; and the architecture’s value at inter-Self and population scope comes from the same principled separation Papers 1 and 2 defend at their respective scopes.

1.2 Contribution statement — extending substrate-mediated coordination across the inter-Self perimeter and population scope

Three things from Papers 1 and 2 are what readers need to read Paper 3 productively. From Paper 1, the substrate/cell two-level structure as foundational architectural commitment and the six commitments — substrate-based hybrid composition with a governance boundary, conflict preservation at the substrate level, human governance over content and rules, AI as substrate mediator, tool-agnosticism at the substrate layer, and linear-cost storage and composition — are what the shared substrate carries within scope at the inter-Self perimeter; the *outside-the-model vs. inside-the-model* spine is the architectural distinction Paper 3 builds on at the inter-Self scope. From Paper 2, the three-level structure (cell, aspect, Self), the DNA-and-action-layer distinction within every cell, the lifecycle and evolution operations within each Self’s home perimeter, and the foundational instinct/reasoning separation define what exchanges through the shared substrate (DNA-layer and action-layer content) and what does not (LLM weights and instinct-layer content). Both papers’ commitments hold within Paper 3’s scope; Paper 3 inherits and extends rather than replacing them.

Paper 3’s contribution is an architectural commitment defended through six claims. The first establishes the **shared substrate** as the architectural object through which two or more CKS-governed Selves coordinate with each other — temporarily constructed for an interaction, perimeter-spanning across distinct Selves’ home governance perimeters, governance-configured-persistence, carrying all six Paper 1 commitments within scope. The second commits **Full Aspect Integration (FAI) as the canonical operation** over the shared substrate — n-ary aspect-level merge with full-merge as architectural default at inter-Self scope, exchange bounded to reasoning-layer content per Paper 2’s instinct/reasoning separation extended at the inter-Self boundary. The third commits **three-tier conflict handling** within the shared substrate — preserve as substrate-level state by default per Paper 1 inheritance; resolve via orchestration with rules as inspectable substrate content under explicit human authority; escalate to humans across joint authority of participating Selves’ governance, with cross-perimeter escalation as architectural commitment rather than as deployment-layer wrapper. The fourth commits a **four-locus evolution-feed mechanism** connecting inter-Self events to home-perimeter Paper 2 evolution machinery — hand-off at FAI dissolution; per-mechanism feed structure with FAI-derived content reaching existing per-Self evolution mechanisms unchanged; layer-routing rule with three-case explicitness (DNA-layer content feeds DNA evolution; action-layer content feeds action-feedback evolution; instinct evolution takes no FAI input by architectural commitment, per Paper 2’s instinct/reasoning separation extended at the inter-Self perimeter); asymmetric ingestion sourced from per-perimeter governance. The fifth commits **configuration as substrate content with recursive applicability** — every dimension of the FAI mechanism is itself substrate content under all six Paper 1 commitments; configuration of configuration substrate is also substrate content; the recursion bottoms at human-authored governance authority per Paper 1 §3.3.

The sixth offers the extension claim that **population-scale collective evolution** is the architectural object Paper 3’s five preceding claims produce at population register when accumulated FAI events across many CKS-governed Selves operate under the configuration-as-substrate-content commitment with recursive applicability. The cumulative coordination structure produced by many FAI events under joint authority across population-level governance perimeters is itself substrate content at population register, with all six Paper 1 commitments holding at population scope and the population substrate-content rather than an agent. Population-scale CKS-governed Self deployments do not yet exist; the architectural commitment is to what the inheritance produces at population scope through composition, with specific population-scale dynamics — the precise trajectories of pattern propagation, the actual rates and shapes of specialization-preserving divergence, the empirical character of collective adaptation — awaiting empirical observation. Capability scaling delivers functional inter-Self coordination at population scope honestly; the differentiation Paper 3 commits to is at architectural-commitment register rather than at the functional-coordination claim. The architectural commitment is rendered confidently; the empirical-verification status of the population-scale extension is rendered with calibrated humility.

The substrate-mediator commitment progresses across the trilogy at six configurations of progressively larger scope. Paper 1 carries rungs 1 (single human $\leftrightarrow$ LLM within one cell, core theory) and 2 (multiple humans $\leftrightarrow$ LLM within one cell, extension); Paper 2 carries rungs 3 (cell $\leftrightarrow$ aspect $\leftrightarrow$ Self within one Self, core theory)

and 4 (the Self as enterprise-scope unified architecture, extension); Paper 3 carries rungs 5 (Self $\leftrightarrow$ Self mediated by shared substrate, core theory across §§4–8) and 6 (population-scale collective dynamics across many Selves, extension at §9). Each paper carries a core-theory rung and an extension-claim rung, with the parallel-extension-claim structure across the trilogy — Paper 1’s Claim 6, Paper 2’s Claim 6, Paper 3’s Claim 6 — making composition rather than fresh emergence theory architecturally honest at trilogy completion. The ladder is observation, not roadmap: Paper 1 did not promise Paper 3, and Paper 2 explicitly bracketed inter-Self dynamics as out of scope. The ladder becomes visible in retrospect because Paper 3 completes it.

The contribution genre is design-pattern writing in the same sense Papers 1 and 2 inhabit: an articulation of a recurring structure of design problem and design response, named so that subsequent work can adopt it, vary it, compose it with adjacent patterns, or argue against it. Paper 3 is not a multi-agent communication protocol specification, not a vendor product, not a novel algorithm, and not an empirical evaluation at scale. Future work may instantiate the pattern as protocols, security-guarantee specifications, message schemas, or deployment architectures; Paper 3 is the architectural commitment such work would inherit from.

1.3 Roadmap

The paper is organized as a single integrated theory paper, following Paper 2’s structural model rather than Paper 1’s three-part core-and-extension structure. Paper 3’s six architectural commitments compose into one design pattern at the inter-Self perimeter and population scope, and the cross-claim work each commitment does in supporting the others reads more naturally as one arc.

Section 2 frames the cross-claim spine, the trilogy positioning, and what Paper 3 is and is not. Section 3 positions Paper 3 within six recognized lineages — multi-agent systems and inter-agent coordination; federated learning and cross-organizational ML; governance-as-code and policy-as-substrate work; multi-coordinator and joint-authority architectures; substrate-as-content and recursive-governance literature; trilogy continuity with Papers 1 and 2 — without re-engaging the per-claim closest-neighbor work the claim sections develop in their own territories. The four-family closest-neighbor structure surfaced at trilogy completion — orchestration and meta-governance blueprints; agent-contracts and governed-reasoning frameworks; enterprise-MAS-governance blueprints; federated-learning consortia at population scope — organizes the most direct contemporary closest neighbors and supplies the structural anchor for paper-level Related Work synthesis.

Sections 4 through 9 defend the six architectural commitments in turn — the shared substrate as the architectural object of inter-Self coordination (§4); Full Aspect Integration as the canonical operation over the shared substrate (§5); the three-tier inter-Self conflict-handling mechanism (§6); the four-locus evolution-feed mechanism at the FAI hand-off boundary (§7); configuration as substrate content with recursive applicability (§8); and population-scale collective evolution as the trilogy-completion architectural object (§9). Each section names the architectural foil at that claim’s scope, engages closest-neighbor work in its specific territory, develops three differentiation axes with primary novelty weight, responds to reviewer objections, and accounts for its inheritance from Papers 1 and 2 jointly and from prior Paper 3 claims.

Section 10 renders Claim 6’s territory concretely through one thought experiment. The thought experiment is domain-agnostic by construction — Paper 3’s architectural claim is about CKS-governed Selves coordinating with each other regardless of what domain any Self serves, and anchoring to a specific industry would shrink the claim to that industry. Section 10 walks through one opening FAI event between two Selves with a conflict surfaced at the preserve tier and carrying through the evolution-feed mechanism to home-substrate annotations, then population scope as accumulated FAI events across the network, calibrated through the vision-claim register Paper 2 §9 establishes. Section 11 reflects on what the composed architecture commits to across the trilogy — six trilogy-level patterns visible across Claims 1 through 6 (the substrate-mediator ladder, six-organizational-pattern coherence, six-FL-boundary pattern, four-inheritance-active-findings compounding, six-foil-parallel pattern, varied calibration landscape), paper-level reviewer-objection responses across the six claims, and trilogy positioning with the calibrated-humility voice anchor preserved at the closing register. Section 12 closes the trilogy at the substrate-mediator ladder’s sixth rung.

2. Framing and Scope

2.1 The cross-claim spine

The cross-claim spine running through Paper 3’s six claims is the **extension of substrate-mediated coordination across the inter-Self perimeter under unified human governance, composed at population scope through accumulated FAI events under joint authority across population-level governance perimeters**. Each claim either establishes the architectural primitive the spine requires, specifies the canonical operation over it, develops the dynamics the construction enables, or articulates what the preceding claims compose to at population register. Claim 1 (§4) establishes the **shared substrate** as the architectural object — temporarily constructed for an interaction, perimeter-spanning across distinct Selves’ home governance perimeters, with all six Paper 1 (Li, 2026a) commitments holding within scope. Claim 2 (§5) specifies **Full Aspect Integration (FAI)** as the canonical operation over the shared substrate, with the unit of exchange the aspect from Paper 2 (Li, 2026b), full merge as the architectural default at inter-Self scope, and exchange bounded to reasoning-layer content per Paper 2’s instinct/reasoning separation extended at the inter-Self boundary. Claim 3 (§6) develops the **three-tier conflict-handling mechanism** within the shared substrate — preserve, resolve via orchestration, escalate to humans — with the cross-perimeter shape of the escalate tier across joint authority of participating Selves’ governance as the genuinely fresh tier shape. Claim 4 (§7) specifies the **four-locus evolution-feed mechanism** connecting inter-Self events to home-perimeter Paper 2 evolution machinery — hand-off at FAI dissolution, per-mechanism feed structure, a layer-routing rule with three-case explicitness routing DNA-layer content to DNA evolution and action-layer content to action-feedback evolution while instinct evolution takes no FAI input by architectural commitment, and asymmetric ingestion sourced from per-perimeter governance. Claim 5 (§8) commits **configuration as substrate content with recursive applicability** across all configurable dimensions, with the recursion bottoming at human-authored governance authority per Paper 1 §3.3. Claim 6 (§9) articulates **population-scale collective evolution** as what the preceding five claims compose to at population register when accumulated FAI events across many CKS-governed Selves operate under joint authority across population-level governance perimeters.

The architectural object — human-governed substrate as coordination medium — operates at progressively larger scopes through the trilogy, with each scope inheriting the prior commitments without replacing them. The substrate-mediator commitment progresses at six configurations: rung 1 single human $\leftrightarrow$ LLM within one cell (Paper 1 core theory); rung 2 multiple humans $\leftrightarrow$ LLM within one cell (Paper 1 extension); rung 3 cell $\leftrightarrow$ aspect $\leftrightarrow$ Self within one Self (Paper 2 core theory); rung 4 the Self as enterprise-scope unified architecture (Paper 2 extension); rung 5 Self $\leftrightarrow$ Self mediated by shared substrate (Paper 3 core theory across §§4–8); rung 6 population-scale collective dynamics across many Selves (Paper 3 extension at §9). Each paper carries a core-theory rung and an extension-claim rung — Paper 1 contributes rungs 1 and 2, Paper 2 contributes rungs 3 and 4, Paper 3 contributes rungs 5 and 6. The ladder is observation rather than roadmap: Paper 1 did not promise Paper 3, and Paper 2 explicitly bracketed inter-Self dynamics as out of scope. The ladder becomes visible in retrospect because Paper 3 completes it, providing readers a clear position for Paper 3 within the series and preempting readings of Paper 3 as multi-agent communication retrofitted with CKS terms. The hybrid commitment Paper 1 §2 establishes at cell scope — substrate handles coordination and governance, LLM handles high-dimensional reasoning, with the division drawn at the governance boundary rather than the capability boundary — extends throughout Paper 3 as the foundation Paper 2’s instinct/reasoning separation rests on at Self scope and Paper 3’s inter-Self coordination rests on at the perimeter. Paper 1’s *outside-the-model vs. inside-the-model* spine extends through Paper 3 unchanged: inter-Self coordination, governance, and conflict-handling live in the shared substrate outside any single LLM; instinct-layer content remains within each Self’s LLM and does not cross the inter-Self perimeter; and the architecture’s value at inter-Self and population scope comes from the same principled separation Papers 1 and 2 defend at their respective scopes.

2.2 Biology vocabulary positioning at Paper 3’s scope

Paper 2’s biology vocabulary engagement is bounded to its intra-Self scope. Paper 2 §2.2 commits biology terms — DNA, action, expression, birth, mating, death, instinct, evolution, lineage — as bounded

load-bearing terminology rather than decorative analogy, with each term carrying architectural content the surrounding sections specify and the structural shape the biology parallel names — cells composing into larger structures, life cycles, evolutionary mechanisms operating on heritable content — earning reader comprehension within Paper 2’s scope. The vocabulary’s interpretation is bounded by the architectural commitments Paper 2’s six claims supply, and Paper 2 explicitly catalogues where CKS differs from biology at multiple specific points where design choices diverge from biology’s evolved constraints. The bounded load-bearing terminology engagement is Paper 2’s commitment at Paper 2’s scope; Paper 3 references that engagement as inheritance rather than extending it.

Paper 3 engages biology lightly and only at one place: Claim 6’s framing. The claim that collective interaction across many participants is a primary driver of population-level evolution has biological resonance — species-level evolution in biology is substantially driven by interactions across organisms, not only by within-organism dynamics. Paper 3 borrows this resonance for Claim 6’s framing, where inter-Self FAI events accumulating across a network constitute a collective evolution mechanism analogous to (not identical to) collective interaction’s role in biological evolution. The core architectural construction Paper 3 develops in §§4–8 has no clean biological parallel. Biology has no governable shared substrate between species; cross-species exchange in biology operates through entirely different mechanisms — horizontal gene transfer, symbiosis, predation, competition for resources — that share neither structure nor governance with the substrate-mediated inter-Self coordination Paper 3 commits to. Paper 3’s substrate-mediated inter-Self coordination is one of the architectural features the trilogy specifies as exceeding biology — a degree of freedom CKS provides because the substrate is engineered rather than evolved. The biology framing therefore stays bounded to Claim 6 and does not extend into the core claims.

The mechanism-names-vs-biology-terms convention §7 establishes is carried forward at Paper 3’s voice register throughout. Paper 2’s intra-Self vocabulary is referenced by inheritance — “Paper 2 §6.3’s intra-Self combination primitive,” “Paper 2 §7’s three evolution mechanisms,” “Paper 2 §8’s per-mechanism governance shapes” — rather than imported as Paper 3 voice register. Where Paper 3 names the three evolution mechanisms feeding from FAI events through Claim 4’s evolution-feed pathway, the convention is to use mechanism names — “DNA evolution,” “action-feedback evolution,” “instinct evolution” — rather than biology terms. Biology terms (“mutation,” “directed selection,” “productive tension”) are reserved for productive-tension-pair contexts where Paper 2 §7.2’s productive-tension subsection’s pair-specific framing applies, surfacing at a single locus in §7. The convention preserves both the inheritance from Paper 2 (the architectural commitments §§5–8 inherit without re-defending) and the bounded scope of Paper 3’s biology engagement (which lives at Claim 6’s framing only).

2.3 What Paper 3 is not

Six disambiguations bound Paper 3’s scope. Paper 3 is not a multi-agent communication protocol specification: protocol specifications would inherit the architectural commitment from Paper 3 rather than the other way around, with message formats, transport layers, security guarantees, and deployment architectures one architectural step downstream of the pattern §§4–9 develop. Paper 3 is not a vendor product or framework: it is a design pattern in the same sense Papers 1 and 2 inhabit, named so subsequent work can adopt it, vary it, compose it with adjacent patterns, or argue against it, with specific implementations of the pattern across organizations each standing on their own deployment claims rather than on Paper 3’s architectural claims. Paper 3 is not a novel algorithm: the commitment is at architectural-pattern register to what the inheritance produces at the inter-Self perimeter and at population scope, not to specific algorithms for FAI execution, conflict resolution, evolution-feed propagation, or population-scale dynamics, with algorithms instantiating the pattern at deployment each standing on their own algorithmic claims. Paper 3 is not about replacing capability scaling: capability scaling delivers functional inter-Self coordination at scale honestly across foundation lab multi-agent systems, agent-protocol ecosystems, and federated-learning consortia, and the differentiation Paper 3 commits to is at architectural-commitment register rather than at the functional-coordination claim, with substrate-content commitment for the medium of coordination, joint authority across distinct governance perimeters as architectural property, and recursive applicability of configuration as substrate content as the commitments capability-motivated architectures do not make at architectural-pattern level. Paper 3 is not biology-as-architecture: as §2.2 develops, the core architec-

tural construction has no clean biological parallel, biology vocabulary engagement is bounded to Claim 6’s framing, and the architectural commitments are CKS’s own at Paper 3’s scope. Paper 3 is not empirical demonstration at scale: it is a design-pattern paper in the genre of Papers 1 and 2, defending architectural commitments at conceptual register, with population-scale CKS-governed Self deployments not yet existing and specific population-scale dynamics — the precise trajectories of pattern propagation, the actual rates and shapes of specialization-preserving divergence, the empirical character of collective adaptation — awaiting empirical observation. The architectural commitment is rendered confidently; the empirical-verification status of the population-scale extension is rendered with calibrated humility, anchored to the substrate-mediator-ladder-completion at the trilogy’s sixth rung as the trilogy’s anticipated landing rather than its empirically demonstrated outcome, calibrated through the vision-claim register Paper 2 §9 establishes.

3. Related Work

Each of Paper 3’s six claims engages closest-neighbor work in its specific territory, and the per-claim engagement is where the detailed differentiation work lives. This section integrates those engagements into paper-level positioning that shows where Paper 3 sits within recognized lineages, without re-engaging the per-claim closest-neighbor work the claim sections develop in their own territories. Six lineages converge on the architectural territory Paper 3 occupies; the subsections that follow name where Paper 3 sits within each, then in §3.7 organize the most direct contemporary closest neighbors at the trilogy-completion register through the four-family structure §9.3 surfaces, and in §3.8 name what configuration the lineages do not jointly occupy.

3.1 Multi-agent systems and inter-agent coordination

The closest research lineage for Paper 3’s inter-Self coordination is contemporary multi-agent systems and inter-agent coordination at scale. Foundation lab multi-agent systems supply the most visible end of this lineage: OpenAI AgentKit, Anthropic Claude multi-agent with MCP, and Google Gemini Enterprise Agent Platform deliver functional coordination across many agents at deployment scope through capability scaling, platform integration, and protocol selection. Kim et al. (2025) derives quantitative scaling principles for multi-agent coordination across more than 180 configurations, with a predictive model selecting optimal coordination architecture for the majority of unseen tasks — explicit articulation of the scaling-science enterprise’s stance toward governance as engineering performance kept outside the scaling claim. Agent-protocol ecosystems supply the standardization layer at protocol-spec register: MCP standardizes contextual exchange under cross-vendor stewardship; A2A (Google) supplies standardized peer-to-peer agent-to-agent communication with a fifty-plus enterprise partner ecosystem; ANP, IBM ACP, and IETF and W3C drafts at the protocol-spec register supply complementary specifications across vendor and standards-body boundaries. Agent frameworks supply the framework-API register for composing agent behavior: LangGraph commits to typed shared state with directed graph routing, per-key reducer logic, and durable execution with checkpointing — the closest contemporary multi-agent framework primitive to typed coordination state at substrate-pattern register; AutoGen, CrewAI, Letta, and MemGPT span agent-memory and orchestration territory with varied commitments to message-list aggregation, hierarchical task delegation, and tiered memory. Decentralized cross-organizational frameworks supply a contemporary point of departure from centralized topologies: AgentNet (Yang et al. NeurIPS 2025) commits to decentralized DAG topology with privacy-preserving cross-organizational collaboration. Swarm and graph-topology architectures supply complementary coordination shapes; the broader 2024–2026 multi-agent-systems-at-scale landscape exhibits functional coordination capabilities at increasingly sophisticated registers.

The same period’s critical literature on multi-agent AI architecture-class limits supplies the structural-assessment register the lineage description requires alongside the functional-coordination register. The Vector Institute’s “Transparency in Agentic AI” survey characterizes most multi-agent frameworks as trajectory-opaque; Raza and colleagues articulate a trajectory-level interpretability gap; institutional-AI work on governing collusion in multi-agent LLM markets introduces separate-governance-graph and append-only governance-log patterns precisely because ungoverned baseline frameworks lack formal representation of origin, authority, or sanctions attached to inter-agent messages; AgentAuditor (2026) introduces dedicated reasoning-tree

mechanisms because existing frameworks discard evidential structure of reasoning traces through majority-vote aggregation; the TechRxiv survey on memory in LLM-based multi-agent systems converges on the same architecture-class assessment at the memory-infrastructure register, observing that multi-agent memory infrastructures store information without persistent queryable links to originating agent or decision point. The cross-source convergence across Vector Institute, Raza et al., institutional-AI work, AgentAuditor, TechRxiv, and Atlan’s 2026 practitioner-literature observation supplies an architecture-class-limits assessment that holds across multiple independent sources rather than resting on a single practitioner-literature voice. This critical literature is what makes the foil §4.2 develops structurally credible at architecture-class register rather than at capability-debt register; §3.1 acknowledges its lineage role at paper-level positioning here without re-engaging the per-claim foil work §4.2 conducts.

Paper 3’s Claim 1 (§4) engages this lineage at the architectural-object level — LangGraph at the architectural-pattern axis, A2A at the inter-vendor-coordination axis, with MCP and the broader protocol-spec landscape at adjacent density — and Claims 2, 3, 4, and 5 each engage specific aspects of contemporary multi-agent systems at their respective angles (FAI’s merge mechanism against per-key reducers; three-tier conflict handling against composite single-perimeter mechanisms; evolution-feed against multi-agent memory and post-deployment-update patterns; configuration as substrate content against framework-API configuration). What Paper 3 adds across the lineage — developed in §§4–9 — is the configuration in which the medium of inter-Self coordination is itself first-class human-governed substrate content carrying Paper 1 (Li, 2026a)’s six commitments within scope, with governance internal to the substrate at architectural-pattern register rather than applied as wrapper around protocol traffic, framework state, or platform deployment. The lineage commits to functional coordination through protocol, framework-API, and capability scaling with governance pushed to deployment-shape and contract/institutional registers; Paper 3’s commitment is at architectural-commitment register rather than at the functional-coordination claim, and capability scaling is honestly granted to deliver functional inter-Self coordination at scale at the registers it occupies.

3.2 Federated learning and cross-organizational ML

The federated-learning literature supplies the most mature lineage for cross-organizational ML coordination at scope, and Paper 3 engages it across all six claims at distinct angles. The aggregation-algorithm family — FedAvg, FedProx, FedAdam, FedMA, secure aggregation, vertical FL, and Byzantine-robust variants (Krum, Trimmed Mean, Median, Bulyan, FLTrust) — commits to coordinator-mediated parameter or gradient aggregation at coordinate-wise granularity over full-model vectors. Model-fusion methods at LLM scale (model soups, task arithmetic, TIES, DARE) extend the aggregation pattern at full-parameter element-wise granularity with merge weights configured per task; mixture-of-experts routing and ensemble preservation methods occupy the other side of the merge spectrum at output-space rather than weight-space. Federated continual learning supplies client-side machinery (replay buffers, gradient projection, importance estimators) for continual adaptation under heterogeneous client trajectories; personalized FL — meta-learning approaches, head-and-trunk separation, prototype-based and clustering-based PFL — commits to client-specific layers and heads with divergence by design. Federated knowledge graphs supply cross-organizational structured-content coordination patterns; FLARE, FATE, and OpenFL consortia at hundreds-of-organizations scale across healthcare, pharma, government labs, and the financial sector supply the population-scope deployment register; Federated Behavioural Planes (Fenoglio et al. NeurIPS 2024) supplies the architectural commitment to representing client trajectory diversity at population scope as first-class objects through error and counterfactual behavioural spaces. Rosendal and Oprea (IEEE Big Data 2025) commits to client-level disagreements as first-class with exclusion policies as configuration and topology adaptation through isolated model-update tracks — the strongest articulation at the disagreement-as-substrate register at federated scope.

Paper 3 engages the FL lineage at six distinct angles across the six claims, and the cross-claim engagement compounds as discipline visible at trilogy level (§11.3 develops the six-FL-boundary pattern). Claim 1 engages at the joint-authority-mechanics angle (federated learning, MPC/TEE, and data-trusts as one of four tied-top closest-neighbor clusters); Claim 2 at the merge-mechanism angle (FedAvg-family aggregation, model-fusion at LLM scale, MoE/ensemble preservation); Claim 3 at the conflict-handling angle (Byzantine-robust aggregation, FedAvg averaging-as-resolution, Rosendal and Oprea (2025) on client-

level disagreements); Claim 4 at the evolution-feed angle (FedProx, FedAdam, FCL client-side machinery, PFL heterogeneity-driven divergence); Claim 5 at the configurable-aggregation-algorithm angle (FL platforms exposing aggregation as configurable algorithm); Claim 6 at the population-scale-dynamics angle (FLARE/FATE/OpenFL consortia, Federated Behavioural Planes). What Paper 3 adds across the lineage — developed in §§4–9 — is the configuration in which substrate content is the architectural object the cross-organizational coordination operates over rather than gradient aggregation with governance attached around it through consortium-governance contracts. The substantive aggregation mechanisms with named defaults across a mature FL literature are not in dispute; the differentiation is at architectural-commitment register at population scope, with substrate-content commitment, joint authority across population-level governance perimeters at architectural-pattern register rather than at contract-and-governance-committee layer, and population-scale dynamics framed as composition of architectural commitments rather than as emergence from FL algorithms and data heterogeneity.

3.3 Governance-as-code and policy-as-substrate work

The governance-as-code and policy-as-substrate literature supplies the lineage in which configuration carries architectural weight rather than living as overlay around opaque substrate. Cedar, OPA/Rego, and Casbin at policy-engine register commit to policy as separate artifact in dedicated language with a dedicated evaluation engine evaluated against substrate state at decision time; Terraform with Sentinel, CloudFormation Guard, and Pulumi at the policy-as-code/IaC register commit to infrastructure configuration as versioned and reviewed source artifact with policy enforcement at apply time. Kubernetes CRDs supply the strongest contemporary instance of dimensions-as-substrate-content within single-cluster scope: CRD spec fields are first-class API content with full inspectability and policy-engine queryability — configuration is substrate content within the cluster substrate rather than separate artifact. Multi-tenant SaaS configuration services persist tenant configuration as core data with substrate-level inspectability across `tenant_id`-keyed routing, authorization, and billing; data mesh implementations treat data-product metadata (classification, sensitivity, access policies, retention rules, regulatory tags) as first-class substrate content under federated computational governance. Agent Contracts (Ye and Tan 2026) supply the resource-budget register at substrate-content commitment through the formal $C=(I,O,S,R,T,\Phi,\Psi)$ tuple with conservation laws on delegated computational budgets — substrate-content commitment over a specific architectural commitment plane.

Paper 3’s Claim 5 (§8) engages this lineage at the configurable-dimensions and recursive-governance angle directly — Kubernetes CRDs, multi-tenant SaaS configuration services, and data mesh implementations as commitment-(a) closest neighbors at the dimensions-as-substrate-content angle; Cedar, OPA, and Casbin as objection-1 contrast at policy-as-separate-artifact register; Agent Contracts as resource-budget-register substrate-content commitment. Claims 1 and 4 engage adjacent ground at perimeter-spanning and asymmetric-ingestion registers respectively. What Paper 3 adds across the lineage — developed in §§4 and §8 — is the configuration in which substrate-content commitment for configuration holds at cross-organizational scope under joint authority across distinct governance perimeters, with recursive applicability through substrate-content commitment rather than through separate hierarchical meta-governance layers. The lineage commits to configuration-as-content at varied registers, but the cross-organizational population-substrate analog at architectural-pattern register — substrate-content commitment for configuration holding all six Paper 1 commitments within scope, joint authority across participating Selves’ governance for the configuration substrate itself, recursion bottoming at human-authored governance authority per Paper 1 §3.3 — is not occupied jointly across the contemporary work the lineage anchors.

3.4 Multi-coordinator and joint-authority architectures

Cross-organizational joint authority supplies a fourth lineage with mature contemporary instances at varied architectural-commitment registers. DAO constitutional architectures (Aragon, Compound, MakerDAO, Arbitrum) commit to self-amending governance with substrate continuity: the same blockchain substrate carries substantive and constitutional decisions, with constitutional and amendment processes implemented as specialized proposal types with additional thresholds and distinct contracts. Distributed-system multi-leader replication (CockroachDB, YugabyteDB) commits to coordination-pattern-as-deployment-choice with

patterns selected at deployment time. Gaia-X federation services with Data Usage Agreements and IDS-RAM with Provenance Storage and Collection Points supply mature reference architectures for cross-organizational data exchange under federated governance through membership, connectors, clearing houses, contracts, and self-descriptions; data trusts and cooperatives instantiate joint authority at organizational and legal layer through trustor-trustee-beneficiary structures and member-controlled governance arrangements. MPC threshold structures and TEE attestation supply joint authority over computation through cryptographic and hardware mechanisms; cross-organizational data-trust governance frameworks supply institutional structures around shared computation. JAMS AI Disputes Clauses and Rules and adjacent dispute-resolution mechanisms supply legal-layer recourse structures; HADA and SettleFL’s crypto-economic coordinator supply technical-mechanism instances at the cross-perimeter register. Fauscette’s April 2026 Arion Research piece on cross-organizational agents articulates bilateral and multilateral governance contracts as machine-readable terms and conditions with agreed decision boundaries, escalation protocols, and dispute-resolution mechanisms.

Paper 3’s Claim 1 (§4) engages this lineage at the perimeter-spanning axis directly — Gaia-X and IDS-RAM jointly as one of four tied-top closest-neighbor clusters at substrate-level perimeter-spanning, the federated-learning / MPC-TEE / data-trusts cluster at the joint-authority-mechanics axis. Claim 3 (§6) engages cross-perimeter joint-authority escalation at the escalate tier, with Fauscette (2026), data trusts, JAMS, HADA, and SettleFL as cross-perimeter-cluster closest neighbors. Claim 5 (§8) engages multi-coordinator coordination as configurable pattern, with multi-leader replication and multi-coordinator FL architectures as commitment-(c) neighbors. What Paper 3 adds across the lineage — developed in §§4, §6, and §8 — is the configuration in which joint authority across distinct governance perimeters operates as architectural commitment of the technical substrate itself rather than as legal-layer or institutional-layer construction with technical systems serving as evidence and enforcement. Cross-perimeter scope is mature in non-AI legal and governance precedent, and Paper 3 does not displace that maturity; the differentiation is on which architectural plane joint authority is encoded — substrate content under joint authority versus contract or institutional layer with technical substrate carrying functional coordination beneath the legal or institutional construction. Decentralizing optimization or computation does not automatically decentralize governance; the architectural-commitment register at substrate content is what Paper 3’s six claims jointly occupy.

3.5 Substrate-as-content and recursive-governance literature

The substrate-as-content and recursive-governance literature supplies the lineage in which governance machinery is itself architectural object rather than overlay. Anthropic’s Constitutional AI commits to governance-as-substrate-content at training-time and objective-shaping register, with the constitution embedded in model substrate via training-time objective shaping. Constitutional AI also extends to runtime applicability through revision-and-critique cycles where the constitution serves as evaluable reference at inference time; the architectural-pattern-register differentiation against configuration-as-substrate-content is that CAI’s runtime evaluation operates over the constitution as embedded model property — with full self-amendment of the constitution itself generally not specified at runtime — while configuration-as-substrate-content commits to the constitution-equivalent as authored substrate content with full self-amendment as architectural property of substrate-content commitment. The CNCF Automated Governance Maturity Model frames Policy → Evaluation → Enforcement → Audit layers with the maturity model itself functioning as meta-governance over how governance is done. Governance-runtime work and governed-reasoning architectures supply contemporary instances of governance as runtime-layer commitment; Cognitive Core (Seck 2026) approaches the territory from architecture-time configuration through typed primitives and multi-tier review architecture; Governance-First Architecture work commits to substrate-content commitment over reasoning-execution operations specifically; AWS Deterministic Governance Engine literature articulates tier-decision logic at single-perimeter scope through rule-based allow/deny, quorum and voting structures, priority policies, and escalation thresholds as human-authored configuration with no LLM calls in the governance loop. Adimulam et al. (2026) “The Orchestration of Multi-Agent Systems: Architectures, Protocols, and Enterprise Adoption” supplies unified orchestration architecture treating coordination across many agents as architectural object rather than as protocol or framework configuration. Oversight-board institutional layers supply complementary institutional-register instances. The CAIA paper (Nguyen 2026, SSRN preprint) commits to regulatory and political-institutional governance for compute, energy, bandwidth, and storage infrastructure

resources at population scope — distinct from Paper 3’s architectural-pattern register and distinct from the training-time-versus-runtime distinction Constitutional AI occupies.

Paper 3’s Claim 5 (§8) engages this lineage at the recursive-governance angle directly — DAO constitutional architectures, the CNCF Automated Governance Maturity Model, Anthropic’s Constitutional AI, and CAIA as commitment-(b) closest neighbors with the explicit register-mismatch flag positioning CAIA outside the closest-neighbor synthesis through its commitment-plane difference. Claims 1, 3, and 6 engage adjacent ground at substrate-content commitment, three-tier conflict handling, and population-scale composition respectively. What Paper 3 adds across the lineage — developed in §§4 and §8 — is the configuration in which configuration-as-substrate-content commits to substrate content all the way down with recursive applicability as architectural property of the substrate-content commitment itself, rather than as separate hierarchical meta-governance layer above a substantive-governance plane. The recursion bottoms at human-authored governance authority per Paper 1 §3.3 rather than at protocol invariants, training-time objective specifications, or institutional procedures. The lineage commits to meta-governance as a separate hierarchical layer or to governance-as-substrate-content at varied registers (training-time, infrastructure-resource, single-perimeter); the runtime substrate-content commitment with full self-amendment of the configuration substrate operating under all six Paper 1 commitments at architectural-pattern register is not jointly specified across the contemporary work the lineage anchors.

3.6 Trilogy continuity with Papers 1 and 2

Paper 3 is one-step extension of Papers 1 and 2 jointly, with the substrate-mediator commitment progressing across the inter-Self perimeter. Paper 1 (Li, 2026a) establishes the substrate as the medium of coordination within a cell, defending six architectural commitments through five core claims and one extension at cell scope. Paper 2 (Li, 2026b) takes the modular cell as starting unit and develops the architectural theory of how cells compose into aspects and Selves through life and evolution under unified human governance, with the foundational instinct/reasoning separation as the cross-claim spine. Paper 3 takes Paper 2’s complete CKS-governed AI Self as starting unit and extends substrate-mediated coordination one further step across the inter-Self perimeter. Trilogy continuity is rendered explicitly across all six claims of Paper 3 — every claim’s §X.1 anchors in Papers 1 and 2 specifically and inherits without redefense, with the inheritance shape varying across claims (heaviest at Claim 6 with seven sources, medium at Claims 2 through 5 at four to six sources, with Paper 1 plus Paper 2 jointly as the trilogy-foundation reference for all six claims). Paper 1 and Paper 2’s prior-art engagements — OIDA, Knowledge Objects, Cognitive Core, MACI, control-plane lineage at Paper 1; dual-process AI, modular AI architectures, biology of evolvability, enterprise AI deployment at Paper 2 — are referenced rather than redone where Paper 3’s territory overlaps with theirs. The substrate-mediator ladder running from intra-cell scope through intra-Self scope through inter-Self scope through population scope, all under unified human governance, is observation rather than roadmap: Paper 1 did not promise Paper 3, and Paper 2 explicitly bracketed inter-Self autonomous dynamics as out of scope. The ladder becomes visible in retrospect because Paper 3 completes it.

3.7 The four-family closest-neighbor structure at trilogy-completion register

Paper 3’s most direct contemporary closest neighbors at trilogy-completion register organize through the family structure §9.3 surfaces. Where Claim 1’s closest-neighbor organization is by territory, Claim 2’s by cluster, Claim 3’s by tier, Claim 4’s by locus, and Claim 5’s by commitment, Claim 6’s organization is by family — four closest-neighbor families plus adjacent territories. The family organization is the sixth distinct organizational shape across Paper 3’s claims; the six-organizational-pattern across Claims 1–6 (territory / cluster / tier / locus / commitment / family) renders the discipline visible at trilogy-level paper structure (§11.2 develops this as a trilogy-level observation). The four-family structure supplies the structural anchor for paper-level Related Work synthesis at the trilogy-completion register: each family operates at a distinct cross-organizational register at population scope, with families clustering by which two of three architectural axes — substrate-content commitment at population scope, joint authority across population-level governance perimeters, trilogy-completion composition versus emergence-as-fresh-property framing — the family holds and which one it falls short on for specific structural reasons.

The first family is **orchestration and meta-governance blueprints**. Adimulam et al. (2026) supplies unified orchestration architecture treating coordination across many agents as architectural object rather than as protocol or framework configuration; meta-governance architectures propose meta-governance layers as explicit architecture objects with the architecture-as-object framing extended to governance machinery. Both commit to substrate-content commitment at intra-organizational scope and to composition rather than emergence as the architectural-honesty register. Where the family falls short: joint authority never crosses into a population substrate across firms — the orchestration-and-meta-governance work operates within one organization’s authority perimeter, with cross-organizational coordination either deferred to contract/institutional register or framed as out of scope.

The second family is **agent-contracts and governed-reasoning frameworks**. Agent Contracts (Ye and Tan 2026) lands at resource-budget register through the formal $C=(I,O,S,R,T,\Phi,\Psi)$ tuple with conservation laws on delegated budgets, supplying substrate-content commitment specifically over delegated computational resources. Cognitive Core and Governance-First Architecture work lands at decision-substrate register through typed primitives and multi-tier review architecture, supplying substrate-content commitment specifically over reasoning-execution operations. Each illustrates one register where substrate-content commitment lands; what each does not extend to is cross-organizational joint authority at architectural-pattern register at population scope, or population-scale dynamics framed as composition rather than as out-of-scope. Substrate-content commitment lands at intra-organizational scope; cross-organizational extension is deferred.

The third family is **enterprise-MAS-governance blueprints**. Governing Multi-Agent AI Systems enterprise blueprints supply enterprise-wide governance substrate at population-of-agents scope through central agent registry as system-of-record (agent identity, ownership, risk tier, prompts, tools, evolution history), policy-driven interaction graphs, and conflict-resolution rules. The blueprint pattern is the closest contemporary instance of population-of-agents-as-architectural-object at single-organization scope — substrate-content commitment over agent registry holds; composition rather than emergence as the architectural-honesty register holds. Where the family falls short: the enterprise as one organization is the single sovereign authority by construction. Cross-organizational extension would require introducing joint authority across distinct governance perimeters — the architectural commitment Family 3’s single-org register does not make.

The fourth family is **federated-learning consortia at population scope**. FLARE-based, FATE-based, and OpenFL consortia at hundreds-of-organizations scale across healthcare, pharma, government labs, and the financial sector supply population-scale federated learning coordinator-mediated under consortium agreements. The cross-organizational scope is real and mature — joint authority across distinct governance perimeters is encoded through consortium charters, governance committees, data-use agreements, and coordinator-selection protocols, and capability scaling delivers functional cross-organizational coordination at scope honestly. Where the family falls short on substrate-content commitment at population scope: joint authority lives in contracts and committees rather than in aggregation substrate — gradient aggregation is the substrate, with governance attached around it through consortium-governance contracts rather than as substrate-content commitment within the aggregation itself. Where the family falls short on composition versus emergence framing: population-scale dynamics framed as emergent from FL algorithms and data heterogeneity rather than as composition of architectural commitments — the convergence, divergence, specialization, and capability-propagation phenomena FL consortia exhibit are framed at the algorithm-and-data-heterogeneity register rather than at the architectural-pattern register. Population-scale CKS-governed Self deployments do not yet exist; the architectural commitment is to what the inheritance produces at population scope, with specific population-scale dynamics — the precise trajectories of pattern propagation, the actual rates and shapes of specialization-preserving divergence, the empirical character of collective adaptation — awaiting empirical observation. The architectural commitment is rendered confidently; the empirical-verification status of the population-scale extension is rendered with calibrated humility, anchored to the substrate-mediator-ladder-completion at the trilogy’s sixth rung as the trilogy’s anticipated landing rather than its empirically demonstrated outcome.

Adjacent families fall further short. **Agent-protocol ecosystems** (MCP, A2A, NIST AI Agent Standards Initiative, Agentic AI Foundation, OpenID Foundation AI Identity Management Community Group) supply strong agent-coordination protocol-and-meta-governance work but fall short on substrate-content commit-

ment and on architectural-pattern joint authority: protocols are substrates for context and messaging not governance objects, and joint authority lives in institutional governance rather than in protocol substrate. **Complex adaptive systems, swarm intelligence, agent-based modeling, and evolutionary game theory work** — including Park et al.’s (2024) thousand-agent generative simulations and the broader collective-intelligence research direction — fall short on substrate-content commitment (modeled population not authored governance substrate) and on composition framing (explicitly emergence-requiring rather than composition-rendering). Across all four families and adjacent territories the failure mode is consistent: substrate-content commitment exists inside single organizations; joint authority across population-level governance exists in legal and institutional space; population-scale dynamics framed as composition exists at intra-organizational scope; population-scale dynamics framed as emergence exists at simulation and analysis scope. The conjunction at cross-organizational population scope at architectural-pattern register is unoccupied across all four families and adjacent territories.

3.8 Where Paper 3 sits

Paper 3 sits at the intersection of these six lineages, not as a single contribution to one lineage. Multi-agent systems and inter-agent coordination supply the architectural shape of agent-to-agent coordination and the contemporary protocol and framework primitives; federated learning and cross-organizational ML supply the cross-organizational coordination scope and aggregation lineage; governance-as-code and policy-as-substrate work supply the configuration-as-content vocabulary; multi-coordinator and joint-authority architectures supply joint-authority instances at varied registers; substrate-as-content and recursive-governance literature supplies the substrate-content commitment vocabulary at varied registers; and trilogy continuity with Papers 1 and 2 supplies the substrate-mediator commitment progressing across the inter-Self perimeter at trilogy completion. Paper 3’s contribution is the configuration in which all of this is articulated as one design pattern at the inter-Self perimeter and at population scope — substrate content as the medium of inter-Self coordination, joint authority across distinct governance perimeters as architectural property of the technical substrate, configuration as substrate content with recursive applicability, and population-scale collective evolution as composition of these commitments under joint authority across population-level governance perimeters. Each claim’s per-claim related-work engagement establishes the contribution within its specific territory; this section’s contribution is to name where Paper 3 sits at the paper level, in the empty configuration the lineages do not jointly occupy.

The contribution genre is design-pattern writing in the same sense Papers 1 and 2 inhabit: an articulation of a recurring structure of design problem and design response, named so that subsequent work can adopt it, vary it, compose it with adjacent patterns, or argue against it. Paper 3 is not a multi-agent communication protocol specification, not a vendor product, not a novel algorithm, and not an empirical evaluation at scale. Future work — formal protocol specifications instantiating the FAI mechanism, security-guarantee specifications at the substrate-content register, message schemas for the substrate-mediator extension across organizational boundaries, deployment architectures for shared-substrate construction and dissolution dynamics, and empirical validation at scale — is downstream of the architectural pattern Paper 3 specifies. Capability scaling delivers functional inter-Self coordination at scale honestly across foundation lab multi-agent systems, agent-protocol ecosystems, and federated-learning consortia at population scope; the differentiation Paper 3 commits to is at architectural-commitment register rather than at the functional-coordination claim. Population-scale CKS-governed Self deployments do not yet exist; the architectural commitment is to what the inheritance produces at population scope, with specific population-scale dynamics awaiting empirical observation. The architectural commitment is rendered confidently; the empirical-verification status of the population-scale extension is rendered with calibrated humility, anchored to the substrate-mediator-ladder-completion at the trilogy’s sixth rung.

4. Claim 1 — The shared substrate as the architectural object of inter-Self coordination

4.1 Claim statement and the substrate-mediator extension across the inter-Self perimeter

Claim 1 establishes Paper 3’s foundational architectural commitment: the **shared substrate** is the architectural object through which two or more CKS-governed Selves coordinate with each other. The medium of inter-Self coordination is itself a CKS substrate, constructed temporarily for an interaction, with all six Paper 1 (Li, 2026a) commitments holding within scope and the distinctive property that its governance perimeter spans more than one Self’s home governance perimeter. Content within carries provenance crossing organizational boundaries; the configuration of the crossing is itself substrate content under joint human authority. The architectural move is a one-step extension of a commitment Papers 1 and 2 (Li, 2026b) make at progressively larger scopes — substrate-mediated coordination under unified human governance — across the inter-Self perimeter. Paper 1 establishes the substrate as the medium of coordination within a cell; Paper 2 extends it as the spine of a Self with instinct (the LLM) and reasoning (the CKS substrate) composing under unified governance through the instinct/reasoning separation; Paper 3 extends it one further step, to the medium between Selves.

The shared substrate is constructed for an interaction and dissolves on completion, with persistence governance-configured per event from nothing retained at one extreme to the full substrate as durable record at the other. The unit of contribution is the aspect from Paper 2 — when a Self contributes content to the shared substrate, the contribution surfaces the constituent cells, DNA-layer content, and action-layer content of contributed aspects. The exchange is bounded to substrate content per Paper 2’s instinct/reasoning separation: DNA-layer and action-layer content exchange; LLM weights and instinct-layer content do not. As §5 develops, the canonical operation over the shared substrate is Full Aspect Integration, with merge semantics, exchange-bounding mechanics, and cardinality cases as that section’s territory. Claim 1’s contribution is the substrate those operations require: a temporarily constructed, perimeter-spanning, governance-configured-persistence shared substrate carrying all six Paper 1 commitments within scope.

The architectural move is an extension rather than a replacement. Paper 1 defended the substrate-as-medium commitment at cell scope, with the substrate/LLM division drawn at the governance boundary rather than the capability boundary. Paper 2 extended that commitment across cells, aspects, and the Self under the instinct/reasoning separation, with the substrate continuing to carry coordination, governance, and reasoning while the LLM continues to carry fast-pattern instinct. Claim 1 carries the same commitment into a third scope. The architectural object — human-governed substrate as coordination medium — operates between Selves as it operates within them, with the same six commitments holding and the perimeter-spanning property added. Papers 1 and 2 properly bracketed inter-Self territory because the architectural commitment for that scope had not yet been established. Claim 1 establishes that commitment.

The named architectural foil is **opaque agent-to-agent communication**, in which inter-AI exchange is conducted through message-passing, tool-calls, or shared-memory access patterns whose substantive content is not human-governable as authored substrate. Paper 3’s alternative is substrate-mediated, aspect-level inter-Self coordination in which the exchange medium itself is first-class human-governed substrate content from the start. The next subsection develops the foil through the 2024–2026 critical literature on multi-agent AI systems and inter-agent protocols.

4.2 The named foil — opaque agent-to-agent communication and its structural limits

The foil Claim 1 positions against is the dominant pattern of inter-AI exchange in 2024–2026 multi-agent and inter-agent-protocol systems: communication conducted through messages, untyped state diffs, tool-call payloads, or memory-access patterns whose substantive content is not first-class human-governable substrate. Articulating the foil requires architectural-pattern-register engagement with the contemporary critical literature, not capability or implementation-specific limits. The literature converges on a structural

assessment.

The Vector Institute’s “Transparency in Agentic AI” survey characterizes most multi-agent frameworks as trajectory-opaque: they expose final outputs and tool traces but do not make multi-agent dialogue and routing decisions first-class queryable objects, leaving what Raza and colleagues describe as a trajectory-level interpretability gap. Institutional AI work on governing collusion in multi-agent LLM markets introduces a separate governance graph and append-only governance log specifically because ungoverned baseline frameworks lack formal representation of origin, authority, or sanctions attached to inter-agent messages. AgentAuditor (2026) introduces a reasoning tree because existing frameworks discard the evidential structure of reasoning traces through majority-vote aggregation. The TechRxiv survey on memory in LLM-based multi-agent systems observes that multi-agent memory infrastructures store information as chunks or messages without persistent queryable links to the originating agent or decision point. Practitioner literature crystallizes the same pattern: Atlan’s 2026 “Agent Memory Architectures” comparison observes that “governance is mostly bolted on, not part of the architecture,” with the qualification that governed catalogs and knowledge graphs can enforce access control but that the governance comes from the data platform rather than the multi-agent framework itself.

The synthesis across these sources distills four structural gaps: typed governance-aware inter-agent substrate; first-class causal traces and distributed provenance; enforceable permission and compartmentalization models; dedicated oversight and control plane. The properties name the foil structurally — the medium of inter-agent exchange is not first-class human-governable substrate content, and governance is treated as a wrapper around an opaque or semi-opaque exchange. What matters for Claim 1 is that these are architecture-class findings rather than capability-debt findings. If the foil’s limits were capability gaps closable by better LLMs or richer protocols, the architectural response would be premature — sharper LLMs and richer protocols would absorb the substrate Claim 1 commits to. The foil literature denies that reading. The limits are properties of the architecture class, where governance is layered around an exchange whose substantive content is not authored substrate. The appropriate response is therefore architectural, and Claim 1 names one: the medium of inter-Self coordination is itself first-class human-governed substrate content with all six Paper 1 commitments holding within scope.

4.3 Closest-neighbor engagement at tied-top across axes

Paper 2 Claim 1’s differentiation engaged a single dominant closest neighbor on the dual-process AI axis — SOFAI (Booch et al. 2021) and the broader S1/S2 architectural family. Paper 3 Claim 1 differs structurally. Four neighbor clusters are tied-top, each closest on a different architectural axis, with the four clusters jointly bounding the territory the shared substrate occupies. Forced ranking would distort the differentiation by elevating one cluster over the others on grounds that don’t track architectural distance. The four-cluster structure is what the territory looks like.

LangGraph multi-agent is closest on the architectural-pattern axis. It commits most explicitly among contemporary multi-agent frameworks to typed shared state with directed graph routing, reducer logic for merging concurrent updates, and durable execution with checkpointing — the closest contemporary multi-agent framework primitive to the typed coordination state Claim 1 commits to extending across the inter-Self perimeter. Where LangGraph stops short: the state contains untyped message lists and user-defined fields rather than substrate content with provenance and conflict-preservation as architectural properties; persistence is a transport/runtime feature rather than substrate content under joint governance configurable per event; the graph operates within a single deployment perimeter rather than across distinct Selves’ home governance perimeters.

A2A (Google) is closest on the inter-vendor-coordination axis. A2A is the most explicit contemporary commitment to cross-vendor agent coordination at protocol-spec register, with Tasks, Messages, and Artifacts as envelopes for agent-to-agent traffic, fifty-plus enterprise partner ecosystem, and explicit inter-organizational deployment scope. Where A2A stops short: the protocol commits to envelope traffic, is stateless by design with persistence pushed to applications, and externalizes governance to the application layer. The architectural register difference is load-bearing — A2A is a message-format-and-transport specification; Claim 1 commits at architectural-pattern register to substrate content as the exchange medium itself. Future

work may specify protocols that instantiate Paper 3’s commitments; A2A and similar specifications are downstream of the architectural commitment Claim 1 names.

Gaia-X and IDS-RAM jointly are closest on the perimeter-spanning axis. Both supply mature reference architectures for cross-organizational data exchange under federated governance. Gaia-X commits to federation services, Data Usage Agreements, and self-descriptions; IDS-RAM commits to Provenance Storage and Collection Points, usage-control policies enforced at connectors and clearing houses, and trust frameworks. Where they stop short: governance is encoded as a layer around exchanges (membership, connectors, clearing houses, contracts) rather than as authored substrate content within the exchange; persistence defaults to infrastructure rather than configurable substrate content per event; provenance is articulated explicitly but is optional in practice and externalized rather than intrinsic to the substrate content the exchange manipulates.

The federated learning, MPC/TEE, and data-trusts cluster is closest on the joint-authority-mechanics axis. Federated learning supplies orchestrator archetypes, aggregation algorithms, and a 2024–2026 governance literature on cross-organizational ML coordination; MPC and TEEs supply joint authority over computation through threshold structures and attestation; data trusts and cooperatives instantiate joint authority at organizational and legal layer through trustor-trustee-beneficiary structures and member-controlled governance arrangements. Where the cluster stops short: federated learning exchanges parameters or model updates with governance applied around the aggregation; MPC and TEEs encode joint authority in protocol parameters and attestation rather than in substrate content the protocols manipulate; data trusts and cooperatives locate joint authority in institutions and contracts rather than in the architecture’s substrate. Federated learning’s substrate is gradient aggregation with governance attached around it; the shared substrate makes provenance, authority, and commitments intrinsic to the inter-Self state itself. The honesty discipline matters here: federated learning embodies non-trivial authority structure, and decentralizing optimization does not automatically decentralize governance. The differentiation against the cluster is on what the joint authority operates over — substrate content versus aggregation protocol — not on whether joint authority exists at all.

Two further notes on closest-neighbor scope. MCP warrants brief acknowledgment as adjacent rather than central: it is agent-to-tool with a “capability bus” framing carrying client-server tool-access traffic, not substrate content shared across coordinating Selves. Solid (Berners-Lee’s project) occupies adjacent personal-data-sovereignty register: pods are personal-data substrates with cross-perimeter coordination through standardized authentication and access-control mechanisms under W3C’s Solid Project standards; the architectural-pattern register difference is that Solid’s substrate is personal-data-as-content rather than coordination-as-content, with the perimeter-spanning property at the access-control layer rather than at substrate authoring. ANP, IBM ACP, IETF and W3C drafts, and emerging UCP-style proposals occupy adjacent protocol-spec territory — the most relevant 2024–2026 standardization activity for inter-agent communication, referenced rather than engaged at depth because they share with A2A the architectural-register difference (protocol-spec versus architectural-pattern) the cluster engagement above developed. The four-family closest-neighbor structure at trilogy-completion register that §3 Related Work develops places these adjacent protocol-spec activities within the broader paper-level positioning.

4.4 Three differentiation axes

The differentiation work above runs on three architectural axes simultaneously. The axes are independent — each holds against neighbors that meet others — and the conjunction is where the architectural contribution lives. The pattern follows Paper 1 §5.2 (OIDA differentiation) and Paper 2 §4.4 (SOFAI differentiation): three axes carrying differentiation jointly, with the *architecturally different from X on Y axis, not X plus Y* framing rendering the conjunction.

The first axis is **exchange-medium-as-substrate-content versus exchange-medium-as-message, tool, or memory-trace**. In the shared substrate, the medium of inter-Self exchange is itself human-governed substrate content from the start, with all six Paper 1 commitments holding within scope. In multi-agent LLM frameworks, the medium is messages, shared state, or tool-call payloads (AutoGen GroupChat, CrewAI task descriptions, LangGraph state dicts, OpenAI Agents SDK handoffs). In inter-agent protocols, the medium is envelope traffic at protocol-spec register (A2A, MCP). In none is the

medium itself first-class human-governable substrate content; governance is applied as a wrapper around an opaque or semi-opaque exchange. This axis is most load-bearing against the named foil and against the multi-agent-AI restatement objection. Claim 1 is architecturally different from LangGraph on the substrate-as-content axis, not LangGraph plus governance.

The second axis is **governance-perimeter-spanning with configuration as substrate content versus opaque cross-perimeter exchange**. The shared substrate’s governance perimeter spans more than one Self’s home governance perimeter, and the configuration of the spanning is itself substrate content under joint human authority. In Gaia-X and IDS-RAM, governance of perimeter-spanning is encoded at the membership, trust-framework, or contract layer rather than as authored substrate within the exchange. In federated learning, joint authority over aggregation is articulated through orchestrator archetypes and aggregation algorithms. In MPC and TEEs, joint authority is encoded in protocol parameters and attestation. In data trusts and cooperatives, it lives in institutions and contracts. This axis is most load-bearing against cross-organizational neighbors and against the data-space-restatement objection. Gaia-X and IDS-RAM occupy perimeter-spanning territory; the differentiation against them on this axis runs on configuration-as-substrate-content. Claim 1 is architecturally different from Gaia-X and IDS-RAM on the configuration-as-substrate-content move within perimeter-spanning, not Gaia-X or IDS-RAM plus inter-Self extension.

The third axis is **temporary construction with governance-configured persistence versus persistent shared infrastructure as default**. The shared substrate is constructed for an interaction and dissolves on completion, with persistence governance-configured per event from nothing retained to the full substrate as durable record. The dominant pattern in adjacent territories treats shared coordination state either as persistent shared infrastructure (Gaia-X federation services, IDS connectors, Solid pods) or as session-bounded with ad-hoc external persistence (AutoGen, CrewAI). LangGraph’s durable execution comes closest as runtime feature, but persistence is treated as transport concern rather than substrate content per event. Federated learning’s global model persists across rounds, but persistence is bound to the algorithm rather than to substrate content. This axis is most load-bearing against the design-choice objection — persistent shared infrastructure is one configuration option (full retention) within the architecture, not a shape the architecture is forced into.

The conjunction carries the architectural contribution. None of the axes alone suffices: Axis 1 without Axis 2 is Paper 2’s intra-Self territory, where substrate-as-content holds within a single Self’s perimeter without crossing into joint authority across perimeters; Axis 2 without Axis 1 is the Gaia-X / IDS-RAM pattern, where perimeter-spanning under federated governance holds without the substrate-content commitment the exchange medium itself would carry; Axis 3 without Axes 1 and 2 is the AutoGen / CrewAI pattern, where session-bounded coordination state holds without either substrate-content or perimeter-spanning commitments. What is unoccupied is the conjunction: substrate-mediated coordination at inter-Self scope, perimeter-spanning under joint governance with configuration as substrate content, temporary by default with persistence as substrate-content-configurable per event, all six Paper 1 commitments holding within scope. **Primary novelty weight is shared across all three axes** per the conjunction logic — Axis 2 specifically carries the configuration-as-substrate-content commitment that §8 will elaborate as recursive applicability across all configurable dimensions of the FAI mechanism.

4.5 Reviewer-objection responses

Four objections anticipate themselves with substantial force; four others surface on specific terms and warrant briefer treatment.

The first objection — *this is just multi-agent AI with governance vocabulary* — has surface plausibility. The response runs on the first axis. The shared substrate’s commitment is that the exchange medium itself is first-class human-governed substrate content with all six Paper 1 commitments holding within scope, while multi-agent LLM frameworks treat inter-agent communication as message traffic, untyped state diffs, or tool-call payloads with governance as wrapper. The reviewer’s surface claim collapses two architectural registers: the framework-API register at which LangGraph, AutoGen, and CrewAI operate, and the architectural-pattern register at which Claim 1 commits. Paper 1 defended the governance-boundary commitment at cell scope, drawing the substrate/LLM division at the governance boundary rather than the capability boundary;

Paper 2 extended it through the instinct/reasoning separation at Self scope. Claim 1 extends that already-defended commitment across the inter-Self perimeter, architecturally distinct from retrofitting governance onto frameworks built around message-passing.

The second objection — *this is just A2A or MCP or similar with extra commitments* — collapses architectural-pattern register with protocol-spec register. A2A is an envelope-and-transport specification, stateless by design and externalizing governance to the application layer; MCP is agent-to-tool with the capability-bus framing rather than agent-to-agent with substrate content as exchange medium; the broader IETF and W3C standardization activity is at protocol-spec register. The architectural-pattern register where Claim 1 commits is upstream of protocol specification. The substrate-as-content commitment, the perimeter-spanning configuration commitment, and the temporary-with-configurable-persistence commitment all bind at architectural-pattern level — upstream of envelope format, transport mechanism, and message schema. Future work may specify protocols that instantiate these commitments; the protocol specifications would be downstream artifacts of the architectural pattern Claim 1 names.

The third objection — *this is just classical blackboard architecture under a new name* — is the strongest lineage objection. HEARSAY-II, BB1, GBB, distributed cognition (Hutchins 1995), common ground in collaborative communication (Clark 1996), and the boundary objects literature (Star and Griesemer 1989) supply conceptual ground for joint coordination through shared informational state. Claim 1 acknowledges the lineage rather than claiming novelty against it. Star and Griesemer’s boundary objects specifically operate through plasticity and robustness — informational structures hold meaning across communities (amateur collectors, professional scientists, administrators, patrons in their original museum case) while preserving local interpretations, with governance encoded in institutional ecology and translation work across the shared structure rather than as authored substrate content within the boundary object itself. The structural parallel to inter-Self FAI events is real: shared informational structure operates across distinct authority perimeters with provenance preserved across communities. The architectural difference is at the locus of governance encoding. What none of the lineage formalizes is governance as first-class architectural property in Paper 1’s specific sense: blackboards are intra-system coordination mechanisms within a single deployment perimeter; distributed cognition and boundary objects treat governance as institutional or contextual rather than as authored substrate content within the shared informational structure. The differentiation against Claim 1: shared substrate’s boundary-object analog is authored substrate content with governance internal to the substrate’s commitments per Paper 1’s six commitments holding within scope, while classical boundary objects rely on translation work across the shared structure under separate institutional authority. Perimeter-spanning with configuration as substrate content under joint authority, temporary construction with governance-configured persistence, and the six Paper 1 commitments holding within scope are not present in the lineage’s design space.

The fourth objection — *this is just federated learning, federated knowledge graphs, or a data space with new vocabulary* — runs against the cross-organizational governance literature. The response runs on three axes simultaneously. Federated learning shares parameters and data spaces share envelopes while the shared substrate exchanges aspect-level substrate content; federated knowledge graphs and data spaces assume schema alignment or rule-based usage control while the shared substrate handles conflicts as first-class state by inheritance from Paper 1; Gaia-X and IDS default to persistent infrastructure while the shared substrate is temporary with configurable persistence. The honesty discipline matters: differentiation against federated learning is not that federated learning lacks governance — federated learning embodies non-trivial authority structure across consortia, and decentralizing optimization does not automatically decentralize governance. The differentiation is that federated learning’s substrate is gradient aggregation with governance attached around it; the shared substrate makes provenance, authority, and commitments intrinsic to the inter-Self state itself.

Four further objections admit briefer treatment. *Why temporary construction rather than persistent infrastructure as default?* Persistent shared infrastructure is one configuration option (full retention) within the architecture, not the default; the temporary-by-default commitment with persistence governance-configured per event is what supports the full configuration range from nothing-retained to durable-record. *What are the mechanics of perimeter-spanning?* Claim 1 commits at architectural-pattern register; the joint-authority-mechanics neighbor cluster surveyed above (federated-learning orchestrator archetypes, MPC threshold struc-

tures, TEE attestation, data-trust trustor-trustee-beneficiary structures) supplies the mechanics inventory under which the architectural pattern operates, with the configuration of the mechanics itself substrate content under the configuration-as-substrate-content dimension §8 will develop. *Isn't this just CKS at a bigger scope?* Paper 3 introduces one new architectural object — the shared substrate with governance perimeter spanning more than one Self's home perimeter — not present at prior-paper scopes; Papers 1 and 2 properly bracketed inter-Self territory because the architectural commitment for that scope had not yet been established, and the disambiguation is master-synthesis-stage work the §2 cross-claim spine renders. *Don't foundation labs solve this anyway through capability scaling?* CKS is governance-motivated, not capability-motivated; the property at stake is whether inter-Self coordination is human-governable as authored substrate content, regardless of model capability. §9 develops the population-scope engagement with capability-motivated work in full; the brief response holds at Claim 1's scope.

4.6 Papers 1 and 2 inheritance and downstream placement

Claim 1 inherits jointly from Papers 1 and 2 (Li, 2026a; Li, 2026b). From Paper 1, all six commitments — substrate-based hybrid composition with a governance boundary, conflict preservation as first-class state, human-governed authority, AI as substrate mediator, tool-agnosticism at the substrate layer, and linear-cost composition — hold within the shared substrate by inheritance. The *outside-the-model vs. inside-the-model* spine extends through Paper 3 unchanged: inter-Self coordination, governance, and conflict-handling live in the shared substrate outside the LLM, while instinct-layer content remains within each Self's LLM and does not cross the perimeter. Paper 1's prior-art engagements (OIDA, Knowledge Objects, Cognitive Core, MACI, control-plane lineage) are referenced rather than redone; Claim 1 does not re-litigate the cell-scope substrate-as-medium commitment Paper 1 defended. The actor-neutrality discipline from Paper 1 §3.3 holds throughout: human-governed does not mean human-authored. Humans hold authority over the shared substrate's structure, configuration, and modification; the labor is available to humans directly or to LLMs operating under human direction.

From Paper 2, the three-level structure (cell, aspect, Self), the DNA-and-action-layer distinction, the life-cycle and evolution operations within each Self's home perimeter, and the instinct/reasoning separation as the trilogy spine all hold without modification. The DNA-layer / action-layer distinction defines what exchanges through the shared substrate; LLM weights and instinct-layer content do not. Paper 2's prior-art engagements (SOFAI and the dual-process AI lineage, modular AI architectures, biology of evolvability, neuro-symbolic-probabilistic architectures, enterprise AI deployment patterns) carry forward without modification. The substrate-mediator ladder Paper 2 articulated within a Self runs from Paper 1 within-cell through Paper 2 within-Self to Paper 3 inter-Self, each scope inheriting prior commitments without replacing them. The §11 Discussion section develops the ladder's six-rung articulation across the trilogy as a paper-level pattern; Claim 1 contributes the inter-Self rung specifically.

Claim 1 establishes the architectural object Claims 2 through 6 operate over. As §5 develops, Full Aspect Integration is the canonical operation over the shared substrate, with merge semantics, exchange-bounding, and cardinality cases as that section's territory. As §6 develops, three-tier inter-Self conflict handling — preserve as first-class substrate state, resolve via configured orchestration as substrate content, escalate to humans across joint authority of participating Selves' governance — operates within the shared substrate. As §7 develops, FAI events feed Paper 2's evolution machinery through governance-configured ingestion at each home perimeter, with a layer-routing rule that holds DNA-layer content feeding DNA evolution at home, action-layer content feeding action-feedback evolution at home, and instinct evolution taking no FAI input by architectural commitment per the exchange-bounding §5 establishes. As §8 develops, the configurable dimensions surfaced across §§5–7 are themselves substrate content with recursive applicability bottoming at human-authored governance authority per Paper 1 §3.3. As §9 develops, FAI events accumulating across a network of CKS-governed Selves constitute a collective evolution mechanism at population scope under joint authority across population-level governance perimeters, with the calibrated-humility voice anchor appropriate to the trilogy-completion register. Claim 1's contribution is the substrate those operations require: a temporarily constructed, perimeter-spanning, governance-configured-persistence shared substrate carrying all six Paper 1 commitments within scope, with the configuration of the perimeter-spanning itself substrate content under joint human authority.

5. Claim 2 — Full Aspect Integration as the canonical operation over the shared substrate

5.1 Claim statement and FAI as the canonical operation over the shared substrate

Claim 2 establishes **Full Aspect Integration (FAI)** as the canonical operation over the shared substrate §4 establishes. The architectural move continues Papers 1 (Li, 2026a) and 2 (Li, 2026b)’s substrate-mediator commitment one further step: Paper 1 establishes the substrate as the medium of coordination within a cell; Paper 2 extends it as the spine of a Self under the instinct/reasoning separation; §4 of Paper 3 establishes the shared substrate as the medium across the inter-Self perimeter; and Claim 2 specifies what operates over that shared substrate. The unit of exchange is the **aspect** from Paper 2 — when a Self contributes content to the shared substrate during an FAI event, the contribution surfaces the constituent cells, DNA-layer content, and action-layer content of contributed aspects. The architectural default is full merge of contributed aspects within the shared substrate, with whether full merge is performed and which aspects each Self contributes governance-configured per event. Cardinality is n-ary, with two-Self FAI as the simplest instance; the cardinality floor at two reflects Paper 3’s scope addressing two-or-more-Self cases, while one-Self self-derived combination remains Paper 2’s intra-Self territory rather than constituting an inter-Self FAI event.

The exchange is bounded to substrate content per Paper 2’s instinct/reasoning separation: DNA-layer content (orchestration substrates, behavior substrates, schemas, rules) and action-layer content (recorded task instances, outputs, lived experience) exchange through the shared substrate; LLM weights and instinct-layer content do not. The boundary is articulated as architectural commitment at the FAI mechanism level for the inter-Self perimeter specifically; the underlying separation is Paper 2 inheritance carrying through the shared substrate §4 establishes.

FAI is the inter-Self analog of Paper 2’s intra-Self combination primitive — the same architectural posture (governable primitive over content combination; three pattern variants under orchestration-configured selection — union, selective merge, lineage-preserved union; multi-cardinality flexibility; cross-lineage compatibility) extended across the inter-Self perimeter. The pattern-variant configurability is Paper 2 inheritance; the architectural-default commitment to full merge specifically is fresh at inter-Self scope, since Paper 2 does not name a default among its three pattern variants at intra-Self scope. The structural analog matters because Claim 2 is **mechanism-defense work**: the architectural commitment must be defended against two specific failure modes that the inheritance density makes available. The first is the *duplicated-with-Paper-2 misreading* — collapsing Claim 2 into Paper 2’s intra-Self combination primitive at bigger scope, missing the perimeter-spanning context Claim 1 inheritance applied and the architectural-default commitment fresh at inter-Self scope. The second is the *biology-as-metaphor reading* — taking the inter-Self analog framing as an analogical extension of Paper 2’s biology engagement rather than as architectural commitment at FAI mechanism level. Paper 2’s biology engagement is bounded to its intra-Self scope per Paper 2 §2.2’s biology-as-bounded-load-bearing-terminology framing and Paper 3 core theory’s biology-positioning subsection; FAI vocabulary at Paper 3’s voice register replaces the intra-Self combination vocabulary at this scope, and the architectural commitment is carried by the substrate-mediated mechanism rather than by the analogical resonance. The mechanism-defense calibration register that the rest of the section operates within takes both failure modes as live and develops the differentiation work that locates Claim 2’s contribution precisely.

The named architectural foil at this section’s scope is **opaque inter-agent merge of outputs without governable substrate** — a sharper rendering, at the merge-mechanism level, of the opaque agent-to-agent communication foil §4 develops. The next subsection develops this merge-mechanism foil through the contemporary multi-agent and inter-agent-protocol literature.

5.2 The named foil — opaque inter-agent merge and the merge-mechanism gap

The foil at the merge-mechanism level sharpens §4.2’s structural assessment to the specific question of what happens when two or more Selves’ outputs need to combine. In contemporary multi-agent and inter-agent-protocol systems, merge of cross-agent content is conducted through arbiter agents that select among candidate outputs, ensemble voters that aggregate at output-space without combining intermediate state,

pipeline aggregators that concatenate or transform without preserving contributor distinction, or framework-API-register reducers that operate at the typed-state-field level within one runtime. The merge logic and the intermediate state across which merge occurs are not first-class human-governable substrate content; governance is applied as a wrapper around an opaque or semi-opaque merge, and conflicts arising during merge are transient artifacts to be eliminated for the merge to complete rather than first-class state the merged result carries forward.

The foil sharpens at the merge-mechanism level in three ways §4.2’s foil articulation does not develop. First, the unit of merge in contemporary systems is heterogeneous and small-grained: typed state fields, message lists, tool-call payloads, parameter vectors, schema elements, entity-edge assertions. None of these units preserves the structural arrangement of cells with DNA and action layers that Paper 2 establishes within a Self. Second, the merge defaults are themselves heterogeneous and architecturally consequential: last-writer-wins per state key, append-to-transcript, coordinate-wise averaging, no-merge-on-partial-alignment, route-without-mixing. The defaults carry implicit commitments about what survives a merge and what is erased; absent first-class substrate, those commitments are not inspectable as substrate content. Third, governance over merge — when it is articulated at all — operates at framework configuration, contract, or protocol-parameter register rather than as authored substrate content within the merge itself. The merge logic is not substrate; the conflicts surfaced during merge are not substrate; the configuration of the merge is not substrate.

What matters for Claim 2 is that these are properties of the architecture class at the merge-mechanism level, not capability gaps that better LLMs or richer protocols would close. Sharper LLMs and richer protocols would absorb the substrate Claim 2 commits to; the foil literature at §4.2 denies that reading at the substrate-content level, and the same denial holds at the merge-mechanism level. The appropriate response is therefore architectural at the merge-mechanism level specifically. FAI’s alternative is substrate-mediated, aspect-level merge with the pattern variant selected by governance-configured orchestration: the merge logic itself is authored substrate content under joint human authority per §4’s perimeter-spanning property; conflicts surfaced during merge are first-class addressable substrate state by inheritance from Paper 1 carrying through §4; the pattern-variant configuration is itself substrate content with recursive applicability §8 will develop.

5.3 Closest-neighbor engagement at three tied-top clusters

§4’s closest-neighbor engagement organized the territory the shared substrate occupies as four tied-top clusters, each closest on a different architectural axis at the substrate-as-architectural-object level. Claim 2’s closest-neighbor structure is different in shape. Three neighbor clusters are tied-top at the merge-mechanism level, each closest on a different aspect of FAI’s architectural shape, with the four substrate-level clusters from §4 referenced as Claim 1 territory rather than re-engaged at depth. Distributed-systems convergence semantics supplements at supporting density. Paper 2’s Git engagement at §6.3’s union-pattern lineage occupies the closest inheritance position and is referenced rather than re-engaged at Paper 3’s voice register.

Multi-agent memory merge primitives are closest on the per-key-default and the artifact-class-configurable-merge framing. LangGraph commits to merge at the field (state-key) level via per-key reducer functions, with overwrite as the default when no reducer is specified — explicit architectural default at fine granularity within one runtime under one governance perimeter. AutoGen aggregates at message-list level with append-to-transcript as effective default. Atlan’s 2026 catalogue treats merge as a design choice rather than a single global default operator; its actor-aware-edges pattern preserves per-agent assertions distinct, sympathetic to FAI’s conflict-preservation posture by Claim 1 inheritance through §4 but operating at entity-edge granularity within one platform. The 2026 position-paper literature on multi-agent memory from a computer-architecture perspective proposes a semantic-consistency model with policy-configurable merging per artifact class and identifies multi-agent memory consistency as a central open problem rather than committing to an architectural answer at substrate scope. Where this cluster stops short: the merge unit is typed state fields, message lists, or entity-edge assertions rather than aspects with three-level structure; the merge perimeter is one runtime under one governance perimeter rather than across distinct Selves’ home governance perimeters; governance over merge is framework-API configuration rather than authored substrate content within the merge itself.

Federated learning aggregation and model fusion are closest on the named-default-aggregation axis at cross-organizational scope. FedAvg and successors (FedProx, FedAdam, FedMA, secure aggregation, Byzantine-robust variants, vertical FL) commit to data-weighted coordinate-wise averaging at full-parameter granularity as canonical default — substantive aggregation mechanism with named default across a mature literature. Model-fusion methods at LLM scale (model soups, task arithmetic, TIES, DARE) operate at full-parameter element-wise granularity with merge weights configured per task, exhibiting substantive architectural variation within the cluster: model soups (Wortsman et al. 2022) and task arithmetic (Ilharco et al. 2023) average task vectors at parameter granularity; TIES-Merging (Yadav et al. 2023) applies trim-elect-disjoint-merge to resolve sign conflicts at parameter level, sympathetic to FAI’s conflict-preservation posture in spirit through sign-conflict-information preservation; DARE (Yu et al. 2024) drops parameters and rescales for sparsity. Mixture-of-experts routing and ensemble preservation methods sit on the other side of the merge spectrum: routers select expert outputs without mixing expert weights; ensembles maintain distinct contributors and aggregate at output-space. Where this cluster stops short — and the contrast against FAI lands on three architectural axes simultaneously, developed in §5.4 directly — the merge unit is parameters or gradients at coordinate granularity rather than aspects; the merge default is averaging-with-erasure rather than full-merge-with-conflict-preservation; the merge layer is instinct-layer content in trilogy terms rather than reasoning-layer content. The TIES-Merging conflict-preservation sympathy holds at parameter-sign level rather than at substrate-state level; the architectural difference at §5.4 Axis 3 is that FAI’s conflict-preservation operates over reasoning-layer substrate content as first-class state, while TIES preserves sign-conflict information at parameter level during the merge to be resolved through trim-elect logic. This cluster also occupies the joint-authority-mechanics neighbor cluster §4.3 engaged at substrate-level perimeter-spanning; the merge-mechanism engagement here is the second of six FL boundaries Paper 3 develops across the six claims, with §3 Related Work synthesizing the cross-claim FL engagement at paper level.

Schema mapping and cross-knowledge-graph integration are closest on the alignment-as-precondition-for-merge axis. COMA, Cupid, LogMap, AML, and AgreementMakerLight produce alignments with confidence scores; merging is a separate downstream phase; unmatched elements remain distinct attributes; conflicting mappings are removed or downgraded during repair rather than forced. The 2024–2026 LLM-assisted alignment literature automates the alignment step but preserves the precondition commitment. Cross-KG subgraph transfer in TRACEKG-style pipelines treats relation instances or small patterns as the unit of transfer, with destination reconstitution under local schema preserving original semantics as metadata. The cluster is sympathetic on conflict-preservation — unaligned elements remain distinct rather than auto-reconciled — but commits to alignment as precondition for any merge, a different architectural commitment than FAI’s full-merge default with conflicts surfacing as first-class state. Where the cluster stops short for FAI’s purposes: schema-mapping operates at schema-element or ontology-term granularity rather than aspect granularity; the alignment-precondition commitment carries an implicit refuse-to-merge default that is architecturally distinct from full-merge; cross-KG integration operates at data-integration register where the layer-bounding question of what reasoning-layer-versus-instinct-layer content travels does not arise.

CRDT and operational-transformation literatures supply the broader convergence-semantics tradition at supporting density. CRDTs embed merge in data-structure type with mathematical convergence properties; OT centralizes resolution in transformation functions at server. Both target convergence over a shared distributed object rather than aspect-level integration across distinct Selves’ home governance perimeters; the architectural object differs at the level the foil articulates. Paper 2’s Git engagement at the union-pattern lineage in §6.3 occupies the closest inheritance position — the same architectural commitment to merge-time conflicts as persistent first-class substrate state, with provenance attached to the conflict and deferral as a terminal recorded outcome — extended across the inter-Self perimeter through §4’s shared substrate rather than within a Self’s home perimeter through Paper 2’s intra-Self combination primitive.

5.4 Three differentiation axes

The differentiation work above runs on three architectural axes simultaneously. The pattern follows Paper 1 §5.2 (OIDA differentiation) and Paper 2 §4.4 (SOFAI differentiation): three axes carrying differentiation

jointly, each named directly, each neighbor characterized within the axis’s territory, with the *architecturally different from X on Y axis, not X plus Y* framing rendering the conjunction. The honest weight across the three axes is asymmetric — Axis 1 carries materially more fresh-at-Claim-2 novelty than Axes 2 and 3 after Paper 2 inheritance is accounted for. The asymmetry is structural rather than stylistic; the section foregrounds Axis 1 accordingly without violating per-axis differentiation crispness.

The first axis is **aspect-as-unit-of-exchange operating across distinct Selves’ home governance perimeters**. The unit of exchange in FAI is an aspect with three-level structure preserved (cells with DNA and action layers) operating across distinct Selves’ home governance perimeters per §4’s perimeter-spanning property. Federated learning exchanges parameters or gradients at coordinate-wise granularity over full-model vectors; multi-agent memory frameworks exchange typed state fields, message lists, task outputs, or entity-edge assertions; schema-mapping operates at schema-element or ontology-term granularity. None operates at aspect-level granularity preserving Paper 2’s three-level architecture across distinct governance perimeters. FAI is architecturally different from federated learning, from LangGraph, and from COMA-family schema mapping on the unit-of-exchange axis, not these neighbors plus aspect-level granularity. This axis is the most load-bearing of the three: aspect-as-the-exchange-unit specifically is fresh at Claim 2, with the perimeter-spanning context as Claim 1 inheritance applied. **Primary novelty weight rests on Axis 1.**

The second axis is **full-merge-as-architectural-default at inter-Self scope**. Contemporary architectures with configurable merge regularly name defaults among their options, but the defaults are heterogeneous: federated learning’s averaging-default destroys contributor distinctions in the merged model; LangGraph’s overwrite-default is last-writer-wins per state key; mixture-of-experts’ preservation-default never mixes expert weights; schema-mapping’s effective default is no-merge-on-partial-alignment; multi-master databases default to last-writer-wins across many products. FAI’s full-merge-with-conflict-preservation default occupies a distinct position in this design space — full merge of contributed aspects with conflicts surfacing as first-class substrate state by inheritance from Paper 1 carrying through §4. FAI is architecturally different from FedAvg on the merge-default axis, not FedAvg plus full-merge: averaging-with-erasure versus full-merge-with-conflict-preservation. The same conjunction logic applies against schema-mapping’s refuse-to-merge default and against LangGraph’s per-key overwrite. Paper 2’s intra-Self combination primitive supplies the configurability frame; the architectural-default commitment specifically is fresh at inter-Self scope, since Paper 2 does not name a default among its three pattern variants at intra-Self scope. The shared substrate is the medium of integration in a way Paper 2’s intra-Self substrate is not.

The third axis is **exchange-bounded-to-reasoning-layer at the inter-Self boundary**. FAI exchanges only DNA-layer and action-layer content per Paper 2’s instinct/reasoning separation; LLM weights and instinct-layer content do not exchange. Federated learning’s contrast is sharpest here — FedAvg-family algorithms aggregate at parameter or gradient level (instinct-layer content in trilogy terms), and model-fusion methods at LLM scale merge at full-parameter granularity. Multi-agent memory frameworks operate within a single Self’s perimeter where the layer separation is internal; schema-mapping is data-integration work where instinct-layer content is not in scope. FAI is architecturally different from federated learning on the layer-bounding axis, not federated learning plus layer-bounding. The articulation as mechanism-level commitment at the inter-Self boundary is fresh at Claim 2; the underlying instinct/reasoning separation is Paper 2 inheritance.

The conjunction carries the architectural contribution. Aspect-as-unit-of-exchange without the merge-default commitment is a granularity claim without a merge mechanism; full-merge-as-default without aspect-as-unit is FedAvg’s architectural shape; layer-bounding without aspect-as-unit is what Paper 2 already establishes within each Self. What is unoccupied is the conjunction — aspect-level merge with full-merge default and reasoning-layer-bounded exchange across distinct Selves’ home governance perimeters within a temporary shared substrate.

5.5 Reviewer-objection responses

Four objections anticipate themselves with substantial force; six others surface on specific terms and warrant briefer treatment. The mechanism-defense calibration register requires substantive response to the two

named failure modes the calibration register operates against: the *duplicated-with-Paper-2 misreading* and the *biology-as-metaphor reading* are not separate objections but readings the differentiation work has to render unavailable, and §5.1’s preempt-at-claim-statement plus the inheritance-vs-fresh accounting that §5.4 and §5.6 develop carry the response across the section.

The first objection — *this is just federated learning aggregation under new vocabulary* — runs on three axes simultaneously. The unit differs (aspects with three-level structure versus parameters or gradients); the default differs (full merge with conflict preservation versus averaging with contributor-distinction erasure); the layer-bounding differs (reasoning-layer content versus instinct-layer content). The honesty discipline matters: federated learning has substantive aggregation mechanisms with named defaults across a mature literature, and decentralizing optimization does not automatically decentralize governance. The differentiation is on specific architectural axes, not on whether federated learning has merge mechanisms. FedAvg and successors define the architectural shape against which Claim 2’s merge commitments differ at unit, default, and layer simultaneously; the conjunction is where the architectural contribution lives.

The second objection — *this is just LangGraph reducers or AutoGen state aggregation at higher abstraction* — runs primarily on the first axis with secondary force on perimeter. Both operate within one runtime under one governance perimeter at field-level or message-list granularity. FAI integrates aspects with three-level architecture preserved across distinct Selves’ home governance perimeters. Multi-agent memory frameworks supply substantive merge primitives within their scope; the architectural differences are on unit-of-exchange and on perimeter-spanning, not on whether the frameworks have merge primitives.

The third objection — *this is just schema mapping or ontology alignment under new vocabulary* — runs primarily on the second axis with sympathetic posture acknowledged. COMA, LogMap, AML, and the LLM-assisted alignment literature treat alignment as precondition for any merge; FAI does not require alignment as precondition because conflicts within the shared substrate are first-class state by Paper 1 inheritance through §4. Schema-mapping operates at schema-element granularity; FAI operates at aspect granularity. The cluster’s commitment to preserving unaligned elements as distinct is sympathetic to Claim 2’s conflict-preservation posture; the architectural difference is what each commitment operates over.

The fourth objection — *why aspect as the unit?* — connects directly to what travels through the substrate during an FAI event. Aspects are purpose-defined arrangements of cells per Paper 2’s structural commitment. Below-aspect granularity (cells alone) surfaces fragments below purpose-coherence; above-aspect granularity (whole Self) either requires absorbing a Self’s identity — architecturally incoherent given Selves are governance-distinct entities per §4’s perimeter-spanning property — or amounts to merging two Selves into one, a different operation than coordination through shared substrate. Cross-KG subgraph transfer and distributed object marshalling supply the broader architectural pattern of purpose-defined-arrangement-reconstituted-in-context that FAI instantiates at aspect scope: structured pattern travels with provenance, the structural-arrangement role is reconstituted at the destination using local context, and the destination reconstitution operates within the receiving party’s governance rather than absorbing the source party’s authority.

Six further objections admit briefer treatment. *Full merge as default seems aggressive — wouldn’t selective merge be safer?* Selective merge is one of three configurable pattern variants inherited from Paper 2; the full-merge default is not aggressive in the trilogy because conflicts surface as first-class substrate state by Paper 1 inheritance through §4 rather than auto-reconciling. *Why bound to DNA and action layers?* The instinct/reasoning separation is foundational trilogy commitment; the Paper 2 inheritance carries through to the inter-Self boundary as architectural commitment at FAI mechanism level, not as an additional commitment Claim 2 adds. *How does merge handle conflicts when aspects conflict during integration?* Conflicts surface as first-class substrate state by Paper 1 inheritance through §4; the three-tier mechanism (preserve / resolve via configured orchestration as substrate content / escalate to humans across joint authority of participating Selves’ governance) is §6’s territory. *How does the receiving Self ingest FAI outputs into its home substrate?* Through governance-configured ingestion at each home perimeter; the four-locus evolution-feed mechanism connecting inter-Self events to home-perimeter Paper 2 evolution machinery is §7’s territory. *How do multiple Selves’ LLMs operate as substrate mediators within one FAI event?* Each operates as substrate mediator per Papers 1 and 2 with coordination configured per event under joint authority per

Paper 3 core theory’s authority-across-the-perimeter passage; the configurable dimensions of multi-mediator coordination are §8’s territory. *Isn’t this just CKS at a bigger scope?* The inheritance-vs-fresh accounting in §5.6 supplies the precise answer: aspect-as-the-exchange-unit specifically, full-merge-as-architectural-default at inter-Self scope, and exchange-bounding as mechanism-level commitment at the inter-Self boundary are all fresh at Claim 2; what Paper 2 establishes at intra-Self scope and what §4 establishes at substrate-as-architectural-object scope are referenced as inheritance rather than re-litigated.

5.6 Inheritance accounting and downstream placement

Claim 2’s commitments compose from a mix of fresh-at-Claim-2 architectural content, fresh inter-Self extension of inheritance frames, and articulation of prior-paper inheritance at FAI mechanism level. The accounting matters because Claim 2 operates at mechanism-defense register: under-claiming would collapse Claim 2 into Paper 2’s intra-Self combination primitive at bigger scope, and over-claiming would obscure the inheritance discipline the trilogy depends on. The differentiation work in §5.4 and the closest-neighbor engagement in §5.3 operate over this accounting.

Aspect-as-the-exchange-unit is fresh at Claim 2. Paper 2 establishes aspects as structural arrangements within a Self; Paper 2 does not commit aspects as an inter-Self exchange unit. The perimeter-spanning context across which the exchange operates is Claim 1 inheritance applied. Full-merge-as-architectural-default at inter-Self scope is fresh at Claim 2. Paper 2’s intra-Self combination primitive specifies three pattern variants without naming a default among them; Claim 2 commits to full merge as the inter-Self-scope default. Exchange-bounding articulated at FAI mechanism level for the inter-Self boundary is fresh at Claim 2; the underlying instinct/reasoning separation is Paper 2 inheritance. The three-pattern configurability — union, selective merge, lineage-preserved union — is pure Paper 2 inheritance carried across the inter-Self perimeter under §4’s substrate medium. Cross-lineage compatibility carries forward under the perimeter-spanning context. Multi-cardinality flexibility above two carries forward, with the cardinality floor at two as inter-Self-scope boundary articulation. Conflict preservation as first-class substrate state during full merge holds by Paper 1 inheritance through §4; the three-tier mechanism for handling conflicts is §6’s territory, not Claim 2’s.

What travels when an aspect is contributed during an FAI event has architectural-pattern precedent in two adjacent traditions. Cross-KG subgraph transfer in TRACEKG-style pipelines treats relation instances or small patterns as the unit of transfer, with destination reconstitution under local schema preserving original semantics as metadata. Distributed object marshalling in the broader systems literature commits to state plus contract rather than full self-contained object — serialized state plus a reference to schema or interface, with the receiving side reconstituting objects from transmitted state using its own code. Both traditions supply the architectural shape: structured pattern travels with provenance; the structural-arrangement role is reconstituted at the destination using local context. FAI instantiates this shape at aspect-level granularity for inter-Self events specifically. When a Self contributes an aspect to the shared substrate, the constituent cells and their layer content travel, the structural arrangement travels, and the provenance travels with carry-over depth governance-configured per §8’s territory; the aspect-as-purpose-defined-coordination role is reconstituted within the shared substrate’s orchestration under joint authority.

Claim 2 establishes the canonical operation §6 through §9 operate over. As §6 develops, the three-tier inter-Self conflict-handling mechanism (preserve / resolve via configured orchestration as substrate content / escalate to humans across joint authority of participating Selves’ governance) handles conflicts surfaced during FAI events within the shared substrate. As §7 develops, FAI outputs feed each participating Self’s evolution machinery from Paper 2 through governance-configured ingestion at each home perimeter, with a layer-routing rule that holds DNA-layer content feeding DNA evolution at home, action-layer content feeding action-feedback evolution at home, and instinct evolution taking no FAI input by architectural commitment per Claim 2’s exchange-bounding. As §8 develops, the configurable dimensions of FAI events — sharing scope, cardinality, persistence policy, cooperation/competition variant, provenance carry-over depth, multi-mediator coordination — are themselves substrate content under recursive governance bottoming at human-authored governance authority per Paper 1 §3.3. As §9 develops, FAI events accumulating across a network of CKS-governed Selves constitute a collective evolution mechanism at population scope under joint authority across population-level governance perimeters, with the calibrated-humility voice anchor appropriate to the

trilogy-completion register. Claim 2’s contribution is the operation those downstream developments require: n-ary aspect-level merge over the shared substrate with full-merge-as-architectural-default at inter-Self scope and exchange bounded to reasoning-layer content, configurable per event under joint human authority, with all six Paper 1 commitments holding within scope by inheritance through §4.

6. Claim 3 — The three-tier inter-Self conflict-handling mechanism

6.1 Claim statement and the three-tier mechanism over the shared substrate

Claim 3 establishes the **three-tier inter-Self conflict-handling mechanism** — preserve, resolve via orchestration, escalate to humans — as what Paper 1 (Li, 2026a) §5.3’s two-level conflict handling produces when those principles compose with §4’s shared substrate and §5’s Full Aspect Integration operation across the inter-Self perimeter. Paper 1 commits at cell scope to substrate-level conflict preservation as architectural default and to cell-level resolution by orchestration rules under the “humans decide resolution logic, not the LLM” axis; Paper 2 (Li, 2026b) extends conflict handling across multi-level operations within a Self. Claim 3 articulates what those principles produce at Paper 3 scope rather than designing fresh conflict-handling architecture.

The architectural register is **principles-extending**. Two implicit failure modes operate against the calibration discipline. The first is the *under-claim* — treating Claim 3 as nothing more than re-citation of Paper 1 §5.3 at bigger scope, with the inter-Self specification dissolving into trilogy-continuity bookkeeping. The second is the *over-claim* — treating Claim 3 as fresh conflict-handling architecture beyond what the inheritance produces, with the three loci of fresh content inflating into a free-standing mechanism that competes with Paper 1 §5.3 rather than extending it. The differentiation work that follows operates against both readings simultaneously: fresh content is precisely localized at three loci — the inter-Self specification of Paper 1 §5.3, the joint-authority shape of the escalate tier, and the substrate-content character of tier-decision logic — and everything else is inheritance Claim 3 articulates rather than commitments Claim 3 introduces. The principles-extending calibration register is the second explicit calibration-register label in Paper 3’s manuscript, after §5’s mechanism-defense register; the varied calibration landscape across the six claims is what §11 Discussion develops as a trilogy-level pattern.

The three tiers each carry inheritance, with cross-perimeter shape applied where the inter-Self scope requires it. The **preserve** tier inherits Paper 1’s substrate-level preservation directly: conflicts surfaced during FAI events within the shared substrate remain first-class addressable substrate state, and as §7 develops, can carry through evolution outputs participating Selves ingest. The **resolve via orchestration** tier inherits Paper 1’s cell-level resolution by orchestration rules: when prebuilt orchestration logic exists for the conflict class, the rules within the shared substrate determine the response, with the rules themselves substrate content authored under joint authority across participating Selves’ governance per Paper 3 core theory’s authority-across-the-perimeter passage. The **escalate to humans** tier inherits Paper 1’s commitment that humans hold final authority where no automated resolution is permitted; the cross-perimeter shape — escalation surfacing to the joint-governance arrangement of participating Selves’ home governance structures — is the genuinely fresh tier shape. Tier-decision logic operates through orchestration substrate that is itself substrate content under joint authority across participating Selves’ governance. The actor-neutrality discipline from Paper 1 §3.3 holds throughout: human-governed does not mean human-authored, and Selves do not “resolve conflicts,” “negotiate,” or “arbitrate” in the architectural register — conflict resolution is the result of orchestration rules under joint authority, with LLMs operating as substrate mediators executing those rules.

The §5 → §6 transition rests on a primitive §5 establishes and §6 inherits without re-defending. FAI is the canonical operation over the shared substrate, with full merge of contributed aspects as the architectural default; conflicts surface during FAI events as first-class substrate state by Paper 1 inheritance through §4. §6 takes up where §5’s brief response on conflict handling deferred — the three-tier mechanism that handles conflicts surfaced during FAI events within the shared substrate — and develops it at the architectural-pattern register Claims 1, 2, 3 commit jointly to. The next subsection develops the named foil at this section’s scope.

6.2 The named foil — opaque conflict resolution and the conflict-handling gap

§4 establishes opaque agent-to-agent communication as the named foil for inter-Self coordination, sharpened at §5 to opaque inter-agent merge of outputs without governable substrate. At Claim 3’s conflict-handling angle, the foil’s contemporary instances are at-merge-resolution architectures committing to immediate conflict resolution at the merge or aggregation point as architectural default, with no first-class persistent state for the conflict itself. Paper 1 §5.4 already engaged detect-resolve-forget at cell scope as named foil for at-merge resolution as architectural default — contradictions detected at inference time, reconciled in-session through clarification, majority voting, or judge-LLM arbitration, and discarded once the session ends. §6 references that engagement at inter-Self scope without re-engaging it.

The contemporary instances at inter-Self scope take three architectural shapes. Consensus protocols, voting mechanisms, and Byzantine-robust aggregators — Krum, Trimmed Mean, Median, Bulyan, FLTrust — resolve at the aggregation point through statistical methods that select trusted updates and discard the rest, treating conflicts as noise to be filtered rather than as substrate content to be preserved. Convergence-via-type-algebra structures — CRDT type-determined merge, OT transformation functions, multi-master database last-writer-wins — embed resolution in data-structure type or in the topology of merge functions, with conflict-free convergence as the structural property the architecture commits to. Arbiter-agent patterns at framework register and reducer-style merge logic within one runtime treat resolution as a transient computational step that completes before downstream processing proceeds.

What the contemporary literature converges on at the conflict-handling level is that conflicts are precondition for completion: the merge or aggregation cannot finish until conflicts are eliminated, with elimination achieved through statistical aggregation, type-algebraic convergence, or arbiter-mediated decision. Governance over the resolution mechanism, when articulated at all, sits at framework configuration, contract, or protocol-parameter register rather than as authored substrate content within the resolution itself. The merge logic is not substrate; the conflicts surfaced during merge are not substrate; the configuration of the resolution is not substrate. As at §4.2 and §5.2, these are properties of the architecture class at the conflict-handling level, not capability gaps that better LLMs or richer aggregation rules would close. The appropriate response is therefore architectural at the conflict-handling level specifically. The contemporary instances surface in §6.3 as Axis 1 closest-neighbor positioning rather than as named-foil re-engagement, since Paper 1 §5.4’s foil engagement at cell scope has done the work; Claim 3 articulates what the inheritance produces at the inter-Self perimeter. Three architectural foils now occupy parallel positions at the foil register Claims 1, 2, 3 commit to jointly — opaque agent-to-agent communication at the substrate-content level, opaque inter-agent merge at the merge-mechanism level, and opaque conflict resolution at the conflict-handling level.

6.3 Closest-neighbor engagement organized by tier

Where §4 organized closest neighbors by territory across four tied-top clusters and §5 organized by cluster across three tied-top clusters at the merge-mechanism level, Claim 3’s three-tier mechanism naturally clusters contemporary closest neighbors by which tier they operate at. The tier-organization is genuine pattern, not template — each tier’s contemporary instances operate at architecturally distinct registers, and the differentiation work that §6.4 develops runs along axes corresponding to the tiers. Most neighbors fall cleanly into one tier; LangGraph operates at all three within a single governance perimeter and serves as composite single-perimeter closest neighbor referenced at each tier in succession.

Preserve-tier neighbors commit architecturally to preservation of conflicts as first-class persistent state rather than at-merge resolution. Atlan’s Pattern 4 actor-aware edges (already referenced at §4.3 within Claim 1 territory) preserve per-agent assertions distinct rather than forcing merged belief, sympathetic to the preserve-tier commitment by inheritance from the substrate-content engagement. Event-sourced systems with conflict events as first-class commit to pending/committed/rejected event states; multi-version concurrency control databases default to version coexistence; Couchbase Mobile retains all conflicting revisions for application inspection; the 2026 conflict-of-laws RAG example holds interpretations under different statutes in separate tagged entries with resolution contextual at query time. CRDT structures supply the broader convergence-via-type-algebra tradition at supporting density — sympathetic on preservation-during-merge but committing to type-algebra convergence as architectural default rather than to substrate-mediated tier-

selection logic. LangGraph’s per-key reducers preserve via merge at field-level granularity, the field-level granularity contrasting with substrate-mediated aspect-level preservation. The architectural commitment converging across these neighbors is preservation-first as default with resolution as higher-tier policy. All operate within single perimeters.

Resolve-via-orchestration-tier neighbors commit to resolution rules as inspectable substrate content with explicit human authority over rule authoring. LangGraph’s explicit merge nodes — separate keys with downstream merge nodes implementing higher-level reconciliation — are the strongest single-perimeter contemporary closest neighbor at this tier. Rosendal and Oprescu’s IEEE Big Data 2025 paper on client-level disagreements in federated learning is the strongest articulation at federated scope: exclusion policies as first-class configuration; the orchestrator builds isolated model-update tracks for mutually excluding clients; the aggregator becomes track-local — disagreements drive topology changes at the orchestration tier rather than being smoothed away at aggregation. The AWS Deterministic Governance Engine literature articulates tier-decision logic specifically — rule-based allow/deny, quorum and voting structures, priority policies, escalation thresholds as human-authored configuration with no LLM calls in the governance loop, single-perimeter typical. AutoGen GroupChat and CrewAI hierarchical occupy the minimal-conflict-architecture register: AutoGen treats conflict as divergent messages with resolution delegated to user-implemented selection functions; CrewAI hierarchical treats manager validation as generic-reviewer prompting at agent-prompt level rather than as inspectable substrate content. The contrast that matters across this tier: contemporary single-perimeter resolve-tier neighbors hold the orchestration-rules-as-substrate-content commitment cleanly within their scope; what is unrealized at this register is the joint-authority extension across distinct Selves’ home governance perimeters, which §6.4 develops on Axis 2.

Escalate-tier neighbors divide into single-perimeter and cross-perimeter clusters. At single-perimeter scope, LangGraph’s interrupt mechanism with state persistence, Okta-style agent orchestration with authority metadata, and the Strwythura ontology pipeline with human curation as non-optional phase commit to human escalation as first-class control-flow tier rather than as control-plane wrapper around an otherwise opaque resolution. The cross-perimeter cluster is the genuinely fresh territory — the third novelty locus where Claim 3’s contribution sits. Data trusts with trustor-trustee-beneficiary structures, joint venture dispute mechanisms, and cross-border privacy recourse frameworks supply the canonical anchor at the cross-perimeter joint-authority escalation register: mature in non-AI legal and governance precedent across multiple jurisdictions with trustees, joint committees, ADR panels, and regulators as authority-articulation mechanics, with technical systems serving as evidence and enforcement layers rather than as the medium through which joint-authority escalation operates. JAMS AI Disputes Clauses and Rules supply AI-specific dispute-resolution framing at the legal-layer register, with adjacent dispute-resolution mechanisms supplying complementary legal-layer recourse structures. Fauscette’s April 2026 Arion Research piece on cross-organizational agents supplies practitioner-literature articulation closer to the architectural-pattern register: bilateral and multilateral governance contracts framed as machine-readable terms and conditions, with agreed decision boundaries, escalation protocols, and dispute-resolution mechanisms — but without formalizing as architectural specification under Papers 1 and 2 commitments. HADA and SettleFL’s crypto-economic coordinator supply mature technical-mechanism instances at the cross-perimeter register. Joint venture dispute mechanisms, multi-organizational data-sharing agreements, and cross-border privacy recourse frameworks supply legal and governance precedent that anchors the cluster’s canonical-source positioning without requiring single-source load-bearing on practitioner-literature anchors.

LangGraph’s status as composite single-perimeter closest neighbor is worth flagging at the tier-organization level. Per-key reducers preserve at the preserve tier; explicit merge nodes resolve at the resolve tier; interrupt-with-state-persistence escalates at the escalate tier — all within one runtime under one governance perimeter. The architectural shape is directly parallel to Claim 3’s three-tier mechanism at the within-perimeter register, with the cross-perimeter joint-authority shape as the fresh extension Paper 3 specifies. The composite single-perimeter neighbor structure is what makes the cross-perimeter axis’s primary novelty weight visible cleanly: at within-perimeter scope, LangGraph already does most of the architectural work the three-tier mechanism specifies; what Claim 3 adds is the joint-authority shape across distinct Selves’ home governance perimeters, with substrate-content tier-decision logic extending across that perimeter shape.

6.4 Three differentiation axes with asymmetric inheritance shape

The differentiation work above runs on three architectural axes simultaneously, following §4.4 and §5.4 in the *architecturally different from X on Y axis, not X plus Y* framing pattern. The honest weight across the three axes is asymmetric — Axis 1 mostly Paper 1 inheritance referenced; Axis 2 Paper 1 §5.3 inheritance with joint-authority extension as fresh Paper 3 content; Axis 3 the genuinely fresh content carrying primary novelty weight. The asymmetry is structural rather than stylistic, reflecting the principles-extending posture: Claim 3 articulates what Papers 1 and 2’s principles produce at Paper 3 scope, and the asymmetric inheritance shape is the calibration register’s defining structural pattern. The differentiation foregrounds Axis 3 accordingly without violating per-axis differentiation crispness.

The first axis is **preserve-as-first-class-substrate-state versus resolve-as-precondition**. Contemporary at-merge-resolution architectures resolve at the merge or aggregation point as architectural default — Byzantine-robust aggregators select trusted updates while discarding the rest; CRDT type algebra makes conflict-free convergence structural; multi-master database last-writer-wins resolves immediately. Claim 3’s preserve tier inherits Paper 1 §5’s preservation-first default at inter-Self scope: conflicts surfaced during FAI events within the shared substrate remain first-class addressable substrate state. Claim 3 is architecturally different from contemporary at-merge-resolution architectures on the preserve-as-first-class-substrate-state axis, not contemporary at-merge-resolution plus preserve. This axis carries materially less fresh-at-Claim-3 novelty than the other two: Paper 1 §5 is the structural source, with Claim 1 inheritance applied through §4’s shared substrate. Recording the inheritance here keeps the asymmetric weight visible at the moment Axis 1 is being argued.

The second axis is **orchestration-rules-as-substrate-content with joint-authority extension versus opaque or single-perimeter resolution logic**. Paper 1 §5.3’s “humans decide resolution logic, not the LLM” axis extends to the inter-Self perimeter. Contemporary arbiter-agent architectures embed resolution logic in agent behavior, treating conflict resolution as something an agent decides rather than as the result of orchestration rules under human authority; the canonical reformulation reads conflict handling via configured orchestration substrate — substrate content authored under joint authority — rather than via arbiter agent. The resolve-via-orchestration tier carries the name “resolve via orchestration” specifically because the orchestration is substrate content under joint authority across participating Selves’ governance, not because an orchestrator-agent makes the decision. LangGraph and Rosendal–Oprescu 2025 occupy the strongest single-perimeter and single-consortium positions at this axis; both hold the orchestration-rules-as-substrate-content commitment within their scope. The joint-authority extension across distinct Selves’ home governance perimeters is the fresh Paper 3 content this axis carries — orchestration rules within the shared substrate are substrate content under joint authority across participating Selves’ governance per Paper 3 core theory’s authority-across-the-perimeter passage. Claim 3 is architecturally different from arbiter-agent reconciliation on the orchestration-rules-as-substrate-content axis with joint-authority extension, not arbiter-agent reconciliation plus substrate.

The third axis is **cross-perimeter joint-authority escalation versus single-perimeter or implicit escalation**. This axis carries the primary novelty weight per the asymmetric inheritance shape. Paper 1’s escalation operates within one governance perimeter; contemporary multi-agent and inter-agent-protocol systems either escalate within one governance domain or treat human-in-the-loop reconciliation as control-plane wrapper around an otherwise opaque resolution rather than as architectural commitment of the technical substrate. Cross-perimeter scope is mature in non-AI legal and governance precedent — joint venture dispute ladders, data-sharing agreements, cross-border privacy recourse — but these encode authority in legal documents with technical systems serving as evidence and enforcement layers. Fauscette’s April 2026 Arion Research piece articulates the cross-perimeter territory at practitioner-literature register through bilateral and multilateral governance contracts as machine-readable terms and conditions with agreed decision boundaries and escalation protocols — but without formalizing as architectural specification under Papers 1 and 2 commitments. Claim 3’s escalate-to-humans tier commits to cross-perimeter joint-authority escalation as architectural specification of the shared substrate. The “X plus Y” misreading would treat contemporary cross-organizational AI dispute resolution mechanisms as assemblable into the cross-perimeter joint-authority escalation tier through supplemental layering. Claim 3 is architecturally different from contemporary cross-organizational AI dispute resolution on the cross-perimeter joint-authority escalation axis, not contemporary

cross-organizational dispute resolution plus joint authority. The architectural commitment is to the technical substrate’s escalation mechanism specifying joint-authority structure, not to legal mechanisms supplying joint authority externally to a substrate that does not commit to joint authority architecturally. **Primary novelty weight rests on Axis 3.**

The conjunction carries the architectural contribution. Axis 1 alone is Paper 1 §5 single-perimeter; Axis 2 alone is LangGraph within one runtime; Axis 3 alone without substrate-content orchestration rules is the legal and governance pattern with technical systems as evidence layer. What is unoccupied is the conjunction operating through the shared substrate during FAI events — preservation as substrate-level architectural default, resolution rules as substrate content under joint authority, and escalation across joint authority of distinct governance perimeters as architectural specification of the technical substrate.

6.5 The inheritance-active finding and reviewer-objection responses

The conjunction §6.4 closes on is what Claim 3 contributes; the cross-cutting research query confirms the conjunction is not realized in contemporary literature. The query asked whether any contemporary architecture commits architecturally to all three clauses simultaneously: preservation of conflicts as first-class persistent state by default; resolution rules as inspectable substrate content under explicit human authority; escalation operating across joint authority of multiple distinct governance perimeters as architectural commitment rather than as deployment-layer wrapper. Five candidates were examined. *Data trusts and ADR clauses* hold cross-perimeter joint authority cleanly at the legal-governance register but project preservation and substrate-content rules into the technical substrate only partially — authority sits in legal documents with the technical systems as evidence and enforcement layer. *Verifiable cross-silo federated learning with consortium governance* holds substrate-content rules and joint authority at consortium scope but tends to embed resolution in coordinator code rather than committing to preservation-first as architectural default. *Event-sourced and MVCC systems under multi-org governance* hold preservation cleanly within one perimeter but typically operate single-perimeter, with cross-organizational joint-authority deployment a configuration choice rather than an architectural commitment. *Contractual AI-specific arbitration including JAMS AI Rules* holds cross-perimeter joint authority with AI-specific dispute mechanisms, but the resolution mechanism sits at legal/contractual layer rather than as substrate-content tier-decision logic. *Cross-organizational AI governance contracts in the Arion Research framing* articulate cross-perimeter joint authority and intend substrate-content rules as machine-readable contracts but realize preservation only conceptually. None holds all three clauses.

The three clauses synthesize patterns currently distributed across domains; the conjunction operating through architectural specification of the shared substrate during FAI events is what Papers 1 and 2’s principles produce when extended at Paper 3 scope. The finding is the closest-neighbor positioning anchor — neither under-claiming (the conjunction is not realized in contemporary literature) nor over-claiming (the conjunction synthesizes patterns rather than introducing fresh architectural commitments beyond what the inheritance produces). This is the first inheritance-active finding in Paper 3’s manuscript; the pattern surfaces again at Claims 4, 5, and 6 with claim-specific shape, and §11 Discussion develops the four-finding compounding pattern across the trilogy.

Three reviewer objections anticipate themselves with substantial force; five others surface on specific terms and warrant briefer treatment.

The first objection — *this is just Paper 1 §5 at bigger scope — what’s new in Claim 3?* — is the internal-trilogy-continuity objection. The inheritance-active finding above supplies the response: the three-clause conjunction operating through architectural specification of the shared substrate during FAI events is not realized in contemporary literature. The fresh content is precisely localized at the three loci named in §6.1 — the inter-Self specification of Paper 1 §5.3, the joint-authority shape of the escalate tier, and the substrate-content character of tier-decision logic; everything else is inheritance. The under-claim and over-claim implicit failure modes operate against this objection on both sides: collapsing Claim 3 into Paper 1 §5.3 plus naming would under-claim the three-clause conjunction the literature does not realize; designing the three tiers as fresh architecture beyond the inheritance would over-claim what the principles-extending register commits to.

The second objection — *this is just arbiter-agent reconciliation under new vocabulary* — runs primarily on Axis 2. Arbiter-agent architectures embed resolution logic in agent behavior; FAI’s resolve-via-orchestration tier commits to resolution rules as substrate content under joint authority across participating Selves’ governance per the canonical reformulation. The joint-authority extension across distinct Selves’ home governance perimeters is fresh Paper 3 content; the architectural difference is a different commitment about where resolution logic lives, not a vocabulary change.

The third objection — *this is just Byzantine-robust aggregation or distributed-systems conflict resolution under new vocabulary* — runs at algorithm-spec register. Byzantine-robust aggregators resolve at aggregation point through statistical methods; CRDT and OT embed resolution in data-structure type or transformation function; multi-master databases default to last-writer-wins. Claim 3’s three-tier mechanism preserves as first-class substrate state by default at architectural-pattern register without committing to algorithm-spec depth, adding the cross-perimeter joint-authority extension to the contemporary convergence on preservation-with-resolution-as-higher-tier-policy. The differentiation is on architectural axes, not on whether contemporary aggregation algorithms have resolution mechanisms.

Five briefer responses follow. *Why three tiers?* The three correspond to Paper 1 §5.3’s decision shapes at inter-Self scope, with the “humans decide resolution logic” axis surfacing as cross-perimeter joint-authority escalation. The number follows inheritance structure rather than fresh architectural design choice. *Mechanics of tier selection?* Tier-decision logic operates through orchestration substrate that is itself substrate content under joint authority across participating Selves’ governance; configurability dimensions are §8’s enumeration territory. *Mechanics of cross-perimeter escalation?* Escalation surfaces to the joint-governance arrangement of participating Selves’ home governance structures; specific authority-articulation mechanics — trustee designation, joint-committee structure, ADR-panel composition, regulator embedding — are configurability dimensions per §8 territory. *Recursive conflicts in orchestration rules?* The orchestration rules are themselves substrate content; conflicts within the rules surface as substrate-level conflicts handleable by the same three-tier mechanism. The recursive applicability extends the substrate-content character of tier-decision logic to the rules governing the rules without architectural addition beyond what the inheritance produces; recursive-governance enumeration is §8 territory while recursive applicability is Claim 3’s. *Joint-authority disagreement on tier selection?* The first-order conflict is between contributed aspects within the shared substrate; the second-order conflict is between participating Selves’ governance positions on which tier should apply. Both surface as substrate-level conflicts handleable by the same three-tier mechanism, with adjacent precedent across data trusts, FL consortia, AI ADR, and Arion Research suggesting that disagreement on tier selection escalates to pre-specified joint-authority structures — trustees, joint committees, ADR panels, regulators. The architectural commitment is recursive applicability of the three-tier mechanism through the cross-perimeter joint-authority escalation locus rather than additional architectural machinery.

6.6 Inheritance accounting and downstream placement

Claim 3 inherits without redefense from four sources. From Paper 1, §5 entire as the architectural antecedent for conflicts as first-class state; §5.3 as the direct structural antecedent for the three-tier mechanism (including the “humans decide resolution logic, not the LLM” axis); §5.4’s detect-resolve-forget foil engagement already done at cell scope; §3.3’s authority-vs-labor distinction supplying the actor-neutrality discipline through the “governance is an authority architecture, not a review workflow” portable phrase. The inheritance is dense and load-bearing — Paper 1 §5 supplies the architectural posture Claim 3 extends, and the “humans decide resolution logic” axis transports unchanged across the inter-Self perimeter.

From Paper 2, §5.6 explicit on Paper 1’s two-level conflict-handling structure extending naturally to multi-level scope across cells, aspects, and Selves; §6.3 carrying that extension into mating’s merge-time conflicts under the union pattern, where merge conflicts become persistent first-class substrate state per Paper 1 §5; §8.3 carrying it into evolutionary dynamics where the boundary between instinct and reasoning is itself governed substrate content with humans holding authority over the orchestration rules under which the boundary’s location is determined. Paper 2’s biology engagement is bounded to its intra-Self scope per Paper 2 §2.2’s biology-as-bounded-load-bearing-terminology framing and Paper 3 core theory’s biology-positioning subsection; FAI vocabulary at Paper 3’s voice register replaces any biology-mating idiom at this scope.

From §4 of Paper 3, the shared substrate as the medium within which conflicts surface during FAI events; the four tied-top closest-neighbor clusters and named foil at substrate-content level referenced as Claim 1 territory rather than re-engaged. From §5, FAI as canonical operation with full-merge default not requiring conflict resolution as precondition; the canonical reformulation phrasing — conflict handling via configured orchestration substrate rather than via arbiter agent — carrying forward as standing reformulation discipline. The four-source inheritance accounting is what makes the principles-extending posture honest at both edges: under-claim drift would treat the inheritance as the entire content with no fresh contribution; over-claim drift would treat the three loci of fresh content as decoupled from the inheritance. The accounting locates each inheritance at its source and renders the fresh content as articulation of what the principles produce at Paper 3 scope.

The federated-learning territory engaged at Claim 3’s conflict-handling angle continues a cross-claim pattern. §4 engaged FL at the joint-authority-mechanics angle, §5 at the merge-mechanism angle; §6 engages at conflict-handling. Byzantine-robust aggregation resolves at aggregation point through statistical methods; FedAvg commits to averaging-as-resolution — a different architectural posture than preserve-as-first-class-state followed by tier-selection logic through substrate content. The third FL boundary makes the cross-claim FL-engagement pattern visible as discipline. The pattern carries forward to §7, §8, and §9 at evolution-feed, configurable-aggregation-algorithm, and population-scale-dynamics angles respectively, with §11 Discussion developing the six-FL-boundary pattern as a trilogy-level observation.

Claim 3 specifies the conflict-handling mechanism that operates over §4 and §5 jointly, supplying the conflict-handling specification §7 through §9 require. As §7 develops, conflicts preserved within the shared substrate carry through as evolution-feed annotations to participating Selves’ home substrate, with the carry-through architecture governance-configured per home perimeter. As §8 develops, the per-event configuration of which tier applies for which conflict class is itself substrate content under recursive governance, with configurability dimensions — tier-selection rules, conflict-class definitions, escalation-path configurations, joint-authority configurations, recursive-governance depth — enumerated as §8’s territory. As §9 develops, accumulated conflict-handling outcomes across many FAI events at population scope feed collective evolution, with the calibrated-humility voice anchor appropriate to the trilogy-completion register applied at that section’s scope. Claim 3’s contribution is the conflict-handling specification those downstream developments require: substrate-mediated conflict handling at the inter-Self scope under joint authority across participating Selves’ governance, with the cross-perimeter shape of the escalate tier as the genuinely fresh tier shape and the orchestration substrate as the medium through which tier-decision logic operates.

7. Claim 4 — The four-locus evolution-feed mechanism at the FAI hand-off boundary

7.1 Claim statement and the four-locus mechanism at the FAI hand-off boundary

Claim 4 establishes the **four-locus evolution-feed mechanism** as what Paper 2 (Li, 2026b) §7’s three evolution mechanisms produce when fed by FAI events through §4’s shared substrate operating under §5’s FAI operation with §6’s three-tier conflict handling. Paper 2 §7 commits within each Self at home perimeter to three evolution mechanisms — instinct evolution, DNA evolution, and action-feedback evolution — operating in productive tension across multiple levels of architectural composition simultaneously, with horizontal and vertical evolution axes, all under unified human governance. Paper 2 §8 extends Paper 1 (Li, 2026a) §3.3’s authority-vs-labor distinction across these dynamics through per-mechanism governance shapes — verification-substrate machinery for instinct integration, authority architecture for DNA evolution, proposal-and-acceptance machinery for action-feedback. Claim 4 articulates what these principles produce at the inter-Self-event-to-home-perimeter boundary rather than designing fresh evolutionary architecture.

The architectural register is **principles-extending**. Two implicit failure modes operate against the calibration discipline. The first is the *under-claim* — treating Claim 4 as nothing more than re-citation of Paper 2 §7 with FAI as input source, with the four loci of fresh content dissolving into trilogy-continuity book-keeping. The second is the *over-claim* — treating Claim 4 as fresh evolutionary architecture beyond what the inheritance produces, with the four loci inflating into a free-standing mechanism that competes with

Paper 2 §7 rather than extending it. The differentiation work that follows operates against both readings simultaneously: fresh content is precisely localized at four loci, and everything else is inheritance Claim 4 articulates rather than commitments Claim 4 introduces. The principles-extending calibration register is the third explicit calibration-register label in Paper 3’s manuscript per the varied-landscape framing — after §5’s mechanism-defense register and §6’s principles-extending register — with §11 Discussion developing the varied calibration landscape across the six claims as a trilogy-level pattern.

Four architectural commitments compose Claim 4. The **hand-off mechanism at FAI dissolution** specifies that when the shared substrate dissolves at the close of an FAI event, content propagates to each participating Self’s home substrate per governance-configured ingestion at the home perimeter — the dissolution event triggers ingestion under home governance, what each home perimeter takes is configurable per home authority, and the absorption itself uses each Self’s existing Paper 2 evolution mechanisms operating at home. The **per-mechanism feed structure** specifies that FAI-derived content reaches Paper 2’s three evolution mechanisms at each home perimeter through the existing per-mechanism governance shapes from Paper 2 §8; the mechanisms operate unchanged, with FAI-feed as additional input source rather than as replacement or modification. The **layer-routing rule** specifies that the layer of FAI-derived content determines which Paper 2 §7 evolution mechanism receives it, with three explicit cases. DNA-layer content feeds DNA evolution operating on home DNA. Action-layer content feeds action-feedback evolution operating within home action layers. Instinct evolution takes no FAI input by architectural commitment, since LLM weights and instinct-layer content do not exchange across FAI per §5’s exchange-bounding and Paper 3 core theory’s instinct-evolution-out-of-scope subsection — the inter-Self extension of Paper 2’s instinct/reasoning separation carries through the exchange-bounding commitment without modification at this scope. The three-case framing is non-trivial architectural specification — not “FAI content reaches mechanisms” but “the layer determines routing,” distinct from learned routing in MoE-style architectures and from task-identity routing in continual-learning architectures. The **asymmetric ingestion property** specifies that different participating Selves may take different things from the same FAI event under their respective home governance authority, with no architectural requirement of symmetric absorption.

Preserved conflicts from §6’s preserve tier carry through as evolution-feed annotations specifically — when an FAI event dissolves with conflicts preserved, those conflicts can become annotations within home substrate flagging boundaries that home governance must address per home authority during evolution-mechanism operation. The fresh content lives precisely at these four loci; everything else is inheritance.

The §6 → §7 transition rests on a primitive §6 establishes and §7 inherits without re-defending. Three-tier conflict handling — preserve, resolve via orchestration, escalate to humans across joint authority — operates over §4 and §5 jointly during FAI events. §7 takes up where §6’s brief response on evolution-feed annotations deferred — the four-locus mechanism by which inter-Self events connect to home-perimeter Paper 2 evolution machinery — and develops it at the architectural-pattern register Claims 1, 2, 3, and 4 commit jointly to. The next subsection develops the named foil at this section’s scope.

7.2 The named foil — opaque post-deployment update and the evolution-feed gap

§4 establishes opaque agent-to-agent communication as the named foil for inter-Self coordination, sharpened at §5 to opaque inter-agent merge of outputs without governable substrate, sharpened further at §6 to opaque conflict resolution at the conflict-handling level. At Claim 4’s evolution-feed angle, the contemporary instances of the foil’s posture are **post-deployment-update architectures introducing fresh joint-event-specific update machinery** — replay buffers as data structures, regularization losses as objective additions, parameter modules as architectural slots, FedProx proximal terms, FedAdam adaptive moments, gradient projection methods, specialized memory-manager subsystems. The architectural posture is “joint coordination requires its own update machinery.”

The foil sharpens at the evolution-feed level in three ways §6.2’s foil articulation does not develop. First, the unit of update in contemporary systems is heterogeneous: gradient updates, adapter weights, replay-buffer entries, retrieval-corpus additions, agent-memory entries, parameter-module weights. None of these units preserves the structural arrangement Paper 2 establishes within a Self of cells with DNA and action

layers being acted on by three architecturally distinct evolution mechanisms. Second, the update mechanism is itself fresh joint-event-specific architecture rather than a pre-existing per-Self evolution mechanism: the architectural pattern is that update machinery is part of system architecture co-deployed with the joint coordination, not just training code added on. Third, governance over the update — when articulated at all — operates at framework configuration, contract, or protocol-parameter register rather than as authored substrate content within the ingestion itself. The update logic is not substrate; the configuration of the ingestion is not substrate; the asymmetry of what each participant absorbs is not substrate.

What matters for Claim 4 is that these are properties of the architecture class at the evolution-feed level, not capability gaps that better LLMs or richer protocols would close. As at §4.2, §5.2, and §6.2, sharper LLMs and richer protocols would absorb the substrate Claim 4 commits to; the foil literature at §4.2 denies that reading at the substrate-content level, and the same denial holds at the evolution-feed level. Paper 2 §7’s foil engagements at intra-Self scope — recursive self-improvement, RLHF, hierarchical evolutionary algorithms, quality-diversity — operate within one Self; the inter-Self foil sharpening at evolution-feed angle is at Paper 3 scope.

The appropriate response is therefore architectural at the evolution-feed level specifically. FAI-feed’s alternative is FAI-derived content reaching each participating Self’s existing Paper 2 evolution mechanisms unchanged through governance-configured ingestion at each home perimeter, with the layer of content determining routing. **Four architectural foils now occupy parallel positions** at the foil register Claims 1, 2, 3, and 4 commit to jointly — opaque agent-to-agent communication at the substrate-content level, opaque inter-agent merge at the merge-mechanism level, opaque conflict resolution at the conflict-handling level, and opaque post-deployment update at the evolution-feed level.

7.3 Closest-neighbor engagement organized by locus

Where §4 organized closest neighbors by territory across four tied-top clusters, §5 by cluster across three tied-top clusters, and §6 by tier across three tiers, Claim 4’s four-locus mechanism naturally clusters contemporary closest neighbors by which locus they operate at. The closest-neighbor density is structurally asymmetric across the loci: locus (a) draws closest neighbors substantially from distributed-systems territory rather than AI/ML; loci (b), (c), and (d) draw from AI/ML at different sub-territories, with locus (d) sitting in the densest closest-neighbor territory of the four. The asymmetry reflects what each locus commits to architecturally — locus (a)’s event-bounded coordination state with explicit dissolution has stronger antecedent in distributed-systems coordination patterns than in contemporary continual or federated learning architectures, while locus (d)’s asymmetric ingestion has dense overlap with multi-tenant and personalized federated work.

Locus (a) — hand-off mechanism at FAI dissolution. Event-sourcing with consumer-driven projections is closest on per-participant-ingestion-governance: producers emit events into a shared log, each consumer builds its own projection under consumer-defined mapping; the gap is explicit dissolution, since the log persists rather than dissolving at coordination close. MPC protocols carry the closest match for explicit dissolution through joint ephemeral computation state — secret shares, partial decryptions, Beaver triples — destroyed after output, with ephemerality treated as security property rather than as substrate-content commitment. Sagas and workflow engines (Camunda, Temporal, Zeebe) carry the closest match for the bounded-coordination-instance plus explicit-lifecycle conjunction, operating within single perimeters at orchestration-runtime register. BPMN choreography supplies a positive vocabulary touchstone for per-participant local logic without central orchestration. Cross-silo federated learning aggregation is the closest AI/ML neighbor at this locus but falls short on discrete-event-lifecycle since the global model persists across rounds rather than dissolving at each round’s close. The convergence across these neighbors: no 2024–2026 architecture commits to the discrete event-bounded coordination state plus explicit dissolution plus per-participant governed post-dissolution ingestion conjunction as a single named architectural pattern.

Locus (b) — per-mechanism feed structure. This locus carries the most direct Paper 2 §7 / §8 inheritance and accordingly draws closest neighbors at the production-deployment register. Production three-channel architectures distinguish channels by update mechanism — fine-tuning via LoRA/adapters, retrieval-layer updates via RAG corpus, agent and memory-layer updates via prompts and long-term memory — but each channel is itself post-deployment-update architecture rather than pre-existing evolution mechanism that

joint-event content feeds into unchanged. Centralized training with decentralized execution in multi-agent reinforcement learning routes per-agent policy updates through a joint critic, which is itself fresh joint-event machinery. Federated continual learning client-side patterns run continual-learning routines with replay buffers, gradient projection methods, and importance estimators that are themselves fresh continual-learning architecture. The 2024–2026 FCL literature illustrates the recurring pattern at family-specific density: Yoon et al.’s (2021) FedWeIT commits to weight-separation patterns routing shared and task-specific updates through architecturally distinct paths at each client; FedCIL commits to class-incremental learning with replay-based methods specifically; FedProto and prototype-based federated approaches commit to client-specific prototypes preserved by design rather than aggregated; FedProx’s proximal regularization terms operate at the loss-objective layer to constrain client drift. Each method introduces fresh joint-event-specific machinery — weight-separation modules, replay buffers, prototype banks, proximal regularization — at its own architectural plane. The recurring pattern: every neighbor close to FAI-content-reaches-existing-mechanisms stops short by introducing fresh joint-event-specific machinery somewhere in the pipeline, with FAI-feed’s commitment to existing per-Self mechanisms unchanged at home perimeter as the architectural difference at locus (b).

Locus (c) — layer-routing rule. Closest neighbors realize routing-by-content at different registers. The T-Box vs A-Box distinction in knowledge-graph systems is closest in spirit but process-defined and authorship-defined rather than automatically routed on content-layer. The RAG-vs-fine-tune split in production LLM systems carries practitioner guidance — factual changes go to RAG, behavioral changes go to fine-tune — without formalization into a generic content-layer router. Rule-based AI gateways with content-based routing commit to architectural rule-routing but at the inference-request level rather than at the post-event-ingestion level. **Letta and MemGPT’s core-vs-archival memory split is not a closest neighbor for layer-routing** despite its salient tiered structure: the routing rule is LLM-judgment-based — the agent’s LLM decides what is trait-like or ephemeral or archival-worthy — rather than an architectural rule keyed on content type, closer to learned routing than to content-layer-keyed routing. The synthesis flags the distinction explicitly because Letta is salient enough that readers will reach for it independently: layer-routing as architectural rule on content layer is distinct from LLM-judgment-driven tiering. The convergence across these neighbors: no 2024–2026 architecture formalizes content-layer-determined routing as architectural property distinct from learned routing and task-identity routing.

Locus (d) — asymmetric ingestion property. This locus sits in the densest closest-neighbor territory across the four. Cross-silo federated learning commits to per-organization adoption choice at the round-by-round register; multi-tenant AI to per-tenant divergence; personalized federated learning with client divergence by design — meta-learning approaches, head-and-trunk separation, prototype-based and clustering-based PFL — commits to client-specific layers and heads that the federated aggregation works around rather than collapsing. Multi-agent memory frameworks reject uniform updates as anti-pattern; federated continual learning with heterogeneous client trajectories confirms aggregated global models hide client diversity; asynchronous federated learning adds timing-driven divergence; data-trust learning and Gaia-X / IDS-RAM commit to symmetric access with asymmetric local adoption. Locus (d)’s closest-neighbor density is what makes the source-of-asymmetry calibration in §7.4 Axis 3 the load-bearing distinction rather than the asymmetric-ingestion property considered alone.

7.4 Three differentiation axes with asymmetric inheritance shape

Three architectural axes carry differentiation simultaneously, each holding against contemporary post-deployment-update neighbors at the evolution-feed angle (distinct from §4’s joint-authority-mechanics, §5’s merge-mechanism, and §6’s conflict-handling angles). The honest weight across the three axes is asymmetric — Axis 1 mostly Claim 1 inheritance applied; Axis 2 carrying primary novelty weight via Paper 2 §7 / §8 inheritance applied; Axis 3 partially fresh Paper 3 content with calibration discipline against widely-shared asymmetric-update territory. The asymmetry is structural, reflecting the principles-extending posture rather than uniform-novelty distribution across axes.

The first axis is **governance-configured-ingestion-at-home-perimeter versus uniform-update-broadcast**. Contemporary continual-learning, federated-learning, and agent-memory-evolution mechanisms typically apply uniform update procedures across participating systems — federated aggregation produces

a global model deployed to all clients, memory consolidation patterns run uniform retention rules across an agent’s lifetime, retrieval-corpus updates apply to all queries — with per-perimeter governance entering at deployment-shape or contract register rather than at architectural-pattern register. FAI’s hand-off mechanism commits to per-home-perimeter governance over what crosses from the dissolved shared substrate at the dissolution of an event-bounded shared substrate, with the configuration of ingestion itself substrate content under joint authority per Claim 1 inheritance. The locus (a) inheritance-active finding supports the differentiation: no 2024–2026 architecture commits to the discrete-event-lifecycle plus explicit-dissolution plus per-participant-governed-ingestion conjunction as a named pattern. Inheritance shape: mostly Claim 1 inheritance applied. Claim 4 is architecturally different from contemporary post-deployment-update on the governance-configured-ingestion-at-home-perimeter axis, not contemporary post-deployment-update plus per-perimeter governance.

The second axis is **existing-Paper-2-mechanisms-at-home versus fresh-update-architecture**. This is the strongest single discriminator. Every contemporary post-deployment-update neighbor introduces fresh update-specific machinery — replay buffers, importance estimators, adapter modules, parameter-isolation logic, FedProx proximal terms, FedAdam adaptive moments, gradient projection methods, memory-manager subsystems — as fresh joint-event machinery added because joint coordination is happening. FAI-feed commits instead to using Paper 2’s three existing evolution mechanisms — instinct evolution, DNA evolution, and action-feedback evolution — at each home perimeter unchanged, with FAI-derived content as additional input source rather than as occasion for new update architecture. The cross-cutting closest-neighbor query confirmed the negative result: no widely recognized AI, multi-agent, federated-learning, or continual-learning architecture explicitly claims all three Claim 4 differentiation properties together. Governance-configured ingestion (Axis 1) and asymmetric absorption (Axis 3) appear often, but once cross-participant coordination is introduced, almost all published architectures add joint-event-specific update machinery, violating Axis 2. Within Axis 2, the layer-routing rule lives as fresh architectural specification distinct from learned routing and task-identity routing — DNA-layer content feeds DNA evolution; action-layer content feeds action-feedback evolution; instinct evolution takes no FAI input by architectural commitment per §5’s exchange-bounding. The locus (c) finding supports this: closest spirit-neighbors are process-defined or informal, not codified, and Letta routes by LLM judgment rather than content layer. Inheritance shape: Paper 2 §7 / §8 inheritance applied, with the layer-routing rule as fresh architectural commitment within FAI’s content-flow shape. Claim 4 is architecturally different from contemporary post-deployment-update on the existing-Paper-2-mechanisms-at-home axis, not contemporary post-deployment-update plus existing-mechanisms. **Primary novelty weight rests on Axis 2.**

The third axis is **asymmetric-ingestion-as-architectural-property versus symmetric-update-by-default — with the source-of-asymmetry calibration as the load-bearing distinction**. The literature treats asymmetry as a *consequence* of heterogeneous data (federated continual learning, asynchronous federated learning), role specialization (multi-agent memory, centralized-training-with-decentralized-execution), personalization objectives (personalized federated learning), or organizational autonomy (cross-silo federated learning, multi-tenant AI). FAI-feed commits to asymmetric ingestion as architectural property *derived from per-perimeter governance* — sourced from Claim 1’s governance-configured ingestion at each home perimeter rather than from heterogeneity, specialization, personalization, or organizational autonomy considered as separate concerns. The differentiation is on the source of the architectural asymmetry, not on whether asymmetric absorption appears in the literature, since locus (d)’s closest-neighbor density confirms that asymmetric absorption appears densely. Inheritance shape: partially fresh Paper 3 content, partially Claim 1 inheritance. Claim 4 is architecturally different from contemporary post-deployment-update on the asymmetric-ingestion-as-architectural-property axis with source-from-per-perimeter-governance, not contemporary post-deployment-update plus asymmetric-ingestion.

The conjunction carries the architectural contribution. Axis 1 alone is per-tenant configuration in multi-tenant AI; Axis 2 alone is unoccupied — once joint coordination is introduced, fresh update machinery follows; Axis 3 alone is widely shared with asymmetry sourced from heterogeneity, specialization, personalization, or organizational autonomy. What is unoccupied is the conjunction with asymmetry sourced from per-perimeter governance, FAI-derived content reaching existing per-Self evolution mechanisms unchanged, at the dissolution of an event-bounded shared substrate.

7.5 The inheritance-active finding and reviewer-objection responses

The conjunction §7.4 closes on is what Claim 4 contributes; three independent supporting findings stack across the axes to support the conjunction’s positioning. The cross-cutting finding — no widely recognized AI, multi-agent, federated-learning, or continual-learning architecture explicitly claims all three Claim 4 differentiation properties together — protects against underclaiming on Axis 2 by confirming the conjunction is unrealized at the architectural-pattern register at this scope. The locus (a) finding — no 2024–2026 architecture commits to the discrete event-bounded coordination state plus explicit dissolution plus per-participant governed post-dissolution ingestion conjunction as a single named pattern — protects against overclaiming locus (a) against AI/ML literature alone, since closest neighbors at locus (a) sit substantially in distributed-systems territory at adjacent register. The locus (c) finding — no 2024–2026 architecture formalizes content-layer-determined routing as architectural property distinct from learned routing and task-identity routing — protects against underclaiming locus (c) by confirming the layer-routing rule lives as fresh architectural specification rather than as articulation of an existing pattern under different vocabulary. The three findings synthesize patterns currently distributed across domains; the conjunction operating through architectural specification at FAI dissolution is what Papers 1 and 2’s principles produce when extended at Paper 3 scope. The finding is the closest-neighbor positioning anchor — neither under-claiming the four loci of fresh content nor over-claiming what the principles-extending register commits to. This is the **second inheritance-active finding** in Paper 3’s manuscript; the pattern surfaces again at Claims 5 and 6 with claim-specific shape, with §6.5 establishing the rendering pattern and §11 Discussion developing the four-finding compounding pattern across Claims 3 through 6 as a trilogy-level observation.

Three reviewer objections anticipate themselves with substantial force; five others surface on specific terms and warrant briefer treatment.

The first objection — *this is just Paper 2 §7 with FAI as input source — what’s new in Claim 4?* — is the internal-trilogy-continuity objection and bears the calibration discipline directly. Paper 2 §7’s three evolution mechanisms operate within each Self at home perimeter; Claim 4 commits to four loci of fresh content at the FAI-feed extension. The hand-off mechanism articulates governance-configured ingestion at FAI dissolution. The per-mechanism feed structure articulates how FAI-derived content reaches existing Paper 2 mechanisms unchanged through the existing per-mechanism governance shapes from Paper 2 §8. The layer-routing rule articulates that the layer determines which mechanism receives content, with the three-case explicitness honoring Paper 3 core theory’s instinct-evolution-out-of-scope subsection. The asymmetric ingestion property articulates that different participating Selves may take different things from the same FAI event under per-perimeter governance. The architectural register is principles-extending, and the cross-cutting negative finding confirms the conjunction: no contemporary post-deployment-update architecture commits to feeding joint-event content into existing per-participant mechanisms unchanged at this scope. The under-claim and over-claim implicit failure modes operate against this objection on both sides: collapsing Claim 4 into Paper 2 §7 plus naming would under-claim the four-locus conjunction the literature does not realize; designing the four loci as fresh evolutionary architecture beyond the inheritance would over-claim what the principles-extending register commits to.

The second objection — *this is just continual learning under new vocabulary* — runs primarily on Axis 2. Contemporary continual learning typically introduces update-specific machinery — replay buffers, regularization losses, parameter modules, importance estimators, slow/fast PEFT layers — and the architectural pattern is that update mechanism is part of system architecture, not just training code. FAI-feed commits to Paper 2’s existing evolution mechanisms unchanged at each home perimeter, with FAI-derived content as additional input source. Vocabulary discipline applies: “evolution” as noun for learning is contemporary continual-learning idiom, while FAI-feed uses Paper 2’s mechanism vocabulary at the architectural-pattern register.

The third objection — *this is just federated continual learning under new vocabulary* — runs on three architectural dimensions. First, persistent-global-model versus dissolving-shared-substrate — federated continual learning’s global model accumulates across rounds, while FAI’s shared substrate is constructed for an interaction and dissolves at completion. Second, FCL-client-side-machinery versus existing-Paper-2-mechanisms-unchanged at the Axis 2 differentiation — FCL clients run their own continual-learning routines

with replay buffers, projection methods, and gradient-balanced replay, with the client-side machinery itself fresh continual-learning architecture. Third, heterogeneity-driven-asymmetry versus governance-configured-asymmetry at the source-of-asymmetry calibration — asymmetric absorption in FCL is a consequence of client heterogeneity and personalization objectives, while in FAI-feed it is an architectural commitment derived from Claim 1’s per-perimeter governance. The fourth federated-learning boundary clarification applies: §4 engaged federated learning at joint-authority-mechanics, §5 at merge-mechanism, §6 at conflict-handling, §7 at evolution-feed.

Five briefer responses follow. *Why asymmetric ingestion?* Asymmetric ingestion is what Claim 1’s perimeter-spanning property and configuration-as-substrate-content commitment produce at the hand-off; the architectural default is per-home governance with the source-of-asymmetry calibration applied. *Hand-off mechanics — push or pull?* Hand-off is governance-configured per home authority; specific push-vs-pull mechanics are configurability dimensions per §8 territory, with adjacent-register neighbors (event-sourcing’s consumer-pull, MPC’s protocol-completion push, Sagas’ event-triggered either) supplying response material. *How do preserved conflicts carry through?* Preserved conflicts become annotations within home substrate during ingestion, flagging boundaries that home governance must address per home authority. Resolved outputs feed evolution mechanisms as content; preserved conflicts feed as annotated content; escalated conflicts already involved human authority before dissolution. *FAI-derived content conflicts with home content?* Home substrate’s existing conflict-handling mechanisms (Paper 1 §5 plus Paper 2’s intra-Self conflict handling) handle these at home perimeter under home governance; Paper 3 does not introduce additional home-conflict architecture, since §6 addresses inter-Self conflicts only. *Can FAI-feed produce fresh capabilities at home?* FAI-feed produces additional input for Paper 2’s evolution mechanisms operating under per-mechanism governance shapes; within those shapes, mutation and directed selection can produce fresh capability per Paper 2 §7’s productive-tension framing. Population-scope emergence across many FAI events at population scale is §9’s territory, where the calibrated-humility voice anchor appropriate to the trilogy-completion register applies.

7.6 Inheritance accounting and downstream placement

Claim 4 inherits without redefense from five sources. The asymmetric inheritance shape across loci tracks the structural-extension posture: locus (a) mostly Claim 1 inheritance applied; locus (b) mostly Paper 2 §7 / §8 inheritance applied; locus (c) Claim 2 plus Paper 2 inheritance applied with the fresh layer-routing rule articulation; locus (d) Claim 1 plus Paper 1 inheritance applied with the fresh source-from-governance articulation.

From Paper 1, all six commitments hold within evolution-feed scope. §5’s conflicts-as-first-class-state commitment carries through §6’s preserve tier and onward to Claim 4’s annotation-carry-through; the “humans decide” commitment holds during ingestion at home perimeter, with LLMs operating as substrate mediators throughout the FAI-feed pathway; §3.3’s authority-vs-labor distinction supplies the actor-neutrality discipline at Claim 4’s scope. From Paper 2, §7 entire — three evolution mechanisms operating in productive tension across multiple levels of architectural composition simultaneously, with horizontal and vertical evolution axes — is the structural-extension anchor; §8 entire establishes the per-mechanism governance shapes that hold at home perimeter when FAI feeds the mechanisms; §6.3’s intra-Self combination primitive over content combination supplies the architectural antecedent for content-flow-through-merge-event posture extended at the inter-Self perimeter through §5’s FAI operation. **Instinct evolution remains out of scope per Paper 2’s instinct/reasoning separation extended through §5’s exchange-bounding — LLM weights and instinct-layer content do not exchange across FAI**, and Paper 3 core theory’s instinct-evolution-out-of-scope subsection grounds this commitment directly. Paper 2’s biology engagement is bounded to its intra-Self scope per Paper 2 §2.2’s biology-as-bounded-load-bearing-terminology framing and Paper 3 core theory’s biology-positioning subsection; FAI-feed vocabulary at Paper 3’s voice register replaces any biology-mating idiom at this scope.

From §4 of Paper 3, the shared substrate dissolves after FAI; the perimeter-spanning property and configuration-as-substrate-content commitment produce per-home-ingestion as the hand-off shape. From §5, FAI as canonical operation with exchange bounded to DNA-layer and action-layer content — FAI outputs are what is available for evolution-feed, with the layer-bounding at the inter-Self boundary inherited as

foundational trilogy commitment. From §6, the three-tier conflict-handling mechanism with preserved conflicts as evolution-feed annotations specifically — the most direct §6 → §7 inheritance point, with §6.6’s closing forward reference to §7 (“conflicts preserved within the shared substrate carry through as evolution-feed annotations to participating Selves’ home substrate, with the carry-through architecture governance-configured per home perimeter”) substantively elaborated at §7.1 and §7.5. The five-source inheritance accounting is what makes the principles-extending posture honest at both edges: under-claim drift would treat the inheritance as the entire content with no fresh contribution at the four loci; over-claim drift would treat the four loci of fresh content as decoupled from the inheritance. The accounting locates each inheritance at its source and renders the fresh content as articulation of what the principles produce at Paper 3 scope.

The federated-learning territory engaged at Claim 4’s evolution-feed angle continues a cross-claim pattern. §4 engaged FL at the joint-authority-mechanics angle, §5 at the merge-mechanism angle, §6 at the conflict-handling angle; §7 engages at evolution-feed. FedAvg, FedProx, FedAdam, federated continual learning’s client-side machinery, and personalized federated learning’s heterogeneity-driven divergence all operate at the evolution-feed register; the source-of-asymmetry calibration applies — FCL clients diverge from heterogeneous data and personalization objectives, while FAI-feed clients diverge from per-perimeter governance configuration. The fourth federated-learning boundary makes the cross-claim FL-engagement pattern visible across all four claims as discipline. The pattern carries forward to §8 and §9 at configurable-aggregation-algorithm and population-scale-dynamics angles respectively, with §11 Discussion developing the six-FL-boundary pattern as a trilogy-level observation.

Claim 4 specifies the FAI-feed mechanism downstream claims operate over. As §8 develops, the per-event configuration of governance-configured ingestion — including which content classes are available for ingestion, ingestion-policy enumeration, conflict-annotation propagation depth, and per-mechanism feed configuration — is itself substrate content under recursive governance, with configurability dimensions enumerated as §8’s territory. As §9 develops, accumulated evolution-feed across many FAI events at population scope drives collective evolution, with the calibrated-humility voice anchor appropriate to the trilogy-completion register applied at that section’s scope. Claim 4’s contribution is the FAI-feed specification those downstream developments require: substrate-mediated coordination at the inter-Self scope under unified human governance, with FAI events feeding each participating Self’s home Paper 2 evolution mechanisms through the four-locus mechanism — hand-off at dissolution, per-mechanism feed structure, layer-routing rule with three-case explicitness, asymmetric ingestion — with the cross-perimeter shape of governance-configured ingestion at each home perimeter as the structural-extension move and the layer-routing rule as the fresh architectural specification within FAI’s content-flow shape.

8. Claim 5 — Configuration as substrate content with recursive applicability

8.1 Claim statement and the enumeration-and-elaboration register at the configurable-dimensions and recursive-governance angle

Claim 5 establishes **configuration as substrate content with recursive applicability** as what Paper 1 (Li, 2026a) §3.2’s substrate-as-content commitment plus Paper 2 (Li, 2026b) §8’s per-mechanism governance shapes plus §4’s configuration-as-substrate-content commitment for the shared substrate produce when applied to FAI events at the inter-Self perimeter. Paper 1 §3.2 commits within a cell to substrate as human-governed content, with structure originating under human authorship rather than as a derived index over a prior corpus. Paper 2 §8 extends this commitment across evolutionary dynamics through per-mechanism governance shapes co-determined with mechanism shape per Paper 2 §8.2 — verification-substrate machinery for instinct integration, authority architecture for DNA evolution, proposal-and-acceptance machinery for action-feedback’s substrate-change cycles, all under the authority-vs-labor distinction Paper 1 §3.3 makes portable. §4 commits configuration to the same substrate-content register as architectural commitment for the shared substrate; the configuration of perimeter-spanning is itself substrate content under joint human authority. Claim 5 articulates what these commitments produce when the cumulative configurable dimen-

sions referenced forward to this section’s territory by §§5–7 are themselves substrate content authored under joint authority across participating Selves’ governance during FAI events, with the recursive applicability of configuration-as-substrate-content as architectural property of §4’s commitment rather than fresh philosophical commitment.

The architectural register is **enumeration-and-elaboration**. Two implicit failure modes operate against the calibration discipline. The first is *catalog-bloat* — treating the enumeration of configurable dimensions as a catalogue with each dimension carrying architectural-commitment weight, with the substantive content dissolving into a list of named knobs. The second is *recursive-overclaim* — treating the recursive applicability of configuration-as-substrate-content as a fresh philosophical commitment requiring meta-layer architecture, with the recursion inflating into a hierarchical-layering pattern that competes with substrate-content commitment rather than being the architectural property of substrate-content commitment when applied to configuration substrate itself. The differentiation work that follows operates against both readings simultaneously: the architectural commitment is the framing — configuration-as-substrate-content with recursive applicability under joint authority — and the dimensions are the substantive consequences of cross-claim development the prior sections produced. The enumeration-and-elaboration calibration register is the fourth explicit calibration-register label in Paper 3’s manuscript per the varied-landscape framing — after §5’s mechanism-defense register and §6 + §7’s principles-extending register — and the first occurrence of enumeration-and-elaboration register pattern. §11 Discussion develops the varied calibration landscape across the six claims as a trilogy-level pattern.

Three architectural commitments compose Claim 5. The **enumeration of configurable dimensions across the FAI mechanism** specifies that the cumulative dimensions referenced forward by §§5–7 are themselves substrate content authored under joint authority across participating Selves’ governance. The architectural status of these dimensions is the same as other substrate content — inspectable, governable, modifiable under authority — and the enumeration is articulation of what is governable as substrate content per §4’s commitment, not a fresh catalogue with each dimension carrying architectural-commitment weight. The **recursive applicability of configuration-as-substrate-content** specifies that configuration substrate is itself substrate content under joint authority; configuration of configuration substrate is also substrate content; the recursion bottoms out at human-authored governance authority per Paper 1 §3.3, with recursive depth itself a configurable dimension. The recursion is architectural property of §4’s commitment — what substrate-content commitment looks like when the substrate carrying configuration is itself substrate — rather than fresh philosophical commitment requiring meta-governance layer architecture. The **multi-mediator coordination as configurable pattern** specifies coordination pattern as runtime substrate dimension under explicit governance: the five contemporary multi-coordinator patterns from §5 — single coordinator, hierarchical, protocol-level joint coordination, independent-plus-merge, locking — surface as configurable instances of this commitment rather than as the commitment itself. The architectural commitment is configurability of coordination pattern under joint authority; the patterns are what’s configured. The fresh content lives precisely in three commitments; everything else is inheritance Claim 5 articulates rather than commitments Claim 5 introduces.

The §7 → §8 transition rests on a primitive §7 establishes and §8 inherits without re-defending. The four-locus evolution-feed mechanism — hand-off at FAI dissolution, per-mechanism feed structure, layer-routing rule with three-case explicitness, asymmetric ingestion — operates over §§4–6 jointly during FAI events and at FAI dissolution. §8 takes up where §7’s brief response on push-vs-pull mechanics deferred and §7’s closing forward reference articulated — the per-event configuration of governance-configured ingestion is itself substrate content under recursive governance — and develops it at the architectural-pattern register Claims 1 through 5 commit jointly to. The next subsection develops the named foil at this section’s scope.

8.2 The named foil — opaque coordination-mechanism configuration and the configuration gap

§4 establishes opaque agent-to-agent communication as the named foil for inter-Self coordination, sharpened at §5 to opaque inter-agent merge of outputs without governable substrate, sharpened further at §6 to opaque conflict resolution at the conflict-handling level, and sharpened at §7 to opaque post-deployment update at

the evolution-feed level. At Claim 5’s configurable-dimensions and recursive-governance angle, the contemporary instances of the foil’s posture are **configuration-as-separate-policy-artifact architectures** — policy languages and engines (Cedar, OPA/Rego, Casbin), policy-as-code frameworks at infrastructure-as-code scope (Terraform/Sentinel, CloudFormation Guard, Pulumi), and AI governance frameworks at compliance-and-management register (NIST AI RMF, ISO/IEC 42001, EU AI Act compliance frameworks). The architectural posture is “configuration is artifact evaluated against substrate at decision time” with governance applied as separate-layer commitment over substrate state.

The foil sharpens at the configurable-dimensions and recursive-governance level in three ways §7.2’s foil articulation does not develop. First, the unit of configuration in contemporary systems is heterogeneous artifact — policy files in dedicated languages, IaC modules with separate state, governance documentation in ISO-aligned formats, compliance specifications in regulatory templates. None of these units carries the substrate-content commitment of being inspectable as substrate content under all six Paper 1 commitments within the operating substrate. Second, the recursive applicability of configuration is treated as separate-layer hierarchy — meta-policy over policy, meta-governance over governance, audit-of-audit, with each layer requiring its own artifact format and evaluation engine. Substrate-content commitment all the way down is not the contemporary architectural pattern. Third, the joint-authority commitment over configuration — when articulated at all in cross-organizational settings — operates at contract, federation-broker, or institutional layer rather than as substrate-content commitment within the configuration substrate itself.

What matters for Claim 5 is that these are properties of the architecture class at the configurable-dimensions and recursive-governance level, not capability gaps that better policy languages or richer compliance frameworks would close. As at §4.2, §5.2, §6.2, and §7.2, sharper policy engines and richer governance documentation would absorb the substrate Claim 5 commits to; the foil literature at §4.2 denies that reading at the substrate-content level, and the same denial holds at the configurable-dimensions and recursive-governance level. EU AI Act’s Articles 9-15 push closer toward embedding governance functionality into the substrate, but the binding-regulation framing keeps even that move at compliance-and-management register rather than at architectural-pattern register for substrate-content commitment. The appropriate response is therefore architectural at the configurable-dimensions and recursive-governance level specifically. Claim 5’s alternative is configuration as substrate content under all six Paper 1 commitments, with the recursion within substrate-content commitment rather than via separate meta-layers, and the joint authority across participating Selves’ governance carried by §4’s perimeter-spanning commitment. **Five architectural foils now occupy parallel positions** at the foil register Claims 1 through 5 commit to jointly — opaque agent-to-agent communication at the substrate-content level (§4.2), opaque inter-agent merge at the merge-mechanism level (§5.2), opaque conflict resolution at the conflict-handling level (§6.2), opaque post-deployment update at the evolution-feed level (§7.2), and opaque coordination-mechanism configuration at the configurable-dimensions and recursive-governance level (§8.2). §11 Discussion develops the trilogy-level foil-parallel pattern across the six claims.

8.3 Closest-neighbor engagement organized by commitment

Where §4 organized closest neighbors by territory across four tied-top clusters, §5 by cluster across three tied-top clusters, §6 by tier across three tiers, and §7 by locus across four loci, Claim 5’s three architectural commitments naturally cluster contemporary closest neighbors by which commitment they engage. The commitment-organization is genuine pattern, not template — each architectural commitment has a distinct closest-neighbor profile operating at a different architectural-pattern register, with neighbors clustering by what the commitment itself engages with rather than by where the synthesis chooses to place them. Commitment-organization is the fifth distinct organizational shape across Paper 3’s claims; the six-organizational-pattern across Claims 1–6 (territory / cluster / tier / locus / commitment / family) renders the discipline visible at trilogy-level paper structure.

Commitment (a) enumeration of configurable dimensions finds closest neighbors at the dimensions-as-substrate-content angle. Kubernetes CRDs are the strongest contemporary instance of dimensions-as-substrate-content, with CRD spec fields as first-class API content carrying full inspectability and policy-engine queryability — the configuration is substrate content within the cluster substrate. Multi-tenant SaaS configuration services persist tenant configuration as core data with substrate-level inspectability across

tenant_id-keyed routing, authorization, and billing pathways — the substrate carries configuration as substantive content, not as separate policy artifact. Data mesh implementations treat data-product metadata (classification, sensitivity, access policies, retention rules, regulatory tags) as first-class substrate content under federated computational governance. The architectural commitment converging across these neighbors is configuration-as-content with substrate-level inspectability; each falls short at a different boundary — CRDs at single-cluster governance with cluster admin authority, SaaS at provider-tenant asymmetric authority, data mesh at per-domain authorship within federated standards. Cedar, OPA/Rego, and Casbin operate at policy-as-separate-artifact register with the policy in its own language and a dedicated evaluation engine evaluated against substrate state at decision time; they serve as objection 1 contrast rather than as commitment (a) anchors.

Commitment (b) recursive applicability finds closest neighbors at the recursive-governance angle. DAO constitutional architectures (Aragon, Compound, MakerDAO, Arbitrum) are the strongest contemporary articulation of self-amending governance with substrate continuity: the same blockchain substrate carries substantive and constitutional decisions, but architecturally separate constitutional/amendment processes are implemented as specialized proposal types with additional thresholds and distinct contracts — the recursion holds at protocol-level register rather than at substrate-content commitment all the way down. Specific architectures within the family illustrate varied recursive-governance shapes: Aragon supports nested governance with on-chain meta-governance over governance contracts; MakerDAO’s Endgame plan articulates SubDAOs with constitutional structure plus delegate authority hierarchies; Optimism Collective’s bicameral structure with Citizens House and Token House supplies dual-authority recursion across token-weighted and citizen-weighted decision domains. Each DAO recursion holds at protocol-substrate-with-specialized-contracts register — different contract types for substantive versus constitutional decisions, with the governance contracts at architecturally distinct contract architectures from substantive contracts. The differentiation against configuration-as-substrate-content’s recursive applicability holds at architectural-plane location: DAO recursion separates governance-contract and substantive-contract architectures; FAI’s recursion holds at substrate-content commitment within the same architectural plane, with no contract-type distinction between governance content and substantive content. The CNCF Automated Governance Maturity Model frames Policy → Evaluation → Enforcement → Audit layers with the maturity model itself functioning as meta-governance over how governance is done; the layered shape is hierarchical-meta-governance rather than substrate-content recursion. Anthropic’s Constitutional AI is closer-in-kind to governance-as-substrate-content register but operates at training-time / objective-shaping register rather than runtime substrate-content register, with the constitution embedded in model substrate via training-time objective shaping and full self-amendment of the constitution itself generally not specified. The CAIA paper (Nguyen 2026, SSRN preprint) is engaged at positioning territory only with **explicit register-mismatch flag** — its commitment plane is regulatory/political-institutional and infrastructure-resource-governance (compute, energy, bandwidth, storage caps), distinct from architectural-pattern register and distinct from the training-time-vs-runtime distinction that applies to Anthropic Constitutional AI. The register-mismatch flag is what positions CAIA outside the closest-neighbor synthesis rather than within it.

Commitment (c) multi-mediator coordination as configurable pattern finds closest neighbors at the runtime-coordination angle. Multi-leader replication patterns in distributed databases (CockroachDB, YugabyteDB) commit to coordination-pattern-as-deployment-choice with patterns selected at deployment time — the pattern is configurable but not as runtime substrate dimension. Multi-coordinator federated-learning architectures including HGAFL expose hierarchical aggregation as configuration, with the hierarchical structure parameterized at framework-API register rather than as substrate-content dimension. Multi-orchestrator workflow patterns in Temporal, Airflow, and Prefect distinguish single-orchestrator from multi-orchestrator deployment topologies with the topology choice at deployment register; the choreography-vs-orchestration distinction in BPMN articulates coordination pattern as architectural choice expressed in process-modeling languages. Across these neighbors the recurring pattern is coordination-pattern-as-deployment-time-or-framework-API-choice; none commits to coordination pattern as runtime substrate dimension under explicit governance with the patterns themselves what’s configured rather than what’s architecturally fixed. Commitment (c) carries fresh-articulation weight at Claim 5 scope.

8.4 Three differentiation axes with asymmetric inheritance shape

Three architectural axes carry differentiation simultaneously, each holding against contemporary configuration-management neighbors at the configurable-dimensions and recursive-governance angle (distinct from §4’s joint-authority-mechanics, §5’s merge-mechanism, §6’s conflict-handling, and §7’s evolution-feed angles). The honest weight is asymmetric — Axis 1 mostly Claim 1 inheritance applied with elaboration; Axis 2 carrying primary novelty weight via §4’s Axis 2 inheritance applied with the recursive-applicability articulation as fresh; Axis 3 mostly Claim 1 inheritance applied at Claim 5 scope. The asymmetry is structural, reflecting the enumeration-and-elaboration posture rather than uniform-novelty distribution across axes.

The first axis is **configuration-as-substrate-content under all six Paper 1 commitments versus configuration-as-policy-language-or-rules-engine**. Contemporary configuration-management architectures commit to configuration as separate policy artifact with its own language and dedicated evaluation engine evaluated against substrate state at decision time. Configuration-as-substrate-content commits to configuration on the same architectural plane as other substrate content, with the same governance shapes, the same inspectability, and the same authority-vs-labor distinction Paper 1 §3.3 commits. Cedar, OPA, and Casbin’s varied deployment shapes reinforce policy-as-separate-artifact at architectural-commitment register even where policies and entity data are richly co-managed at the deployment layer; the differentiation holds at architectural-commitment level rather than at deployment-shape level. Inheritance shape: mostly §4’s Axis 2 inheritance applied; the enumeration at Claim 5 elaborates rather than re-defends. Claim 5 is architecturally different from contemporary configuration-management on the substrate-content axis, not contemporary configuration-management plus substrate.

The second axis is **recursive applicability through substrate-content commitment versus separate meta-governance layer**. This is the strongest single discriminator and carries primary novelty weight at Paper 3 scope. Contemporary AI governance frameworks (NIST AI RMF, ISO/IEC 42001, EU AI Act compliance) and contemporary meta-governance work typically commit to a separate meta-governance layer — governance over governance lives in different artifacts than substantive governance, with the cluster pattern being hierarchical-layering across distinct planes. DAO constitutional architectures carry the same separate-layer commitment despite operating on shared substrate, since the constitutional/amendment processes are specialized proposal types with distinct contracts. Configuration-as-substrate-content’s recursive applicability commits to substrate-content all the way down — configuration substrate is substrate content, configuration of configuration substrate is substrate content, with no architectural distinction between governing and meta-governing. The recursion is architectural property of §4’s substrate-content commitment, not fresh philosophical commitment requiring meta-layer architecture; it bottoms out at human-authored governance authority per Paper 1 §3.3 rather than at protocol invariants or institutional procedures. Inheritance shape: §4’s Axis 2 inheritance applied, with the recursive-applicability articulation as fresh. **Primary novelty weight rests on Axis 2**. Claim 5 is architecturally different from contemporary meta-governance on the recursive-applicability-through-substrate-content-commitment axis, not contemporary meta-governance plus substrate.

The third axis is **joint authority across participating Selves versus single-perimeter configuration authority**. Contemporary configuration-management within single organizations operates under single-perimeter authority. Cross-organizational configuration architectures (data trust governance, FL consortia, multi-tenant configurations, IDS-RAM, Gaia-X) typically encode joint authority at contract or institutional layer rather than as substrate-content architectural commitment, with federation-broker mediation patterns, consortium governance bodies expressed in contracts, or single-perimeter author with cross-perimeter consumers as the architectural shape. None commits joint authority at substrate-content architectural-commitment level. Inheritance shape: mostly Claim 1 inheritance applied at Claim 5 scope. Claim 5 is architecturally different from contemporary cross-organizational configuration on the joint-authority-across-participating-Selves-at-substrate-content-register axis, not contemporary cross-org configuration plus joint authority.

The conjunction carries the architectural contribution. Axis 1 alone is Kubernetes-CRD territory at single-cluster scope; Axis 2 alone with separate-layer commitment is the CNCF maturity model and DAO consti-

tutional pattern; Axis 3 alone at federation-broker register is Gaia-X and IDS-RAM. What is unoccupied is the conjunction with all three holding at architectural-pattern register simultaneously — configuration as substrate content under all six Paper 1 commitments, recursive applicability through substrate-content commitment, joint authority across participating Selves’ governance, all operating through the shared substrate during FAI events.

8.5 The third inheritance-active finding and reviewer-objection responses

The conjunction §8.4 closes on is what Claim 5 contributes; the cross-cutting research query confirms the conjunction is not realized in contemporary literature. The query asked whether any contemporary architecture commits architecturally to all three Claim 5 axes simultaneously: configuration as substrate content under all six Paper 1 commitments; recursive applicability through substrate-content commitment; joint authority across participating Selves’ governance. Five candidates were examined. *Kubernetes CRDs* come closest at Axis 1 with substrate-level inspectability and policy-engine queryability, but commit to single-cluster governance with cluster admin as authority — the recursive-applicability commitment requires governance over CRD-governance to be substrate content rather than separate-layer policy (Kyverno, OPA Gatekeeper), and the joint-authority commitment requires architecture-level commitment to multiple distinct governance perimeters jointly authoring the substrate, neither commitment Kubernetes makes at architectural-pattern level. *DAO constitutional architectures* come closest at Axis 3 within their voting-mechanism register, with token-holder governance over constitutional amendments approximating joint authority at protocol level; but DAOs separate constitutional/amendment processes from substantive governance as specialized proposal types, the recursive-applicability commitment to substrate-content all the way down is not made, and the configuration-as-substrate-content commitment is partially held at smart-contract substrate level but with type-boundary distinction between governance contracts and protocol contracts. *Anthropic’s Constitutional AI* comes closest at Axis 2 in spirit, with constitution embedded in model substrate via training-time objective shaping; but CAI operates at training-time / objective-shaping register rather than runtime substrate-content register, full self-amendment of the constitution itself is generally not specified, and the joint-authority commitment is single-developer rather than across-Selves. *Multi-tenant SaaS configuration* partially holds Axis 1 with asymmetric provider-tenant authority. *Data mesh implementations* hold Axis 1 cleanly under federated computational governance but treat shared standards as separate-layer policy. None holds all three at architectural-pattern register at the scope required.

The three-axis conjunction operating through architectural specification of configuration as substrate content under FAI is what Papers 1 and 2’s principles produce when extended at Paper 3 scope plus §4’s configuration-as-substrate-content commitment elaborated. **This is the third inheritance-active finding across Claims 3, 4, 5** — §6’s three-clause conjunction at the conflict-handling angle, §7’s three-axis conjunction at the evolution-feed angle, and now Claim 5’s three-axis conjunction at the configurable-dimensions and recursive-governance angle — with the pattern compounding across claims as discipline carrying forward to Claim 6. §11 Discussion develops the four-finding compounding pattern across the trilogy.

Three reviewer objections anticipate themselves with substantial force; five others surface on specific terms and warrant briefer treatment.

The first objection — *this is just configuration management or policy-as-code under new vocabulary* — runs primarily on Axis 1. Cedar, OPA/Rego, Casbin, and IaC governance frameworks commit to configuration as separate policy artifact with its own language and dedicated evaluation engine; AI governance frameworks commit to configuration as compliance-and-management artifacts in governance documentation. Configuration-as-substrate-content commits to configuration on the same architectural plane as other substrate content under all six Paper 1 commitments — same governance shapes, same inspectability, same authority-vs-labor distinction. The differentiation holds at architectural-commitment register rather than at deployment-shape register. EU AI Act’s Articles 9-15 push closer toward embedding governance functionality into the substrate, but the binding-regulation framing keeps even that move at compliance-and-management register rather than at architectural-pattern register for substrate-content commitment.

The second objection — *this is just meta-governance or recursive policy under new vocabulary* — runs primarily on Axis 2. The pattern across contemporary meta-governance work, DAO constitutional architec-

tures, and oversight-board institutional layers is hierarchical-layering of meta-governance over substantive governance. Configuration-as-substrate-content’s recursive applicability commits to substrate-content all the way down — the recursion is architectural property of §4’s substrate-content commitment, not fresh philosophical commitment requiring meta-layer architecture. The recursion bottoms out at human-authored governance authority per Paper 1 §3.3, not at protocol invariants or institutional procedures. The CAIA paper engaged at positioning territory only with explicit register-mismatch flag — its commitment plane is regulatory/political-institutional and infrastructure-resource-governance register, not architectural-pattern register for substrate-content commitment.

The third objection — *this is just an enumeration catalogue, not architectural commitment* — is the deepest defensive challenge for Claim 5 and runs on the calibration discipline directly. Adjacent enumerations (Kubernetes CRDs, multi-tenant SaaS, data mesh metadata, FL hyperparameter taxonomies, NIST AI RMF and ISO 42001 dimension structures) share the appearance of configurable-dimensions enumeration but commit at different architectural registers — substrate-content under single-cluster or asymmetric authority, federated-standards-as-separate-layer policy, framework-API and governance-documentation register respectively. Claim 5’s enumeration carries architectural commitment beyond catalogue precisely because the three architectural axes hold the conjunction the adjacent enumerations do not realize: the architectural commitment is the framing — configurability under joint authority with recursive applicability — and the dimensions are the substantive consequences of cross-claim development rather than a free-standing list of named knobs. The catalog-bloat failure mode is what the calibration discipline guards against; the enumeration-and-elaboration register is what makes the architectural commitment honest.

Five briefer responses follow. *How does the recursive applicability bottom out — doesn’t this risk infinite regress?* Recursion bottoms out at human-authored governance authority per Paper 1 §3.3; recursive depth itself is a configurable dimension, with the bottom human rather than architectural. The recursive-overclaim failure mode is what the bottom-out commitment guards against — the recursion is architectural property of substrate-content commitment, not separate philosophical commitment requiring meta-layer architecture. *Why these dimensions specifically — where do they come from?* Dimensions originate from cross-claim development of §§5–7; the enumeration is not exhaustive; the architectural commitment is to configurability under joint authority. *How does configuration-as-substrate-content interact with home substrate’s existing configuration?* Home substrate’s existing configuration mechanisms operate at home perimeter under home governance per Paper 1; configuration substrate for FAI events lives within the shared substrate during the event, with home substrate’s configuration applying to home-perimeter ingestion per §7. *Can configuration substrate be authored by LLMs rather than humans?* Paper 1 §3.3’s authority-vs-labor distinction holds — LLMs draft, suggest, modify configuration substrate operating as substrate mediators; humans hold authority over what is approved, modified, or revoked. *Does configuration substrate during FAI events synchronize across participating Selves’ home substrates?* Configuration substrate for an FAI event lives within the shared substrate during the event and dissolves with substrate dissolution per §4; configuration-substrate effects on home substrate operate per §7’s governance-configured ingestion, with no cross-Self synchronization architecture introduced.

8.6 Inheritance accounting and downstream placement

Claim 5 inherits without redefense from six sources. The inheritance-shape pattern is heavy at §4’s Axis 2 and Paper 2 §8 (architectural antecedents); medium-heavy at §§5–7 (cumulative dimensions plus deferrals plus precedents); light at Paper 1 (foundational commitments hold within scope without re-engagement).

From Paper 1, all six commitments hold within configuration-substrate scope: configuration substrate is substrate content. §3.2’s substrate-as-content commitment is foundational — structure originates under human authorship at the cell scope Paper 1 defends, and the same commitment carries forward into configuration substrate at FAI scope. §3.3’s authority-vs-labor distinction supplies the actor-neutrality discipline at Claim 5’s scope and underwrites the recursive-governance bottom-out at human-authored governance authority rather than architectural invariants. The portable phrase Paper 1 §3.3 renders — governance is an authority architecture, not a review workflow — holds across configuration substrate at FAI scope: humans hold authority over configuration substrate; LLMs draft, populate, and modify configuration substrate as substrate mediators.

From Paper 2, §8 entire is the most load-bearing antecedent: per-mechanism governance shapes co-determined with mechanism shape per Paper 2 §8.2 hold at home perimeter when configuration is substrate content, and the inter-Self scope inherits these shapes through the home perimeter, adding joint authority across participating Selves' governance for cross-perimeter configuration substrate. Paper 2 §8.5 supplies the primary differentiation-passage tone anchor where the contrast against mechanism-agnostic AI governance frameworks is articulated; Paper 2 §8.3 supplies the architectural antecedent for boundary-as-substrate-content treatment that Claim 5 extends to configuration boundaries. Paper 2's biology engagement is bounded to its intra-Self scope per Paper 2 §2.2's biology-as-bounded-load-bearing-terminology framing and Paper 3 core theory's biology-positioning subsection; FAI-feed and configuration-substrate vocabulary at Paper 3's voice register replaces any biology idiom at this scope.

From §4 of Paper 3, §4.4 Axis 2's configuration-as-substrate-content commitment is the most load-bearing inheritance — Claim 5 elaborates this commitment at the configurable-dimensions and recursive-governance angle rather than re-defending. The shared substrate carries configuration as substrate content within scope; the elaboration at Claim 5 extends this commitment recursively and across joint authority of participating Selves' governance. From §§5–7, the cumulative configurable dimensions referenced forward to this section's territory carry the substantive content of commitment (a). §5 contributes six dimensions at the FAI-event level — sharing scope, cardinality, persistence policy, cooperation/competition variant, provenance carry-over depth, multi-mediator coordination — plus the five contemporary multi-coordinator patterns supplying commitment (c) inheritance material. §6 contributes six dimensions at the conflict-handling level — tier-selection rules, conflict-class definitions, escalation-path configurations, joint-authority configurations, recursive-governance depth, authority-articulation mechanics — plus the explicit deferrals at §6.5 brief responses where the recursive-governance enumeration itself is referenced forward to Claim 5 territory. §7 contributes five dimensions at the evolution-feed level — hand-off push-vs-pull, per-mechanism feed configurability, content-class availability at home, ingestion-policy enumeration, conflict-annotation propagation depth. The seventeen cumulative dimensions across §§5–7 carry forward at Claim 5 scope as substrate content authored under joint authority. The six-source inheritance accounting is what makes the enumeration-and-elaboration posture honest at both edges: catalog-bloat drift would treat the enumeration as a free-standing list of named knobs decoupled from architectural commitment; recursive-overclaim drift would treat the recursive applicability as fresh philosophical commitment decoupled from §4's substrate-content commitment. The accounting locates each inheritance at its source and renders the fresh content as articulation of what the principles produce at Paper 3 scope.

The federated-learning territory engaged at Claim 5's configurable-aggregation-algorithm angle continues a cross-claim pattern. §4 engaged FL at the joint-authority-mechanics angle, §5 at the merge-mechanism angle, §6 at the conflict-handling angle, §7 at the evolution-feed angle; §8 engages at configurable-aggregation-algorithm. Contemporary FL platforms expose aggregation as configurable algorithm — FedAvg, FedProx, FedAdam as configurable aggregation algorithms; secure-aggregation parameters as configurable; Byzantine-robust thresholds as configurable; asynchronous-FL timing as configurable; personalized-FL configuration as configurable. The aggregation-algorithm-as-configurable parallel is at the architectural-pattern shape level; the substrate-content register and joint-authority register are the differentiation distance from FL's framework-API register and consortium-coordinator authority register. The fifth FL boundary makes the cross-claim FL-engagement pattern visible across all five claims as discipline. The pattern carries forward to §9 at the population-scale-dynamics angle, with §11 Discussion developing the six-FL-boundary pattern as a trilogy-level observation.

Claim 5 specifies the configurable-dimensions enumeration plus recursive-governance framing for FAI events that the cumulative cross-claim development produces under §4's configuration-as-substrate-content commitment elaborated. As §9 develops, accumulated configured FAI events across many participating Selves at population scope drive collective evolution under the recursive-governance framing — population-scale dynamics inherit the configurability commitment, with **configuration of population-scale dynamics also substrate content under joint authority and recursion bottoming at population-level human governance per Paper 1**, with the calibrated-humility voice anchor appropriate to the trilogy-completion register applied at §9's scope. Claim 5's contribution is the enumeration-and-elaboration specification those downstream developments require: substrate-mediated coordination at the inter-Self scope under unified

human governance, with FAI events configured per event through configuration substrate authored under joint authority across participating Selves' governance, and the recursive applicability of configuration-as-substrate-content as architectural property of §4's commitment elaborated at Claim 5's enumeration register.

9. Claim 6 — Population-scale collective evolution as the trilogy-completion architectural object

9.1 Claim statement and population-scale collective evolution as the trilogy-completion architectural object

Claim 6 establishes **population-scale collective evolution** as the architectural object Paper 3's five preceding claims produce at population register when accumulated FAI events across many participating CKS-governed Selves operate under the configuration-as-substrate-content commitment with recursive applicability. Paper 1 (Li, 2026a) §3.2 commits within a cell to substrate as human-governed content. Paper 2 (Li, 2026b) §6 establishes lifecycle as governed primitives within a Self — particularly §6.3's mating primitive with multi-cardinality and §6.4's two architectural results for death; §7 commits to three evolution mechanisms in productive tension across multiple composition levels; §8 specifies per-mechanism governance shapes co-determined with mechanism shape. §4 of Paper 3 establishes the shared substrate as the architectural object at the inter-Self perimeter; §7 specifies the evolution-feed mechanism through which FAI events reach each home Self's Paper 2 evolution machinery; §8 elaborates the recursive applicability of configuration-as-substrate-content. Claim 6 articulates what these compose into when many CKS-governed Selves accumulate FAI events under cross-population governance — population-scale dynamics carrying through §7's evolution-feed mechanism extended at the population perimeter under §8's recursive-applicability framing.

Three architectural commitments compose Claim 6. The **population-scale collective evolution as architectural object** commitment articulates that the cumulative coordination structure produced by many CKS-governed Selves engaging in FAI events under joint authority is itself an architectural object — substrate content at population register under cross-population governance, with all six Paper 1 commitments holding at population scope and the population substrate content rather than an agent. The **cross-event evolution dynamics under joint authority across population-level governance** commitment articulates that accumulating FAI events produce collective evolution dynamics: population-scale mating as Paper 2 §6.3's multi-cardinality primitive extended at the inter-Self perimeter at population scope; population-scale death-types as §6.4's two architectural results extended; population-scale productive tension between mutation and directed selection as §7 extended through §7's evolution-feed mechanism. Joint authority extends at population scope to joint authority across population-level governance perimeters; the recursion bottoms at population-level human governance per Paper 1 through §8's recursive applicability. The **architectural-commitment differentiation against capability-motivated population-scale coordination** commitment articulates that governance-motivated population-scale architecture commits to substrate-content commitment at population scope, joint authority across population-level governance, and recursive applicability — commitments capability-motivated architectures do not make at architectural-pattern level even when they scale functional coordination. This commitment engages §4.5's brief redirect of foundation-labs-solve-this-anyway to §9 territory through §9.5 below.

The §8 → §9 transition rests on a primitive §8 establishes and §9 inherits without re-defending. The configurable dimensions across the FAI mechanism with recursive applicability of configuration-as-substrate-content — configuration substrate as substrate content under all six Paper 1 commitments, recursion bottoming at human-authored governance authority per Paper 1 §3.3, multi-mediator coordination as configurable pattern — operates over §§4–7 jointly during FAI events. §9 takes up where §8's closing forward reference articulated — accumulated configured FAI events across many participating Selves at population scope drive collective evolution under the recursive-governance framing, with **configuration of population-scale dynamics also substrate content under joint authority and recursion bottoming at population-level human governance per Paper 1**, with the calibrated-humility voice anchor appropriate to the trilogy-completion register applied at §9's scope — and develops it at the architectural-pattern register Claims 1 through 5 commit jointly to.

Claim 6 lands in **trilogy-completion / extension-claim register**, not fresh-architectural-commitment register. Per Paper 3’s core theory, the substrate-mediator commitment progresses across the trilogy at six configurations of progressively larger scope: rung 1 single human $\leftrightarrow$ LLM within one cell (Paper 1 core theory); rung 2 multiple humans $\leftrightarrow$ LLM within one cell (Paper 1 extension at its Claim 6); rung 3 cell $\leftrightarrow$ aspect $\leftrightarrow$ Self within one Self (Paper 2 core theory); rung 4 Self as enterprise-scope unified architecture (Paper 2 extension at its Claim 6); rung 5 Self $\leftrightarrow$ Self mediated by shared substrate (Paper 3 core theory across §§4–8); rung 6 population-scale collective dynamics across many Selves (Paper 3 extension at this section). Each paper carries a core-theory rung and an extension-claim rung; Claim 6 of Paper 3 occupies the sixth rung structurally parallel to Paper 1’s Claim 6 and Paper 2’s Claim 6. The **parallel-extension-claim structure** is what makes composition rather than fresh emergence theory architecturally honest — three extension claims at progressively larger scope, each operating within the same substrate-mediator commitment, each acknowledging calibrated humility at its own register. The ladder is observation rather than roadmap: Paper 1 did not promise Paper 3, and Paper 2 explicitly bracketed inter-Self dynamics as out of scope; the ladder becomes visible in retrospect because Paper 3 completes it.

The calibration discipline holds in **three poles** distinct from Claims 2–5’s two-failure-mode patterns. The first pole is *emergent dynamics yes, fresh emergence theory no* — substantive emergent characteristics per Paper 3 core theory (coordinated handling of problems beyond what any single Self accumulated independently, capability propagation across organizational and domain boundaries, collective adaptation to novel conditions) are what the inheritance produces through composition rather than what fresh emergence-theoretic apparatus is required to explain. The substantive distinction Pole 1 carries lies between emergence-as-descriptor and emergence-as-load-bearing-theoretical-frame. The architecture commits to emergence-as-descriptor — population-scale dynamics named as emergent characteristics arise from composition of the architectural commitments §§4–8 establish, with those architectural commitments themselves the explanation for what the inheritance produces. The architecture rejects emergence-as-load-bearing-frame — fresh ontological commitments distinct from the architectural commitments themselves (Holland-style complex-adaptive-systems apparatus as the load-bearing frame, weak-versus-strong-emergence taxonomy, autopoiesis-style frames committing to irreducibility or downward causation as required theoretical apparatus) are not required to explain what the architectural inheritance produces at population scope, since the architectural commitments composing forward at population register are the explanation. The criterion of distinction is whether the explanation runs through architectural commitments composing or through fresh theoretical apparatus added at population scope. The second pole is *capability scaling delivers functional inter-Self coordination at scale honestly, with the differentiation at architectural-commitment register rather than at the functional-coordination claim*. The third pole is *population-scale CKS-governed Self deployments do not yet exist; the architectural commitment is to what the inheritance produces at population scope, with specific population-scale dynamics awaiting empirical observation* — this is the trilogy’s anticipated landing rather than its empirically demonstrated outcome. The **calibrated-humility voice anchor** activates from §9.1 onward and applies throughout §9, distinguishing the architectural commitment (rendered confidently) from the empirical-verification status (rendered with calibrated humility). The trilogy-completion / extension-claim calibration register is the fifth explicit calibration-register label in Paper 3’s manuscript per the varied-landscape framing — after §5’s mechanism-defense register, §6 + §7’s principles-extending registers, and §8’s enumeration-and-elaboration register — and the first occurrence of trilogy-completion register pattern. §11 Discussion develops the varied calibration landscape across the six claims as a trilogy-level pattern. The next subsection develops the named foil at this section’s scope.

9.2 The named foil — opaque population-scale coordination and the population-scale-dynamics gap

§4 establishes opaque agent-to-agent communication as the named foil, sharpened at §5 to opaque inter-agent merge at the merge-mechanism level, sharpened further at §6 to opaque conflict resolution at the conflict-handling level, sharpened at §7 to opaque post-deployment update at the evolution-feed level, and sharpened at §8 to opaque coordination-mechanism configuration at the configurable-dimensions and recursive-governance level. At Claim 6’s population-scale-collective-evolution angle, three contemporary foil postures hold simultaneously, and the foil is the composite each posture contributes to.

The first posture is **capability-motivated population-scale coordination architectures**. Foundation lab multi-agent systems at scale (OpenAI AgentKit, Anthropic Claude multi-agent + MCP, Google Gemini Enterprise Agent Platform), agent-protocol ecosystems standardizing communication mechanics with governance pushed out of the protocol (MCP under cross-vendor stewardship, A2A across enterprise partners), and large-scale agent swarm architectures with governance at graph topology all treat coordination at scale as capability/architecture/protocol selection with governance offloaded to deployment-shape and contract/institutional registers. The second posture is **emergence-as-fresh-property architectures**. Complex-adaptive-systems work treating population-scale dynamics as emergent property requiring fresh theoretical framing, persona-based multi-agent simulations, swarm-intelligence and collective-intelligence frameworks all commit to emergence-theoretical apparatus as the load-bearing frame for explaining what populations of interacting agents produce. The third posture is **federated-learning-at-population-scope architectures**. FLARE-based and FATE-based consortia at hundreds-of-organizations scale, with governance at coordinator-mediated aggregation under consortium-governance contracts, treat large-scale FL as fulfilling cross-organizational coordination at population scale.

The foil sharpens at the population-scale-collective-evolution level in three ways §4.2’s foil articulation, and its sharpenings at §§5.2–§8.2, do not develop. First, the unit of coordination at population scope in contemporary systems is heterogeneous — protocol message exchange, model-fusion outputs, inter-agent dialogue traces, simulation event records — none of which carries the substrate-content commitment of being inspectable as substrate content under all six Paper 1 commitments at population register. Second, governance over population-scale coordination, when articulated at all, operates at framework-API, deployment-shape, contract/institutional, or simulation-analysis register rather than as substrate-content commitment within the population-scale dynamics themselves. Third, the recursive applicability of configuration that §8 commits to at the FAI-event level is not extended at population scope by any contemporary architecture; configuration of population-scale dynamics, when configurable at all, is configurable at meta-governance-layer or consortium-contract register rather than as substrate content under all six Paper 1 commitments.

What matters for Claim 6 is that these are properties of the architecture class at the population-scale-collective-evolution level, not capability gaps that better protocols or richer consortium contracts would close. As at §4.2, §5.2, §6.2, §7.2, and §8.2, sharper protocols and richer governance frameworks would absorb the substrate Claim 6 commits to; the foil literature at §4.2 denies that reading at the substrate-content level, and the same denial holds at the population-scale-collective-evolution level. The trilogy-completion alternative is articulation of what the cumulative architecture produces at population scope through composition: substrate-mediated coordination at the population scope under unified human governance, with FAI events accumulating across many participating Selves under joint authority across population-level governance perimeters and the recursive applicability of configuration-as-substrate-content extended at population register. **Six architectural foils now occupy parallel positions** at the foil register Claims 1 through 6 commit to jointly — opaque agent-to-agent communication at the substrate-content level (§4.2), opaque inter-agent merge at the merge-mechanism level (§5.2), opaque conflict resolution at the conflict-handling level (§6.2), opaque post-deployment update at the evolution-feed level (§7.2), opaque coordination-mechanism configuration at the configurable-dimensions and recursive-governance level (§8.2), and opaque population-scale coordination at the population-scale-collective-evolution level (§9.2). §11 Discussion develops the trilogy-level foil-parallel pattern across the six claims.

9.3 Closest-neighbor engagement organized by family

Where §4 organized closest neighbors by territory across four tied-top clusters, §5 by cluster across three tied-top clusters, §6 by tier across three tiers, §7 by locus across four loci, and §8 by commitment across three commitments, Claim 6’s closest-neighbor structure organizes by **family** — four closest-neighbor families plus adjacent territories — at the trilogy-completion register. The family-organization is genuine pattern, not template: each family operates at a distinct cross-organizational register at population scope, with families clustering by which two of Claim 6’s three differentiation axes the family holds and which one it falls short on for specific structural reasons. Family organization is the sixth distinct organizational shape across Paper 3’s claims; the **six-organizational-pattern across Claims 1–6** (territory / cluster / tier / locus / commitment / family) renders the discipline visible at trilogy-level paper structure. §11 Discussion develops

the six-organizational-pattern as a trilogy-level observation.

Family 1 — orchestration and meta-governance blueprints holds Axes 1 and 3 at intra-organizational scope and falls short on Axis 2. Adimulam et al. (2026) “The Orchestration of Multi-Agent Systems: Architectures, Protocols, and Enterprise Adoption” supplies unified orchestration architecture treating coordination across many agents as architectural object rather than as protocol or framework configuration; meta-governance architectures propose meta-governance layers as explicit architecture objects with the architecture-as-object framing extended to governance machinery itself. Both commit to substrate-content commitment at intra-organizational scope and to composition rather than emergence as the architectural-honesty register. Where the family falls short: joint authority never crosses into a population substrate across firms — the orchestration-and-meta-governance work operates within one organization’s authority perimeter, with cross-organizational coordination either deferred to contract/institutional register or framed as out of scope.

Family 2 — agent-contracts and governed-reasoning frameworks holds Axis 1 at varied registers and falls short on Axes 2 and 3 at population scope. Agent Contracts (Ye and Tan 2026) lands at resource-budget register through the formal $C=(I,O,S,R,T,\Phi,\Psi)$ tuple with conservation laws on delegated budgets, supplying substrate-content commitment specifically over delegated computational resources. Cognitive Core and Governance-First Architecture work lands at decision-substrate register through typed primitives and multi-tier review architecture, supplying substrate-content commitment specifically over reasoning-execution operations. Each illustrates one register where substrate-content commitment lands; what each does not extend to is cross-organizational joint authority at architectural-pattern register at population scope (Axis 2) or population-scale dynamics framed as composition rather than as out-of-scope (Axis 3) — the substrate-content commitment lands at intra-organizational scope, with cross-organizational extension deferred.

Family 3 — enterprise-MAS-governance blueprints holds Axes 1 and 3 at single-organization population-of-agents scope and falls short on Axis 2 by construction. Governing Multi-Agent AI Systems enterprise blueprints supply enterprise-wide governance substrate at population-of-agents scope through central agent registry as system-of-record (agent identity, ownership, risk tier, prompts, tools, evolution history), policy-driven interaction graphs, and conflict-resolution rules. The blueprint pattern is the closest contemporary instance of population-of-agents-as-architectural-object at single-organization scope — substrate-content commitment over agent registry holds; composition rather than emergence as the architectural-honesty register holds. Where the family falls short: the enterprise as one organization is the single sovereign authority by construction. Cross-organizational extension would require introducing joint authority across distinct governance perimeters — the architectural commitment Family 3’s single-org register does not make.

Family 4 — federated-learning consortia at population scope holds Axis 2 at contract-and-governance-committee register and falls short on Axes 1 and 3. FLARE-based, FATE-based, and OpenFL consortia at hundreds-of-organizations scale across healthcare, pharma, government labs, and financial sector supply population-scale federated learning coordinator-mediated under consortium agreements. Specific consortia commit at varied technical registers within the family: NVIDIA FLARE supports federated XGBoost, vertical FL, secure aggregation, and Byzantine-robust aggregation as architectural primitives within the consortium framework; FATE supports homomorphic-encryption-based secure aggregation as substrate-level cryptographic commitment; OpenFL supports Trusted Execution Environment-based aggregation through hardware-attested computation. These technical commitments are real and operate at aggregation-substrate-with-cryptographic-or-hardware-guarantees register, not at substrate-content commitment register where the aggregation substrate itself carries provenance, authority, and Paper 1’s six commitments within scope as architectural property. The cross-organizational scope is real — joint authority across distinct governance perimeters is encoded through consortium charters, governance committees, data-use agreements, and coordinator-selection protocols. Where the family falls short on Axis 1: joint authority lives in contracts and committees rather than in aggregation substrate — gradient aggregation is the substrate, with governance attached around it through consortium-governance contracts rather than as substrate-content commitment within the aggregation itself. Where the family falls short on Axis 3: population-scale dynamics framed as emergent from FL algorithms and data heterogeneity rather than as composition of architectural commitments — the convergence, divergence, specialization, and

capability-propagation phenomena FL consortia exhibit are framed at the algorithm-and-data-heterogeneity register rather than at the architectural-pattern register Claim 6 commits to.

Adjacent families fall further short. **Agent-protocol ecosystems** (MCP, A2A, NIST AI Agent Standards Initiative, Agentic AI Foundation, OpenID Foundation AI Identity Management Community Group) supply strong Axis 3 protocol-and-meta-governance work but fall short on Axes 1 and 2: protocols are substrates for context and messaging not governance objects, and joint authority lives in institutional governance rather than in protocol substrate. **Complex adaptive systems, swarm intelligence, agent-based modeling, and evolutionary game theory work** fall short on Axis 1 (modeled population not authored governance substrate) and Axis 3 (framing explicitly emergence-requiring rather than composition-rendering). Across all four families and adjacent territories the failure mode is consistent: substrate-content commitment exists inside single organizations; joint authority across population-level governance exists in legal/institutional space; population-scale dynamics framed as composition exists at intra-organizational scope; population-scale dynamics framed as emergence exists at simulation/analysis scope. The conjunction at cross-organizational population scope at architectural-pattern register is unoccupied.

9.4 Three differentiation axes with asymmetric inheritance shape

Three architectural axes carry differentiation simultaneously, each holding against contemporary capability-motivated and emergence-frame architectures at the trilogy-completion angle. The honest weight is **asymmetric** — Axis 1 carries primary novelty via redirected-§4.5-objection-8 engagement; Axis 2 is mostly Claim 1 inheritance at population scope; Axis 3 anchors architectural honesty for the trilogy-completion register through the parallel-extension-claim structure.

The first axis is **substrate-content commitment at population scope versus capability-driven population-scale coordination without architectural-commitment register for governance**. Contemporary capability-motivated population-scale architectures commit to functional coordination through model-capability, infrastructure, and protocol scaling, with governance applied at deployment-shape or contract/institutional register. Foundation lab multi-agent systems at scale derive quantitative scaling principles across many configurations with governance treated as “out of scope” for the scaling-science enterprise (Kim et al. 2025); agent-protocol ecosystems hold this pattern at architectural-commitment level, with MCP intentionally agnostic about what is allowed and A2A leaving policy to implementers. Single-organization substrate-content-commitment architectures supply the pattern at varied registers per §9.3: Agent Contracts at resource-budget register; Cognitive Core and Governance-First Architecture work at decision-substrate register; enterprise MAS governance blueprints at single-org population-of-agents register. Each illustrates one register where substrate-content commitment lands; the cross-organizational population-substrate analog at architectural-commitment register is unoccupied. The inheritance shape is heavy §4.4 Axis 2 (configuration-as-substrate-content for the shared substrate) plus §8’s recursive applicability applied at population scope, with the population-scope extension fresh through the redirected-objection-8 engagement at §9.5 obj 1. Claim 6 is **architecturally different from capability-motivated coordination on the substrate-content commitment axis at population scope, not capability-motivated coordination plus substrate-content commitment**.

The second axis is **joint authority across population-level governance perimeters versus single-organizational or capability-organizational authority at scale**. Contemporary population-scale architectures typically operate under single-organizational or capability-organizational authority. Cross-organizational coordination at population scale exists but typically encodes joint authority at contract or institutional layer. Decentralized cross-organizational frameworks supply the closest neighbor for cross-organizational framing — fully decentralized DAG-structured frameworks with explicit cross-organizational privacy-preserving collaboration commitment exist (AgentNet, Yang et al. NeurIPS 2025) — but joint authority is enabled by privacy-preserving design rather than encoded as multi-party control over shared substrate; governance sits at framework/API register. FL consortia per §9.3 Family 4 encode joint authority through consortium agreements and coordinator-mediated aggregation. Claim 6’s joint authority across population-level governance perimeters extends §4’s joint-authority commitment at population scope, with the recursion bottoming at population-level human governance per Paper 1 through §8’s recursive applicability extended at population scope. The inheritance shape is mostly Claim 1 inheritance applied

at population scope. Claim 6 is **architecturally different from contract-layer joint authority on the architectural-pattern-register-encoding axis, not contract-layer joint authority plus architectural encoding.**

The third axis is **trilogy-completion composition versus emergence-as-fresh-property framing.** Contemporary emergence theory in its complex-adaptive-systems, multi-agent-emergence, swarm-intelligence, and agent-based-modeling registers treats population-scale dynamics as emergent property requiring fresh theoretical framing. Indian PMEAC’s CAS framework, Molt Dynamics, Park et al.’s (2024) thousand-agent generative simulations, and collective-intelligence work all commit to emergence-theoretical apparatus as load-bearing frame for explaining what populations of interacting agents produce. Claim 6 commits to population-scale dynamics as composition of Paper 3’s architectural commitments — articulation rather than fresh emergence theory. Specifically, the architecture rejects fresh-emergence-theory-as-load-bearing-frame: the requirement for ontological commitments such as Holland-style complex-adaptive-systems apparatus, weak-versus-strong-emergence taxonomy, irreducibility commitments, downward-causation requirements, or autopoiesis-style frames as the apparatus through which population-scale dynamics are explained. What the architectural commitment carries instead is composition: population-scale dynamics are explained by the composition of substrate-content commitment, joint authority across population-level governance perimeters, and recursive applicability of configuration-as-substrate-content extended at population scope, with substantive emergent characteristics following from that composition rather than requiring fresh theoretical apparatus added at population scope. The criterion of distinction is whether the explanation runs through architectural commitments composing forward or through fresh theoretical apparatus introduced at population scope. The parallel-extension-claim structure articulated in §9.1 is the structural anchor for “composition rather than fresh emergence theory architecturally honest”: Paper 1’s Claim 6 plus Paper 2’s Claim 6 plus Paper 3’s Claim 6 are parallel extension claims at progressively larger scope, each operating within the same substrate-mediator commitment. Substantive emergent characteristics are part of the architectural commitment per the calibration discipline rendered at §9.1; what the discipline rejects is fresh-emergence-theory-as-load-bearing-frame for explaining what the inheritance produces. The inheritance shape is composition of Paper 1’s substrate-mediator commitment, Paper 2’s evolution machinery (§§6–8), and §§4–8 of Paper 3 at population register. Claim 6 is **architecturally different from emergence-theory-as-load-bearing-frame architectures on the composition-register axis, not emergence-theory plus composition.**

The conjunction carries the architectural contribution. Axis 1 alone is single-organization substrate-content territory across varied registers (per §9.3 Families 1, 2, 3). Axis 2 alone with contract-layer joint authority is the FL consortium pattern (§9.3 Family 4), or with framework-API decentralization, the AgentNet pattern. Axis 3 alone at composition-register operates at intra-organizational scope through orchestration and meta-governance blueprints (§9.3 Family 1). What is unoccupied is the conjunction with all three holding simultaneously at cross-organizational population scope.

9.5 The fourth inheritance-active finding and reviewer-objection responses

The conjunction in §9.4 is what Claim 6 contributes; the cross-cutting research finding confirms it is unrealized as a named, analyzed pattern at architectural-pattern register at cross-organizational population scope across the 2024–2026 landscape. The four families plus adjacent territories examined at §9.3 each hold two of the three axes and fall short on the third for specific structural reasons, with the failure-mode pattern across all four families consistent: substrate-content commitment exists inside single organizations; joint authority across population-level governance exists in legal/institutional space; population-scale dynamics framed as composition exists at intra-organizational scope; population-scale dynamics framed as emergence exists at simulation/analysis scope. The conjunction at cross-organizational population scope at architectural-pattern register is unoccupied.

This is the **fourth inheritance-active finding compounding across Claims 3, 4, 5, 6.** §6 surfaced the three-clause conjunction at the conflict-handling level (preserve as first-class persistent state by default; resolution rules as inspectable substrate content under explicit human authority; escalation operating across joint authority of multiple distinct governance perimeters as architectural commitment). §7 surfaced the three-axis conjunction at the evolution-feed level (asymmetry sourced from per-perimeter governance;

FAI-derived content reaching existing per-Self evolution mechanisms unchanged; at the dissolution of an event-bounded shared substrate). §8 surfaced the three-axis conjunction at the configurable-dimensions and recursive-governance level (configuration as substrate content under all six Paper 1 commitments; recursive applicability through substrate-content commitment; joint authority across participating Selves’ governance). Claim 6 surfaces the three-axis conjunction at population scope (substrate-content commitment at population scope; joint authority across population-level governance perimeters; trilogy-completion composition rather than emergence-as-fresh-property framing). The four-finding compounding pattern is genuine — each finding represents the consistent unrealized-conjunction at architectural-pattern register at the claim’s scope. **The pattern compounds across claims as discipline carrying through to trilogy completion;** §11 Discussion develops the four-finding compounding pattern as a trilogy-level observation.

Three objections warrant substantive treatment; five others briefer.

The first objection — *foundation labs solve this anyway; capability scaling delivers functional inter-Self coordination at scale, so why does architectural commitment matter?* — is the redirected fuller engagement of §4.5’s brief response on capability scaling at population scope. Both halves require substantive treatment. The first half is honestly granted: capability scaling does deliver functional coordination at population scale. Kim et al. (2025) derives quantitative scaling principles across 180-plus configurations with a predictive model selecting optimal coordination architecture for 87% of unseen tasks; MCP under cross-vendor stewardship standardizes contextual exchange across foundation lab platforms; A2A with 50-plus enterprise partners supplies standardized peer-to-peer communication; OpenAI’s Agents SDK and AgentKit supply model-capability plus platform-capability plus developer-capability scaling. The second half is the architectural-commitment differentiation: governance in capability-motivated architectures sits at framework/API, deployment-shape, and contract/institutional registers rather than at substrate-content commitment register — Kim et al. treats governance as engineering performance; MCP is intentionally agnostic about what is allowed; A2A leaves policy to implementers; NIST AI Agent Standards Initiative shapes how substrates must behave rather than authoring substrates; Agentic AI Foundation supplies institutional governance of open technical projects but does not dictate deployment substrate. The differentiation is at architectural-commitment register: governance-motivated population-scale architecture commits to substrate-content commitment at population scope, joint authority across population-level governance, and recursive applicability — commitments capability-motivated architectures do not make at architectural-pattern level even when they scale functional coordination, and the calibrated-humility voice anchor holds the differentiation open at the architectural-commitment register rather than at the functional-coordination claim itself.

The second objection — *this is just emergence theory or complex adaptive systems under new vocabulary* — runs primarily on Axis 3. Specific architectures committed to CAS or emergence-theoretical apparatus as load-bearing frame include CAS frameworks at policy-research register, agent-based-modeling and simulation work, persona-based simulations, and collective-intelligence frameworks. These treat population-scale dynamics as emergent property requiring fresh theoretical framing. Per the emergence-discipline sharpening rendered at §9.1 and §9.4 Axis 3: the discipline is not “no emergent dynamics” but emergent dynamics yes, fresh emergence theory no. The architecture produces population-scale dynamics with substantive emergent characteristics per Paper 3 core theory — coordinated handling of problems beyond what any single Self accumulated independently, capability propagation across organizational and domain boundaries, collective adaptation to novel conditions; what the discipline rejects is fresh-emergence-theory-as-load-bearing-frame for explaining what the inheritance produces. The substrate-mediator ladder articulated in §9.1 — Paper 1’s Claim 6 plus Paper 2’s Claim 6 plus Paper 3’s Claim 6 as parallel-extension-claim structure across the trilogy — is what makes composition rather than fresh emergence theory architecturally honest.

The third objection — *this is just large-scale federated learning or multi-agent systems under new vocabulary* — runs on the FL/MAS-restatement angle and triggers the sixth-FL-boundary clarification per §9.6. FLARE/FATE/OpenFL consortia at hundreds-of-organizations scale and foundation lab multi-agent systems at scale supply functional population-scale coordination — large-scale FL exhibits convergence, divergence, specialization-preservation, and capability-propagation; foundation lab MAS exhibits population-scale coordination at scale. The differentiation is at architectural-commitment register: contemporary large-scale FL operates at framework-API plus coordinator-mediated-aggregation plus consortium-governance-contract registers; contemporary MAS at scale operates at framework-API plus deployment-shape plus protocol-spec

registers; Claim 6 commits at substrate-content architectural-commitment register at population scope under joint authority across population-level governance perimeters with recursive applicability extended at population scope. The differentiation tracks the architectural-pattern register Claim 6 occupies rather than functional-coordination capabilities the contemporary architectures deliver.

Five briefer responses follow. *Why three commitments — where do they come from?* The three commitments correspond to Paper 3’s full inheritance composing at population scope: substrate-content commitment (§4 inheritance applied at population scope); evolution-dynamics composition (Paper 2 §§6–8 through §7’s evolution-feed mechanism); capability-motivated differentiation (redirected §4.5 brief response on capability scaling). The number follows inheritance structure rather than fresh architectural design choice. *Does Claim 6 require population-scale empirical data to defend?* Claim 6 commits at architectural-pattern register; population-scale CKS-governed Self deployments do not yet exist; the commitment is to what the inheritance produces, with specific dynamics awaiting empirical observation per the calibrated-humility framing for the extension-claim register. *How does population-scale governance bottom out — doesn’t this risk infinite regress?* Population-scale governance bottoms at population-level human governance per Paper 1 §3.3 through §8’s recursive applicability extended at population scope; recursive depth is itself a configurable dimension under joint authority. *How does cross-population governance handle cross-cultural or cross-jurisdictional differences?* Cross-population governance perimeters are themselves substrate content per §8’s recursive applicability extended at population scope; differences are encoded as substrate content under joint authority across population-level governance, configurable per perimeter. *Why is Claim 6 in this paper rather than a separate vision paper?* Claim 6 is the canonical extension claim per Paper 3 core theory — the trilogy-completion sixth rung of the substrate-mediator ladder. Separating it would leave Paper 3 with five claims deferring their composition to elsewhere; the trilogy is structurally complete only when the population-scale composition is articulated within the same architectural register as the preceding claims. Calibrated humility is the appropriate voice anchor for the extension-claim register, not a reason for separation.

9.6 Inheritance accounting and trilogy closing

Claim 6 inherits without redefense from seven sources — the heaviest inheritance density across the six claims of Paper 3. The inheritance shape is heavy at Paper 2 §§6–8 jointly (the most direct architectural antecedent for population-scale evolution dynamics) and at §4 plus §8 (extending substrate-content commitment, joint authority, and recursive applicability at population scope); medium at Paper 1 (foundational commitments hold within scope) and §§5–7 of Paper 3 (cumulative FAI, conflict-handling, and evolution-feed inheritance composing at population register).

From Paper 1, all six commitments hold within population scope per §9.1. §3.2’s substrate-as-content commitment is foundational — structure originates under human authorship at the cell scope Paper 1 defends, and the same commitment carries forward into population-scale collective-evolution substrate. §3.3’s authority-vs-labor distinction supplies the actor-neutrality discipline at Claim 6’s scope; humans hold authority over population-scale dynamics, and the labor is available to humans directly or to LLMs operating under human direction. The portable phrase Paper 1 §3.3 renders — governance is an authority architecture, not a review workflow — holds across population-scale dynamics as it holds across configuration substrate at FAI scope. Paper 1 §5’s conflict preservation as first-class state carries through to population-scale conflict structure as preserved conflicts at the FAI level carry through §7’s evolution-feed mechanism into participating Selves’ home substrates as boundary annotations. Paper 1 §6’s linear-cost commitment underwrites the cumulative-coordination-as-composition framing — population-scope coordination is composition of FAI events at the architectural-pattern register, not a fresh joint-coordination architecture. Paper 1’s governance-motivated-vs-capability-motivated framing underwrites the redirected-§4.5-objection-8 engagement at §9.5 obj 1; Paper 1 §5.2’s OIDA differentiation supplies the secondary differentiation-passage tone anchor for the architecturally-different-from-X-on-Y-axis-not-X-plus-Y discipline §9.4 holds across all three axes.

From Paper 2, §6 entire is the architectural antecedent for lifecycle primitives at population scope. §6.3’s mating primitive with multi-cardinality applies within a Self per Paper 2; Claim 6 extends at population scope through the inter-Self perimeter — population-scale mating across many CKS-governed Selves’ lifecycles is architectural extension of §6.3 at population scope. §6.4’s two architectural results for death (lineage

supersession and functional obsolescence) extend at population scope as population-scale death-types operating across many Selves’ lifecycles. §7 entire is the structural-extension anchor for evolution dynamics at population scope — three evolution mechanisms in productive tension across multiple composition levels extend at population scope through §7’s evolution-feed mechanism, with the productive-tension framing between mutation and directed selection appropriate at population scope per Paper 2 §7.2’s productive-tension subsection where the biology vocabulary applies. §8 entire is the structural-extension anchor for governance shapes at population scope — per-mechanism governance shapes co-determined with mechanism shape per Paper 2 §8.2 hold at home perimeter and compose at population-level governance perimeters. Paper 2 §6.5, §7.5, and §8.5 supply the primary differentiation-passage tone anchors. Paper 2’s biology engagement is bounded to its intra-Self scope per Paper 2 §2.2’s biology-as-bounded-load-bearing-terminology framing and Paper 3 core theory’s biology-positioning subsection; the population-scale-mating-extension framing renders as architectural extension of Paper 2 §6.3 across many CKS-governed Selves’ lifecycles, not as biology vocabulary infiltration at population register.

From §4 of Paper 3, the shared substrate as architectural object is the structural antecedent for §9.1’s population-as-substrate-content construction; configuration-as-substrate-content and joint authority commitments compose forward at population scope (§4.4 Axis 2 and Axis 3); §4.5’s brief response on capability scaling redirects the fuller engagement to §9.5 obj 1. From §5, FAI as canonical operation accumulates across many events at population scope. From §6, three-tier conflict handling outcomes accumulating across many FAI events feed collective evolution. From §7, the four-locus evolution-feed mechanism is the per-event hand-off Claim 6 develops at the population-scale-dynamics register; §7.5’s brief response 5 (“Can FAI-feed produce fresh capabilities at home?”) explicitly redirected population-scope emergence to Claim 6’s territory. From §8, configuration-as-substrate-content with recursive applicability is the most direct Paper 3 inheritance; configuration of population-scale dynamics is itself substrate content under joint authority, with recursion bottoming at population-level human governance, and the seventeen cumulative configurable dimensions across §§5–7 carry forward at population scope. The seven-source inheritance accounting is what makes the trilogy-completion posture honest at all three poles of the calibration discipline: the inheritance density makes population-scale collective evolution composition rather than fresh emergence theory; the inheritance positions the architectural commitment at the differentiation register against capability-motivated architectures; the inheritance positions empirical-verification status as appropriate humility at the extension-claim register rather than as a hedge on the architectural commitment itself.

The federated-learning territory engaged at Claim 6’s population-scale-dynamics angle continues the cross-claim pattern. §4 engaged FL at the joint-authority-mechanics angle, §5 at the merge-mechanism angle, §6 at the conflict-handling angle, §7 at the evolution-feed angle, §8 at the configurable-aggregation-algorithm angle; §9 engages at population-scale-dynamics. Large-scale FL consortia exhibit population-scale dynamics — convergence, divergence and specialization through personalized FL, capability propagation as new members inherit current global model state — across accumulated rounds; Federated Behavioural Planes (Fenoglio et al. NeurIPS 2024) supplies the architectural commitment to representing client trajectory diversity at population scope as first-class objects through error and counterfactual behavioural spaces. The contrast is at architectural-commitment register at population scope: contemporary large-scale FL commits to coordinator-mediated aggregation under consortium-governance contracts; Claim 6 commits to substrate-content architectural commitment at population scope under joint authority across population-level governance perimeters. **The sixth FL boundary closes the cross-claim FL-engagement pattern at trilogy completion;** §11 Discussion develops the six-FL-boundary pattern as a trilogy-level observation.

Claim 6 specifies population-scale collective evolution as what Paper 3’s five preceding claims compose into when their architectural commitments operate at population register through accumulated FAI events under joint authority across population-level governance perimeters and §8’s recursive applicability extended at population scope. Claim 6 closes Paper 3. The substrate-mediator commitment progresses across the trilogy at six configurations of progressively larger scope, with Claim 6 occupying the sixth rung as the population-scale extension parallel to Paper 1’s Claim 6 and Paper 2’s Claim 6 — the parallel-extension-claim structure across the trilogy is what makes composition rather than fresh emergence theory architecturally honest at trilogy completion. Papers 1 and 2 plus Paper 3’s six claims jointly specify substrate-mediated coordination from intra-cell scope through intra-Self scope through inter-Self scope through population scope, all

under unified human governance — the trilogy’s architectural object extending to progressively larger scopes through the series, each scope inheriting the prior commitments without replacing them. As §10 develops, the thought experiment renders Claim 6’s territory through one concrete domain-agnostic instance: opening FAI event with conflict surfacing and preserved-conflict carry-through to home substrate, then population scope as accumulated FAI events across the network, with population-level capability emerging through governed substrate composition rather than monolithic scaling. As §11 develops, the trilogy-level patterns surfaced across Claims 1–6 — six-organizational-pattern, six-FL-boundary, four-finding compounding, foil-parallel pattern, varied-calibration-landscape, substrate-mediator-ladder — render as discipline visible across the trilogy. As §12 develops, the paper-level closing positions the trilogy at the substrate-mediator ladder’s sixth rung with the calibrated-humility voice anchor preserved across the closing prose. Future work — formal protocol specifications, security guarantees, message schemas, deployment architectures, empirical validation at scale — is downstream of the architectural pattern the trilogy specifies. Population-scale CKS-governed Self deployments do not yet exist; the architectural commitment is to what the inheritance produces at population scope, with specific dynamics awaiting empirical observation, and the calibrated-humility voice anchor holds for the trilogy’s closing as it has held for Claim 6’s opening.

10. Thought Experiment

This section renders Claim 6’s territory through one concrete instance — an opening FAI event between two CKS-governed Selves, then population scope across many such events accumulating across the network, then alignment with the prior claims, and what the thought experiment is not. The rendering is domain-agnostic by construction. Paper 3’s architectural claim is about CKS-governed Selves coordinating with each other regardless of what domain any Self serves; anchoring the thought experiment to a specific industry would shrink the claim to that industry. The two Selves the rendering names — Self A and Self B — are described by the architectural shape of their respective coordination situations rather than by their domain content. The contrast against Paper 2’s healthcare-anchored thought experiment is not coincidental. Paper 2’s enterprise-brain claim was about within-organization architecture and supported domain anchoring without scope-shrinkage, since one organization in one domain is one instance of the within-organization architecture; Paper 3’s inter-Self claim is about cross-organizational coordination across distinct authority structures and resists domain anchoring without artificially constraining scope, since anchoring to one cross-domain pair (e.g., healthcare ↔ pharma) would render the thought experiment as illustrative of one cross-organizational instance rather than of the architectural pattern. The domain-agnostic posture is therefore architecturally derived from Paper 3’s scope rather than chosen for stylistic divergence from Paper 2. In the manner of Paper 2 (Li, 2026b) §10, the rendering is offered as architectural vision — concrete rendering of the architecture at the scope Paper 3 claims, calibrated through the vision-claim register Paper 2 §9 establishes. Population-scale CKS-governed Self deployments do not yet exist; the architectural commitment is to what the inheritance produces at the scopes the rendering describes, with specific dynamics awaiting empirical observation.

10.1 The opening FAI event

Two CKS-governed Selves, each constituted per Paper 2’s three-level structure with cells composing into aspects under unified human governance within a Self, with DNA and action layers within every cell, and with instinct and reasoning composing under the instinct/reasoning separation, exist within distinct organizations under distinct authority structures. Self A has developed mature handling for a particular class of coordination problem — its work over time has produced an aspect within A’s home substrate covering specialized protocols, edge-case logic, and exception pathways accumulated through action-feedback evolution under A’s governance. The aspect carries the structural shape Paper 2 §5 specifies: cells composing into the aspect through relational-role membership, DNA-layer content carrying stabilized orchestration substrates and policy substrate, action-layer content carrying recorded execution. Self B faces a related problem without comparable institutional history. B has cells covering its operating context’s general infrastructure, but lacks the specialized accumulation A’s aspect represents.

Under governance configurations defined within each organization’s authority structure, an FAI event is

initiated. Per §8’s commitment, the configuration is itself substrate content under joint authority: it specifies which aspects each Self contributes, the depth of provenance carry-over at the perimeter, and the persistence policy after the event dissolves. The configuration substrate is authored within whichever participating Self holds origination authority for the event — typically the initiating Self under its home governance, with the other participating Self’s governance approving the configuration before substrate construction begins. The approval mechanics are themselves governance-configurable. Both organizations’ humans hold authority over the configuration; the labor of authoring it is available to humans directly or to LLMs operating under human direction per Paper 1’s (Li, 2026a) authority-vs-labor distinction extended at the inter-Self perimeter.

The shared substrate is constructed for the event per §4. Within its scope, the six Paper 1 commitments hold by inheritance: the substrate carries coordination, conflict-preservation, and governance content; humans hold authority over its content and the orchestration rules within it; LLMs operate as substrate mediators per Paper 2’s instinct/reasoning separation; the substrate is tool-agnostic; composition stays linear-cost. The distinctive perimeter-spanning property is added: the shared substrate’s governance perimeter spans both A’s and B’s home governance perimeters, with provenance carrying organizational origins explicitly. A’s specialized aspect and B’s relevant infrastructure aspect occupy the shared substrate together. Per §5, the unit of contribution is the aspect; the contributions surface their constituent cells, DNA-layer content, and action-layer content. LLM weights and instinct-layer content do not exchange — the inter-Self extension of Paper 2’s instinct/reasoning separation carries through the exchange-bounding commitment without modification at this scope.

Within the shared substrate, a conflict surfaces. A’s specialized protocols embed assumptions about its operating environment that diverge from B’s operating context — assumptions about what counts as edge-case, about which exception pathways apply, about how certain ambiguities resolve. The conflict is not noise to be filtered for the merge to complete. Per §6’s three-tier mechanism, it is a first-class addressable substrate object, and the task does not require unification of those assumptions. The preserve tier applies: the conflict is preserved in the shared substrate, with both A’s protocol assumptions and B’s operating-context assumptions retained distinct, the divergence point identified, and provenance attached. Preservation is the architectural default Paper 1 §5 commits to within a cell, extended through §4 across the inter-Self perimeter and operationalized through §6 at the inter-Self conflict-handling level.

The event proceeds. A’s contribution and B’s contribution have done architectural work in the shared substrate even with the conflict preserved; what each Self can do with the substrate’s content is governed within each home authority structure. As the event dissolves, per §7’s four-locus mechanism, content propagates to each participating Self’s home substrate per governance-configured ingestion at home. The layer-routing rule applies per Paper 3 core theory’s instinct-evolution-out-of-scope commitment: DNA-layer content reaches DNA evolution operating on home DNA per Paper 2 §7’s mechanism; action-layer content reaches action-feedback evolution operating within home action layers; instinct evolution takes no FAI input. Asymmetric ingestion holds: A and B may take different things from the same FAI event under their respective home governance. B’s evolution mechanisms internalize content from A’s contribution as it flows through B’s authorized governance — schemas, rules, and orchestration patterns absorbed selectively per B’s home authority via DNA evolution, recorded execution from the event reaching B’s action-feedback evolution machinery via the proposal-and-acceptance machinery Paper 2 §8 specifies.

The preserved conflict carries through specifically as evolution-feed annotation per §7. When B’s evolution mechanisms internalize the FAI content into B’s home substrate, the preserved conflict converts into an annotation within B’s home substrate — flagging the boundary B’s coordination work must address per B’s home governance. Whether B treats the boundary as a place to do further work, as a place to leave the divergence in place, or as a place to escalate is determined within B’s home authority structure as the home-substrate work proceeds. The same applies symmetrically at A’s perimeter to whatever A absorbs.

The event dissolves. The shared substrate exits operation. What persists is configurable per §8: the FAI record itself remains as substrate content auditable per the persistence policy the configuration specified — at one extreme nothing retained beyond what each Self ingested into home substrate, at the other extreme the full shared substrate retained as durable inter-organizational record. The configuration substrate is itself substrate content remaining inspectable per §8’s recursive applicability.

10.2 Population scope

Many such FAI events accumulate across the network of CKS-governed Selves over time. Selves operating in distinct domains, owned by different organizations, governed under distinct authority structures, engage in events under configurations their respective governance structures author. Common patterns propagate where governance configurations enable propagation — schemas, orchestration substrates, and protocol logics flowing as DNA-layer content from Selves with mature accumulation to Selves whose home governance authorizes absorption. Specializations preserve where governance configurations protect them; an organization whose Self carries content the organization treats as strategic does not contribute that content to the shared substrate, or contributes under provenance-carry-over policies that bound how downstream Selves can use it. Asymmetric ingestion at every event compounds across the population: what each Self ingests from any given event is governance-configured within home authority, and what propagates across the network is the cumulative result of many home-governance decisions rather than centrally-determined population-level update.

The population-level capability that emerges across many Selves over many events exhibits characteristics anchored specifically to the architectural commitments §§4–8 establish: coordinated handling of coordination-substrate problems beyond what any single Self accumulated independently through its home perimeter; specialization-preserving propagation of substrate-content patterns — schemas, orchestration substrates, protocol logics — across organizational and domain boundaries; collective adaptation through cumulative configuration-substrate authoring under joint governance across population-level governance perimeters. The architectural commitment is that these emerge through governed substrate composition rather than through monolithic scaling. Authority over what propagates, what diverges, what persists, and what dissolves remains substrate content at every event, configurable within each Self’s governance perimeter, with cross-population governance perimeters themselves substrate content under joint authority per §8’s recursive applicability extended at population scope. Population-scale CKS-governed Self deployments do not yet exist; the architectural commitment is to what the inheritance produces at population scope through accumulated FAI events, with specific population-scale dynamics — the precise trajectories of pattern propagation, the actual rates and shapes of specialization-preserving divergence, the empirical character of collective adaptation — awaiting empirical observation.

10.3 Alignment with prior claims

The thought experiment exercises all five Paper 3 core claims in a single coherent narrative. §4’s shared substrate is the medium within which the opening FAI event is conducted, with all six Paper 1 commitments holding within scope and the perimeter-spanning property carrying provenance across A’s and B’s home governance perimeters. §5’s FAI is the canonical operation: the unit of contribution is the aspect, the architectural default of full merge applies, the cardinality-two case is the simplest n-ary instance, and the exchange is bounded to substrate content per Paper 2’s instinct/reasoning separation. §6’s three-tier conflict handling operates at the conflict that surfaces — the preserve tier applies because the task does not require unification, and the inheritance from Paper 1 §5 carries through to the inter-Self perimeter directly. §7’s four-locus evolution-feed mechanism operates as the event dissolves: hand-off at FAI dissolution, per-mechanism feed structure into Paper 2’s three evolution mechanisms operating unchanged at home, the layer-routing rule with the three-case explicitness Paper 3 core theory’s instinct-evolution-out-of-scope commitment requires, and asymmetric ingestion as A and B take different things from the same event. §8’s configuration as substrate content with recursive applicability operates throughout: the FAI configuration is substrate content under joint authority before the event begins, the configuration of how the configuration is authored is itself substrate content, and the recursion bottoms at human-authored governance authority per Paper 1.

The narrative preserves all six Paper 1 commitments within the shared substrate. The hybrid-with-governance-boundary commitment holds: the substrate handles coordination and governance, LLMs operate as substrate mediators, and the division between substrate and LLM is drawn at the governance boundary rather than at the capability boundary. The conflict-preserving commitment holds: the divergence between A’s protocol assumptions and B’s operating-context assumptions is preserved as a first-class addressable object. The human-governed-not-human-generated commitment holds: humans across both organizations hold authority, while the labor of contributing aspects, of authoring configuration, and of operating during

the event is available to humans directly or to LLMs operating under human direction. AI-as-substrate-mediator holds: LLMs read, write, and compose substrate content within the shared substrate, executing operations in accordance with the orchestration rules the configuration substrate carries. Tool-agnosticism holds: any environment supporting persistent structured state, human read/write access, and LLM access to substrate content can instantiate the construction. Linear-cost composition holds: a third Self joining the event under its own governance, or a different pair of Selves engaging in their own event elsewhere in the network, expands cumulative coordination linearly.

The narrative preserves Paper 2’s instinct/reasoning separation: only DNA-layer and action-layer content exchange across FAI; LLM weights and instinct-layer content do not exchange across the inter-Self perimeter. Each Self’s three-level structure stays intact within home perimeter throughout — cells, aspects, and the Self itself remain Paper 2’s primitives within A and within B before, during, and after the event. Paper 2’s lifecycle and evolution operations operate unchanged at home: Paper 2 §6’s birth, mating, and death-types operate within each Self’s home perimeter; Paper 2 §7’s three evolution mechanisms in productive tension operate within each home perimeter, with FAI-feed as additional input source rather than as replacement or modification of those mechanisms.

The thought experiment surfaces possibilities consistent with Paper 3 core theory’s possibilities-not-prescriptions principle. The narrative does not prescribe convergence over diversity — preservation is the architectural default at the conflict surfaced, and B’s home governance determines whether the boundary becomes work to do or stays as encoded divergence. The narrative does not prescribe specific FAI configurations — the configurations the event proceeds under are the kinds of configurations A’s and B’s governance structures might author, not configurations Paper 3 commits to as preferred. The narrative does not claim AGI as such — population-level capability is rendered through the calibrated-humility framing of characteristics often associated with broadly-capable AI systems, articulating what governed substrate composition produces at population scope without committing to AGI as an architectural claim. The surface is possibilities at every locus where the architecture would otherwise tempt prescription.

10.4 What the thought experiment is not

The rendering offered in this section is not a deployment design. It does not specify how shared substrates are physically implemented, how cross-organizational governance is administered operationally, how authority structures are coordinated across organizations at organizational pace, or how the FAI configurations are administered across firms. These are deployment-architecture and protocol-specification questions the design pattern leaves to deployments rather than thought-experiment work the rendering would over-specify by pursuing. The rendering is not an industry case study; the domain-agnostic posture is constitutive — Paper 3’s architectural claim is about CKS-governed Selves coordinating with each other regardless of what domain any Self serves, and the rendering preserves that posture rather than committing to a specific domain context that would shrink the claim. The rendering is not a proposed product or protocol specification; future work — formal protocol specifications, security guarantees, message schemas, deployment architectures, and empirical validation at scale — is downstream of the architectural pattern Paper 3 specifies. The rendering is not a prediction about what AI systems will do; the population-scope passage at §10.2 names characteristics governed substrate composition can produce at scale, with calibrated humility that population-scale CKS-governed Self deployments do not yet exist and specific dynamics await empirical observation.

What the rendering offers is concrete instantiation of the architecture at the scope Paper 3 claims. The opening FAI event renders Claims 1, 2, 3, 4, and 5 in a single coherent narrative, exercising the architectural commitments §§4–8 develop separately. The population-scope passage renders Claim 6’s territory through accumulated events, with the calibrated-humility voice anchor that activates from §9.1 throughout §9 carrying through §10 unchanged. As §11 develops, the trilogy-level patterns surfaced across Claims 1–6 — six-organizational-pattern, six-FL-boundary, four-finding compounding, foil-parallel pattern, varied-calibration-landscape, substrate-mediator-ladder — render as discipline visible across the trilogy. As §12 develops, the paper-level closing positions the trilogy at the substrate-mediator ladder’s sixth rung with the calibrated-humility voice anchor preserved across the closing prose. The thought experiment is calibrated through the vision-claim register Paper 2 §9 establishes — architecturally coherent as a design pattern, pending debate and observation in real-world deployment.

11. Discussion and Implications

This section steps back from the per-claim architectural defense §§4–8 develop and the trilogy-completion claim §9 develops and the concrete-instance rendering §10 offers, and reads what the six claims jointly produce as pattern visible across Claims 1–6. Six trilogy-level patterns are visible across the trilogy that compose into discipline renderable as deliberate at paper level — the substrate-mediator ladder across six rungs at paper-level scopes; the six-organizational-pattern coherence across closest-neighbor organizations; the six-FL-boundary pattern across cross-claim federated-learning engagement; the four-inheritance-active-findings compounding pattern across Claims 3, 4, 5, 6; the six-foil-parallel pattern across §§4.2–§9.2; and the varied calibration landscape across the six claims. These patterns are observation across Claims 1–6 rather than fresh architectural commitments requiring fresh defense. They emerge from the cross-claim consistency the architectural commitments produce when composed; reading them at trilogy-level register completes the architectural picture the per-claim sections develop in their own registers. Three load-bearing reviewer-objection types recur across the six claims with claim-specific shape, and reading them at paper-level register supplies the cross-claim positioning the per-claim reviewer-objection responses individually defer or reference. The closing positioning then composes the substrate-mediator-ladder-completion framing with the trilogy-as-design-pattern-paper-across-three-scopes framing and gestures generically toward future work, with the calibrated-humility voice anchor activated at §9.1 and threaded through §9 and §10 carrying through §11 unchanged.

11.1 The substrate-mediator ladder — six rungs at paper-level scopes

The most load-bearing trilogy-level pattern is the substrate-mediator ladder per Paper 3’s core theory articulation. The substrate-mediator commitment progresses across the CKS series at six configurations of progressively larger scope. Rung 1 is single human $\leftrightarrow$ LLM within one cell, the configuration Paper 1 (Li, 2026a) defends as the architectural object of its core theory. Rung 2 is multiple humans $\leftrightarrow$ LLM within one cell, the configuration Paper 1’s Claim 6 develops as its extension. Rung 3 is cell $\leftrightarrow$ aspect $\leftrightarrow$ Self within one Self, the configuration Paper 2 (Li, 2026b) defends as the architectural object of its core theory across its six core-theory claims. Rung 4 is the Self as enterprise-scope unified architecture, the configuration Paper 2’s Claim 6 develops as its extension claim. Rung 5 is Self $\leftrightarrow$ Self mediated by shared substrate, the configuration Paper 3’s §§4–8 jointly defend as Paper 3’s core theory. Rung 6 is population-scale collective dynamics across many Selves, the configuration Paper 3’s Claim 6 (§9) develops as Paper 3’s extension claim. Each paper carries a core-theory rung and an extension-claim rung — Paper 1 contributes rungs 1 and 2; Paper 2 contributes rungs 3 and 4; Paper 3 contributes rungs 5 and 6.

The structural shape is a parallel-extension-claim pattern across the trilogy: Paper 1’s Claim 6, Paper 2’s Claim 6, and Paper 3’s Claim 6 each occupy the extension-claim position at their paper’s scope, each operating within the same substrate-mediator commitment, each acknowledging calibrated humility at its own register. This parallel structure is what makes composition rather than fresh emergence theory architecturally honest at trilogy completion. The architectural object — human-governed substrate as coordination medium — extends to a larger coordination scope at each rung without changing its fundamental commitments, and each scope inherits the prior rung’s commitments without replacing them. The same-architectural-object claim warrants substantive articulation at the rung-1 versus rung-6 contrast specifically, since authority structure (single user versus joint authority across population-level governance perimeters), temporality (within-cell session versus cumulative coordination structure across many events), and substrate cardinality (one substrate versus many composing) differ across the scope range. The hybrid-with-governance-boundary commitment from Paper 1 §2 is the load-bearing identity: at rung 1, the substrate handles coordination and governance while the LLM handles high-dimensional reasoning, with the division drawn at the governance boundary rather than at the capability boundary; at rung 6, the population-scale substrate carries cumulative coordination state and governance content while LLMs operating across many Selves handle per-Self reasoning, with the population-level division at the same governance boundary. Conflict-preservation holds identically across the contrast — first-class addressable substrate state at rung 1 carries forward to population-scale substrate state at rung 6 through accumulation across many events. Human governance over content and rules holds at single-user authority at rung 1 and at joint authority across population-level governance perimeters at rung 6, with the authority-versus-labor distinction from Paper 1 §3.3 portable

across both scopes. AI-as-substrate-mediator, tool-agnosticism at the substrate layer, and linear-cost composition each carry through with within-scope rendering at the larger rung — LLMs operating as substrate mediators across many Selves at rung 6 rather than within one cell at rung 1; tool-agnosticism holding at the population substrate’s structured-state plus access requirements rather than at one cell’s; linear-cost composition holding at population-scale composition of FAI events rather than at within-cell composition of operations. The six rungs are configurations of one architectural pattern at progressively larger scopes, not six separate architectural patterns composed into a sequence.

The ladder is observation rather than roadmap. Paper 1 did not promise Paper 3, and Paper 2 explicitly bracketed inter-Self dynamics as out of scope. The ladder becomes visible in retrospect because Paper 3 completes it — articulating the sixth rung supplies the structural shape that makes the prior five visible as a series. The series is not a forward-prediction of where the architectural commitment was always heading; it is a backward-reading of what the commitment composes to when extended at progressively larger scopes. The calibrated-humility voice anchor activated at §9.1 applies at trilogy-level register at this pattern in particular: the substrate-mediator-ladder-completion at the sixth rung is the architectural commitment to what the inheritance produces at population scope through the parallel-extension-claim structure, with population-scale CKS-governed Self deployments not yet existing and specific population-scale dynamics awaiting empirical observation. The ladder’s architectural shape is rendered confidently; the empirical-verification status of the sixth rung is rendered with calibrated humility that does not collapse into general hedging on the architectural commitments themselves.

11.2 Six-organizational-pattern coherence — territory through family

Closest-neighbor organization across the six claims of Paper 3 produces six structurally distinct organizational shapes — territory at §4.3 (four tied-top neighbor clusters, each closest on a different architectural axis); cluster at §5.3 (three tied-top neighbor clusters at the merge-mechanism level); tier at §6.3 (neighbors organized by which tier they operate at — preserve, resolve, escalate); locus at §7.3 (neighbors clustered by which of four loci they operate at); commitment at §8.3 (neighbors organized by which of three architectural commitments they engage); family at §9.3 (four closest-neighbor families plus adjacent territories at the trilogy-completion register). The six organizational shapes complete a pattern visible across the six claims jointly.

The pattern is genuine rather than mechanical. Each claim’s contribution shape determines its closest-neighbor organization rather than each organization fitting a slot pre-allocated at paper outline. Claim 1’s four-axis-tied-top closest-neighbor density at the architectural-pattern register — LangGraph multi-agent, A2A, Gaia-X and IDS-RAM jointly, the federated-learning / MPC-TEE / data-trusts cluster — justifies territory because four neighbors hold tied-top closeness on four distinct architectural axes and no single cluster carries the load. Claim 2’s three-cluster merge-mechanism positioning — FedAvg-family aggregation, model fusion at LLM scale, MoE and ensemble preservation — justifies cluster because the merge-mechanism level admits three structurally similar approaches. Claim 3’s three-tier mechanism justifies tier organization because the architectural mechanism itself partitions the closest-neighbor space into preserve-tier, resolver-tier, and escalate-tier work. Claim 4’s four-locus mechanism similarly partitions into four locus territories for closest-neighbor clustering. Claim 5’s three-commitment articulation — substrate-content commitment, recursive applicability, joint authority across participating Selves — partitions closest-neighbor literature into three commitment-anchored groupings. Claim 6’s four-family closest-neighbor structure at the trilogy-completion register — orchestration and meta-governance blueprints; agent-contracts and governed-reasoning frameworks; enterprise-MAS-governance blueprints; federated-learning consortia at population scope — operates at the most cross-cutting register and admits the family-organization shape.

What the pattern makes visible at trilogy level is that closest-neighbor engagement at architectural-pattern register has a structural shape that depends on the claim’s contribution shape. Each organizational shape is the load-bearing structure for the per-claim closest-neighbor differentiation at that claim’s register; the cross-claim diversity of organizational shapes is itself a discipline anchor — each claim’s closest-neighbor engagement reads as deliberate rather than as template-fitted because the organization shape is the contribution shape of the claim itself. Reviewer engagement with each claim’s closest-neighbor differentiation at the per-claim level inherits the structural shape Paper 3 commits to at that claim’s scope; reading the six

shapes jointly at trilogy level establishes that no single closest-neighbor template would have served all six claims, which is itself a finding about the structural diversity of the six contributions Paper 3’s architectural commitments compose.

11.3 Six-FL-boundary pattern — joint-authority-mechanics through population-scale-dynamics

Cross-claim federated-learning engagement compounds across all six claims as discipline visible across the trilogy. Claim 1 engages FL at the joint-authority-mechanics angle — the federated-learning / MPC-TEE / data-trusts cluster as one of four tied-top neighbors at §4.3, with the differentiation at architectural-commitment register against FL’s coordinator-mediated aggregation under consortium-governance contracts. Claim 2 engages FL at the merge-mechanism angle — FedAvg-family aggregation, model fusion at LLM scale (model soups, task arithmetic, TIES, DARE), MoE and ensemble preservation as three tied-top clusters at §5.3, with the differentiation against weight-space averaging or activation-space fusion at the merge-mechanism level. Claim 3 engages FL at the conflict-handling angle — Byzantine-robust aggregation (Krum, Trimmed Mean, Median, Bulyan, FLTrust), FedAvg averaging-as-resolution, and Rosendal and Oprescu 2025’s client-level disagreements as conflict-tier closest neighbors, with the differentiation against averaging-as-resolution at the architectural-pattern register. Claim 4 engages FL at the evolution-feed angle — FedProx, FedAdam, federated continual learning client-side machinery, personalized FL heterogeneity-driven divergence as evolution-feed closest neighbors at §7.3, with the differentiation against client-side adaptation as built-in pattern. Claim 5 engages FL at the configurable-aggregation-algorithm angle — FL platforms exposing aggregation as configurable algorithm at §8.3, with the differentiation against aggregation-as-framework-API. Claim 6 engages FL at the population-scale-dynamics angle — FLARE-based, FATE-based, and OpenFL consortia at hundreds-of-organizations scale plus Federated Behavioural Planes (Fenoglio et al. NeurIPS 2024) at §9.3 Family 4, with the differentiation against coordinator-mediated aggregation under consortium-governance contracts at population scale.

The sixth boundary closes the pattern at trilogy completion. Each claim names what was engaged at which angle in prior claims and engages FL at its own specific angle, with the cross-claim boundary work visible as deliberate discipline rather than incidental. The discipline produces a six-boundary partition of FL engagement across the trilogy where each claim’s boundary corresponds to the architectural primitive that claim contributes. The pattern reads at trilogy level as architectural-commitment differentiation against FL operating at six distinct registers — none of which the contemporary federated-learning literature occupies as substrate-content commitment at architectural-pattern register at the corresponding scope.

What the six-boundary pattern makes visible is that the federated-learning territory is six neighbors rather than one. Each claim’s FL boundary is at a different angle of the architectural primitive that claim establishes, with the differentiation at architectural-commitment register precisely localized at that claim’s scope. The cross-claim discipline establishes that the architectural-commitment differentiation against FL is not at one boundary but at six — joint-authority-mechanics, merge-mechanism, conflict-handling, evolution-feed, configurable-aggregation-algorithm, population-scale-dynamics — each load-bearing for a different reviewer concern about whether Paper 3 reduces to large-scale FL under new vocabulary. Reading the six boundaries at trilogy level supplies the cross-claim shape of the FL-restatement defense the per-claim reviewer-objection responses individually engage at their respective scopes.

Concretely, the architectural-commitment register makes operational distinctions visible against contemporary FL-with-consortium-governance at three points the cross-claim FL-restatement defense rests on. First, governance configuration occupies different architectural planes. In FL-with-consortium-governance, the consortium charter is a separate document evaluated against substrate state at decision time, with the charter and the gradient-aggregation substrate on architecturally distinct planes; in FAI, the equivalent governance content is queryable substrate with the same inspectability and authority structure as substantive substrate content, with no architectural distinction between governance content and substantive content. Second, change procedures differ in authority structure. In FL, modifying an aggregation algorithm or governance constraint typically requires consortium re-charter at legal or institutional layer with the technical substrate following as implementation; in FAI, the equivalent change is substrate-content modification under

the same joint authority across participating Selves' governance that authors substantive substrate content per Claim 5's recursive applicability. Third, conflict-handling occupies different architectural-pattern shapes. In FL with Byzantine-robust aggregation, conflicts among contributed updates are filtered at the aggregation point through statistical methods that select trusted updates and discard the rest, with conflict elimination as precondition for the aggregation to complete; in FAI, conflicts surfaced during merge become first-class addressable substrate state under §6's three-tier mechanism, with preservation as architectural default and resolution-via-orchestration-as-substrate-content plus escalation-across-joint-authority as higher-tier policy. These three operational distinctions illustrate what the architectural-commitment register at substrate-content commitment with joint authority across governance perimeters and recursive applicability differs in operationally — architectural-plane location, change-authority structure, and conflict-handling shape — rather than at register-language alone.

11.4 The four-inheritance-active-findings compounding pattern

Claims 3, 4, 5, 6 each surface a conjunction-unrealized-at-architectural-pattern-register finding at the claim's scope, and the four findings compound as discipline carrying through to trilogy completion. §6.5 surfaced the three-clause conjunction at the conflict-handling level — preserve as first-class persistent state by default; resolution rules as inspectable substrate content under explicit human authority; escalation operating across joint authority of multiple distinct governance perimeters as architectural commitment. §7.5 surfaced the three-axis conjunction at the evolution-feed level — asymmetry sourced from per-perimeter governance; FAI-derived content reaching existing per-Self evolution mechanisms unchanged; at the dissolution of an event-bounded shared substrate. §8.5 surfaced the three-axis conjunction at the configurable-dimensions and recursive-governance level — configuration as substrate content under all six Paper 1 commitments; recursive applicability through substrate-content commitment; joint authority across participating Selves' governance. §9.5 surfaced the three-axis conjunction at population scope — substrate-content commitment at population scope; joint authority across population-level governance perimeters; trilogy-completion composition rather than emergence-as-fresh-property framing.

The compounding pattern is genuine. Each finding represents the consistent unrealized-conjunction at architectural-pattern register at the claim's scope — examined across multiple candidates with negative results recorded in the per-claim closest-neighbor analysis. §8.5 and §9.5 explicitly named the compounding pattern at the time the findings were rendered; §6 and §7 surfaced findings that fit the pattern without naming it as compounding because the compounding is only visible from §8 onward. The compounding pattern's independence-of-findings claim rests on the substantively distinct closest-neighbor literatures at each of the four scopes. §6 conflict-handling literature (Byzantine-robust aggregators, CRDTs, multi-master databases, LangGraph composite single-perimeter neighbor) does not overlap with §7 evolution-feed literature (FedProx, FedAdam, FCL family methods including FedWeIT and FedCIL, personalized FL, multi-tenant AI), which does not overlap with §8 configurable-dimensions literature (Cedar, OPA, Casbin, Kubernetes CRDs, DAO constitutional architectures, NIST AIRMF and ISO 42001 frameworks), which does not overlap with §9 population-scale-dynamics literature (FLARE/FATE/OpenFL consortia, foundation lab multi-agent systems at scale, complex-adaptive-systems frameworks, agent-based modeling). The conjunctions are unrealized at each closest-neighbor literature independently — no closest-neighbor at any scope holds all clauses of its conjunction even when the closest-neighbor literatures do not overlap across scopes. This independence is what makes the compounding genuine at trilogy register rather than reflective of one architectural-commitment-register property surfaced four times across one literature. Reading the four findings jointly at trilogy level establishes that the architectural-commitment register Paper 3 occupies at cross-organizational substrate-content commitment with joint authority across governance perimeters is not a single finding at one scope but a compounding pattern at four progressively larger scopes — conflict-handling, evolution-feed, configurable-dimensions and recursive-governance, and population-scale-dynamics.

What the compounding pattern says about the cross-organizational substrate-content commitment at architectural-pattern register is that the conjunction is unoccupied not at one scope but at every scope Paper 3 examines. The pattern is the consistent shape of the unoccupied territory across the four scopes the four findings compound across; the shape is not coincidental, and Paper 3's architectural commitment is precisely at the conjunction the four findings progressively localize. The trilogy-level rendering surfaces

the four-finding compounding as architectural-commitment evidence at register-level discipline rather than as four separately-defended claims about closest-neighbor literature. The reviewer-objection map at paper level inherits this discipline: the restatement-objection pattern at each claim’s scope is engaged through the closest-neighbor literature’s failure to hold the conjunction at the corresponding scope, and the compounding pattern across four claims is itself the discipline that anchors the architectural-commitment register at trilogy level.

11.5 The six-foil-parallel pattern across §§4.2–§9.2

Each per-claim section opens its closest-neighbor engagement by naming the architectural foil at that claim’s scope. §4.2 named opaque agent-to-agent communication as the foil at the inter-Self perimeter scope, with content carried as messages, function calls, or per-call retrieval rather than as substrate. §5.2 sharpened the foil at the merge-mechanism level: opaque inter-agent merge — content-fusion or weight-space averaging or activation-space fusion or schema mapping as the merge mechanism, with the architectural-commitment register for the merged content not occupied. §6.2 sharpened further at the conflict-handling level: opaque conflict resolution — disagreements filtered, averaged, suppressed, or escalated to humans without architectural-pattern-register persistence as substrate state. §7.2 sharpened at the evolution-feed level: opaque post-deployment update — content from inter-agent events reaching home systems through retraining, fine-tuning, or context-window injection without architectural-pattern-register specification of the hand-off mechanism. §8.2 sharpened at the configurable-dimensions and recursive-governance level: opaque coordination-mechanism configuration — coordination policies, governance rules, and orchestration logic at framework-API or policy-language register rather than as substrate content under the six Paper 1 commitments. §9.2 sharpened at the population-scale-dynamics level: opaque population-scale coordination — capability-motivated population-scale coordination, emergence-as-fresh-property architectures, and federated-learning-at-population-scope architectures with governance pushed out of the substrate at population scale.

The six foils occupy parallel positions at trilogy completion. Each foil is the per-claim sharpening of the prior at the claim’s specific architectural angle, with the cross-claim foil discipline making the architectural-commitment register at substrate-content commitment visible against six structurally parallel oppositions. The architectural derivation of the six-foil-parallel pattern is the substrate-content commitment’s negation at each claim’s scope: substrate-content commitment at the substrate-as-architectural-object level produces opaque agent-to-agent communication as the foil; substrate-content commitment at the merge-mechanism level produces opaque inter-agent merge; substrate-content commitment at the conflict-handling level produces opaque conflict resolution; substrate-content commitment at the evolution-feed level produces opaque post-deployment update; substrate-content commitment at the configurable-dimensions and recursive-governance level produces opaque coordination-mechanism configuration; substrate-content commitment at the population-scale-dynamics level produces opaque population-scale coordination. The six foils are not enumeration of six independent architectural choices but six negations of one architectural commitment at six progressively more specific angles — the per-claim foil sharpening discipline traces each foil to the corresponding claim’s substrate-content commitment, with the cross-claim foil discipline making the architectural-pattern register’s commitment visible against its six structurally parallel oppositions. The cross-claim discipline reads as deliberate at paper level: the foil at every claim’s scope is what the architectural-pattern register would otherwise be at that scope when the substrate-content commitment is replaced with content-carriage-as-message, content-merge-as-fusion, conflict-handling-as-filtering, evolution-feed-as-retraining, configuration-as-policy-language, or population-scale-dynamics-as-emergence-or-coordinator-mediated-aggregation.

What the six-foil-parallel pattern makes visible is that opaque-X-at-six-angles is the consistent architectural shape Paper 3’s substrate-content commitment differentiates against, with each claim’s foil sharpening the differentiation at the specific architectural angle that claim engages. Reviewer engagement with the per-claim foil at the per-claim level inherits the architectural shape the corresponding claim establishes; reading the six foils jointly at trilogy level makes the cross-claim architectural-shape discipline visible as the consistent substrate-content-vs-opaque-content commitment at six progressively more specific architectural angles across the trilogy.

11.6 Calibration registers across the trilogy — varied landscape, not closed taxonomy

The trilogy exhibits a varied calibration landscape rather than a clean N-register taxonomy. Claim 1 calibrates implicitly through architectural-pattern-register anchoring without an explicit calibration-register label in synthesis prose, operating through the architecturally-different-from-X-on-Y-axis-not-X-plus-Y framing inherited from Paper 1 §5.2's OIDA differentiation. Claim 2 calibrates at mechanism-defense register, with the under-claim and over-claim discipline against duplicated-with-Paper-2 misreading and biology-as-metaphor reading explicit at §5.1. Claims 3 and 4 calibrate at principles-extending register, with the under-claim and over-claim discipline against architectural-commitment-extension misreading at §6.1 and §7.1. Claim 5 calibrates at enumeration-and-elaboration register, with the catalog-bloat and recursive-overclaim discipline at §8.1. Claim 6 calibrates at trilogy-completion / extension-claim register, with the three-pole calibration discipline — emergent dynamics yes / fresh emergence theory no; capability scaling honestly granted with differentiation at architectural-commitment register; population-scale CKS-governed Self deployments do not yet exist with calibrated humility on empirical verification — explicit at §9.1.

Each claim's calibration register matches the claim's contribution shape rather than each claim adopting a register pre-allocated at paper outline. The implicit-calibration register for Claim 1 reflects the architectural-anchor position Claim 1 occupies — the architectural-pattern-register anchoring carries the calibration through the conjunction-logic differentiation discipline at §4.4 rather than through an explicit calibration-register label that would underspecify the conjunction logic itself. The mechanism-defense register for Claim 2 reflects the mechanism-articulation position Claim 2 occupies at the merge-mechanism level. The principles-extending register for Claims 3 and 4 reflects the inherited-mechanism-with-articulation position Claims 3 and 4 occupy at the conflict-handling and evolution-feed levels. The enumeration-and-elaboration register for Claim 5 reflects the catalog-articulation position Claim 5 occupies across configurable dimensions with recursive applicability. The trilogy-completion register for Claim 6 reflects the extension-claim position Claim 6 occupies at population scope with the calibrated-humility voice anchor activated.

The contribution-shape-determining-calibration-register relationship has substantive architectural derivation across four specific claims that anchor the varied-landscape framing concretely. Claim 1's architectural-anchor position determines its implicit-calibration register because the conjunction-logic differentiation discipline at §4.4 — three independent axes with the conjunction carrying the contribution — itself carries the calibration: the conjunction logic specifies what Claim 1 commits to and what each axis on its own does not commit to, articulating the under-claim-and-over-claim discipline structurally without a separate calibration-register label. Claim 2's mechanism-articulation position at the merge-mechanism level determines mechanism-defense register because the inheritance density from Paper 2's intra-Self combination primitive plus the biology engagement Paper 2 §2.2 establishes makes both the duplicated-with-Paper-2 misreading and the biology-as-metaphor reading available; the calibration discipline must defend the inheritance-vs-fresh accounting at FAI mechanism level explicitly to render either reading unavailable, and the mechanism-defense register is what that defense calls for. Claim 5's enumeration position determines enumeration-and-elaboration register because the configurable-dimensions enumeration plus the recursive-applicability commitment together would otherwise admit catalog-bloat drift or recursive-overclaim drift; the calibration discipline must defend the architectural-commitment-as-framing posture against catalogue-reading and against meta-layer-fresh-philosophical-commitment-reading, and the enumeration-and-elaboration register is what that defense calls for. Claim 6's extension-claim position at population scope determines trilogy-completion register because population-scale CKS-governed Self deployments do not yet exist, making the architectural-commitment-rendered-confidently-vs-empirical-verification-rendered-with-calibrated-humility distinction load-bearing in a way that does not arise at Claims 2 through 5's scopes; the calibration discipline must defend three poles distinct from Claims 2–5's two-failure-mode patterns at the extension-claim closing register, and the trilogy-completion register is what that three-pole discipline calls for. The varied calibration landscape across the six claims is therefore architecturally derived from the contribution-shape of each claim rather than presentation-derived after the fact, with each register the discipline that the contribution-shape requires at the corresponding scope.

The varied-landscape framing — explicit calibration-register labels at Claims 2 through 6 with the pattern stabilizing from Claim 3 onward, and Claim 1 calibrating implicitly through the architectural-anchor framing

— reads as deliberate at paper level. The five distinct calibration registers across the six claims make the cross-claim calibration discipline visible: each register is what the claim’s contribution shape calls for, and the cross-claim diversity is itself the discipline that prevents the calibration from collapsing into either uniform under-claim drift or uniform over-claim drift. The cumulative quote-ledger discipline composes with this calibration landscape at register level — two sources closed against direct quotation across the trilogy (Atlan at §4.2 carrying the practitioner-literature voice on architecture-vs-bolted-on governance; Kim et al. 2025 at §9.4 carrying the explicit governance-deferral commitment within the scaling-science enterprise), with all other engagement with closest-neighbor literature operating via paraphrase under attribution-soft framing. Paraphrasing as default discipline is itself part of the cross-claim calibration: direct quotation reserved for cases where the original wording substantively affects meaning, with the one-quote-per-source ceiling preventing the cumulative engagement from drifting toward source-reproduction at scale.

11.7 Restatement objections — paper-level response across the six claims

The first load-bearing reviewer concern recurs at every claim’s scope as a restatement objection: this is just X under new vocabulary, where X is the closest-neighbor literature engaged at the claim’s level. Claim 1 engages restatement against multi-agent-AI-with-governance-vocabulary, A2A-or-MCP-with-extra-commitments, classical-blackboard-architecture-under-new-name, and federated-learning-or-data-spaces-under-new-vocabulary. Claim 2 engages restatement against federated-learning-aggregation, LangGraph-reducers-or-AutoGen-state-aggregation-at-higher-abstraction, and schema-mapping-or-ontology-alignment. Claim 3 engages restatement against §5-of-Paper-1-at-bigger-scope, arbiter-agent-reconciliation, and Byzantine-robust-aggregation-or-distributed-systems-conflict-resolution. Claim 4 engages restatement against §7-of-Paper-2-with-FAI-as-input-source, continual learning, and federated continual learning. Claim 5 engages restatement against configuration-management-or-policy-as-code, meta-governance-or-recursive-policy, and enumeration-catalogue-not-architectural-commitment. Claim 6 engages restatement against emergence-theory-or-CAS and large-scale-FL-or-MAS.

The paper-level response composes across the six claims at architectural-commitment register. The architectural-commitment register Paper 3 occupies — substrate-content commitment, joint authority across governance perimeters, all six Paper 1 commitments holding within scope, configuration as substrate content with recursive applicability — is what contemporary work in each closest-neighbor territory does not occupy at architectural-pattern level even when the contemporary work delivers functional coordination at the relevant scope. The differentiation at architectural-commitment register is not a claim about the closest-neighbor literature’s failure to deliver functional outcomes at scale; it is a claim about the architectural-pattern register at which Paper 3’s commitments are made and at which the closest-neighbor literature operates. The inheritance-vs-fresh accounting per per-claim §X.6 inheritance accounting subsections locates precisely what each claim establishes that is not inherited from prior claims and not present at the closest-neighbor literature, with the four-finding compounding pattern across Claims 3–6 (§11.4 above) supplying the cross-claim discipline anchor for the architectural-commitment register at conjunction logic. The restatement objection at each claim’s scope is engaged through the conjunction logic at that claim’s three differentiation axes — Axis 1 without Axis 2 is one neighbor’s territory; Axis 2 without Axis 1 is another’s; the conjunction is what each claim occupies. The cross-claim conjunction logic operates as six per-claim conjunctions (one per claim) of three axes each, with primary novelty weight identified per axis at each claim. Reading the six conjunction-logic engagements at trilogy level establishes that the architectural-commitment register at substrate-content commitment is not a uniform commitment defended six times but a register-stable commitment articulated at six progressively more specific architectural angles, with the trilogy-level commitment as the conjunction of the six per-claim conjunctions held jointly across the architectural primitives Paper 3 establishes.

11.8 Capability-motivated objection — foundation labs solve this anyway

The second load-bearing reviewer concern is the capability-motivated objection. The objection originated at §4.5 objection 8 with brief response gesturing at the governance-vs-capability framing inherited from Papers 1 and 2; redirected fuller engagement was concentrated at §9.5 obj 1 dual-half engagement at population

scope. Reading the engagement at paper level surfaces the architectural-commitment-vs-capability-scaling discipline as the cross-claim load-bearing differentiation Paper 3 commits to at trilogy level.

The first half is honestly granted at paper level. Capability scaling does deliver functional inter-Self coordination at scale. Kim et al. (2025) derives quantitative scaling principles across 180-plus configurations with a predictive model selecting optimal coordination architecture for 87 percent of unseen tasks; MCP under cross-vendor stewardship standardizes contextual exchange across foundation lab platforms; A2A with 50-plus enterprise partners supplies standardized peer-to-peer communication; OpenAI’s Agents SDK and AgentKit supply model-capability plus platform-capability plus developer-capability scaling at deployment scope; the broader contemporary multi-agent-systems-at-scale landscape exhibits functional coordination capabilities at increasingly sophisticated registers. The second half is the architectural-commitment differentiation at paper level. Governance in capability-motivated population-scale architectures sits at framework/API, deployment-shape, and contract/institutional registers rather than at substrate-content commitment register. Kim et al. treats governance as engineering performance with the scaling-science enterprise’s governance treatment kept out of scope; MCP is intentionally agnostic about what is allowed; A2A leaves policy to implementers; the NIST AI Agent Standards Initiative shapes how substrates must behave rather than authoring substrates; the Agentic AI Foundation supplies institutional governance of open technical projects but does not dictate deployment substrate. The differentiation is at architectural-commitment register: governance-motivated population-scale architecture commits to substrate-content commitment at population scope, joint authority across population-level governance perimeters, and recursive applicability — commitments capability-motivated architectures do not make at architectural-pattern level even when they scale functional coordination. The operational distinctions §11.3 articulates against contemporary federated learning — architectural-plane location of governance configuration, change-authority structure for governance modification, and conflict-handling shape under joint authority — render the register-language differentiation concretely at the FL-specific surface and apply at paper-level register against capability-motivated architectures more broadly.

The calibrated-humility voice anchor activated at §9.1 applies at paper level at this objection in particular. The differentiation is not whether capability scaling works at the functional-coordination level (it does), but whether the architectural-commitment register for governance is occupied at substrate-content commitment with joint authority and recursive applicability. The architectural-commitment register Paper 3 commits to is rendered confidently — what the inheritance produces at scope when Papers 1 and 2’s principles compose with Paper 3’s six claims — while empirical-verification status of the commitment at population scope is rendered with calibrated humility, since population-scale CKS-governed Self deployments do not yet exist and specific population-scale dynamics await empirical observation. The anchor distinguishes the architectural commitment from empirical-verification status without collapsing into general hedging on the architectural commitments themselves; capability scaling is honestly acknowledged without the architectural-commitment register being conflated with the functional-coordination claim. The paper-level response at the capability-motivated objection sits at this distinction precisely.

11.9 Scope-and-framing objections — why these specific scopes

The third load-bearing reviewer concern is the scope-and-framing objection. Each claim’s scope and framing carry their own answer that composes into paper-level positioning. Why the inter-Self perimeter as Claim 1’s scope? Paper 3 introduces one new architectural object — the shared substrate with governance perimeter spanning more than one Self’s home governance perimeter — not present at prior-paper scopes; the inter-Self perimeter is what the substrate-mediator commitment extends to at Paper 3 core theory’s rung 5. Why FAI as canonical operation at Claim 2? FAI is the inter-Self analog of Paper 2’s mating primitive at the granularity Paper 2’s three-level structure makes coherent — the unit of contribution is the aspect rather than the cell or the Self because the aspect is the architectural granularity at which coordination work surfaces between Selves. Why three-tier conflict handling at Claim 3? The three tiers — preserve, resolve via orchestration, escalate to humans — are what Paper 1 §5.3’s two-level conflict handling produces when extended at Paper 3 scope where joint authority across distinct governance perimeters becomes architecturally required at the escalate tier; the fresh content is precisely localized at three loci. Why the four-locus evolution-feed mechanism at Claim 4? The four loci — hand-off at FAI dissolution, per-mechanism feed structure, layer-routing rule

with three-case explicitness, asymmetric ingestion — are what Paper 2 §7’s three evolution mechanisms produce when fed by FAI events; the layer-routing rule’s three-case explicitness honors Paper 3 core theory’s instinct-evolution-out-of-scope commitment by rendering the rule explicitly at architectural-pattern register. Why configuration-as-substrate-content at Claim 5? Configuration on the same architectural plane as other substrate content is what Paper 1 §3.2 plus Paper 2 §8 plus §4.4 Axis 2 produce when applied to FAI events; the recursive applicability is the architectural property Claim 1’s substrate-content commitment composes to under Paper 1 §3.2’s same-plane architectural commitment, not a fresh philosophical commitment. Why is Claim 6 in this paper rather than in a separate vision paper?

The last scope-and-framing objection warrants the paper-level register’s most direct response. Claim 6 is the canonical extension claim per Paper 3 core theory — the trilogy-completion sixth rung of the substrate-mediator ladder. Separating it would leave Paper 3 with five claims deferring their composition to elsewhere; the trilogy is structurally complete only when the population-scale composition is articulated within the same architectural register as the preceding claims. The parallel-extension-claim structure across the trilogy (Paper 1’s Claim 6 + Paper 2’s Claim 6 + Paper 3’s Claim 6) is itself the architectural honesty anchor for the trilogy-completion register. Calibrated humility is the appropriate voice anchor for the extension-claim register, not a reason for separation. Reading the six scope-and-framing answers at paper level establishes that Paper 3’s scope is determined by what the substrate-mediator commitment composes to at the inter-Self perimeter and at population scope, with each claim’s specific scope load-bearing for the architectural-pattern register the trilogy occupies at that scope rather than open to alternative scope choices that would relocate the architectural commitment.

11.10 Trilogy positioning, future work, and the calibrated-humility voice anchor

Paper 3 closes the substrate-mediator ladder at its sixth rung. Papers 1 and 2 plus Paper 3’s six claims jointly specify substrate-mediated coordination from intra-cell scope through intra-Self scope through inter-Self scope through population scope, all under unified human governance. The parallel-extension-claim structure across Paper 1’s Claim 6, Paper 2’s Claim 6, and Paper 3’s Claim 6 is architecturally derived rather than narratively imposed across three papers. Each paper’s core theory at its scope produces an extension claim at the next-larger configuration under the same substrate-mediator commitment: Paper 1’s cell-scope core theory at single-human substrate-mediator produces multi-humans-within-cell as the extension at the same cell scope (rung 2 inheriting cell-scope commitments at multi-human cardinality); Paper 2’s Self-scope core theory at intra-Self substrate-mediator produces enterprise-brain-Self as the extension at extended Self scope (rung 4 inheriting Self-scope commitments at enterprise cardinality); Paper 3’s inter-Self core theory at substrate-mediator across the inter-Self perimeter produces population-scale collective evolution as the extension at extended inter-Self scope (rung 6 inheriting inter-Self commitments at population cardinality). The structural parallel is a property of the substrate-mediator commitment composing forward at each paper’s scope, with the calibrated-humility voice anchor appropriate to the extension-claim register at each scope as a calibration discipline that follows from the inheritance shape rather than from trilogy framing. The architectural object — human-governed substrate as coordination medium — extends to progressively larger scopes through the series, with each scope inheriting the prior commitments without replacing them. The trilogy is a design-pattern paper across three scopes: Paper 1 establishes the design pattern at cell scope with proof-of-concept grounding; Paper 2 extends the pattern to Self scope with the enterprise brain Self as architectural vision; Paper 3 extends the pattern to inter-Self scope and population scope with the calibrated-humility voice anchor at the extension-claim register. The contribution at each paper’s scope is design-pattern writing in Paper 1’s genre, with the design pattern’s structured hypothesis available at each scope to be adopted, varied, composed with adjacent patterns, or argued against.

Future work is downstream of the architectural pattern the trilogy specifies. Formal protocol specifications instantiating the FAI mechanism, security guarantees at the substrate-content register, message schemas for the substrate-mediator extension across organizational boundaries, deployment architectures for the shared-substrate construction and dissolution dynamics, and empirical validation at scale of the population-scale collective evolution Paper 3 commits to are research directions the architectural pattern opens rather than research the architectural pattern leaves incomplete. Each direction is an extension theory the trilogy acknowledges; each would stand on its own claim set rather than on arguments the trilogy develops. The

trilogy does not commit to specific protocols, specific deployment architectures, or specific scaling outcomes at population scope; what it commits to is the architectural pattern those research directions would instantiate, vary, or argue against.

Calibrated humility is the appropriate voice anchor for the trilogy’s closing as it has been for Claim 6’s territory throughout. Population-scale CKS-governed Self deployments do not yet exist; the architectural commitment is to what the inheritance produces at population scope, with specific population-scale dynamics — the precise trajectories of pattern propagation, the actual rates and shapes of specialization-preserving divergence, the empirical character of collective adaptation — awaiting empirical observation. The architectural commitment is rendered confidently; the empirical-verification status is rendered with calibrated humility that does not collapse into general hedging on the architectural commitments themselves and does not conflate capability-scaling acknowledgment with architectural-commitment differentiation. The substrate-mediator-ladder-completion at the sixth rung is the trilogy’s anticipated landing rather than its empirically demonstrated outcome, calibrated through the vision-claim register Paper 2 §9 establishes. As §12 develops, the paper-level closing positions the trilogy at this register with the calibrated-humility voice anchor preserved across the closing prose.

12. Conclusion

This paper has extended substrate-mediated coordination across the inter-Self perimeter and at population scope, taking up the territory Papers 1 (Li, 2026a) and 2 (Li, 2026b) jointly bracket and developing it through six architectural commitments in turn. The first establishes the **shared substrate** as the architectural object through which two or more CKS-governed Selves coordinate — temporarily constructed for an interaction, perimeter-spanning across distinct Selves’ home governance perimeters, with all six Paper 1 commitments holding within scope and the configuration of the perimeter-crossing itself substrate content under joint human authority. The second commits **Full Aspect Integration (FAI)** as the canonical operation over the shared substrate — n-ary aspect-level merge with full-merge as architectural default at inter-Self scope, exchange bounded to reasoning-layer content per Paper 2’s instinct/reasoning separation extended at the inter-Self boundary. The third commits **three-tier conflict handling** within the shared substrate — preserve as substrate-level state by default per Paper 1 inheritance; resolve via orchestration with rules as inspectable substrate content under explicit human authority; escalate to humans across joint authority of participating Selves’ governance, with cross-perimeter escalation as architectural commitment rather than as deployment-layer wrapper. The fourth commits a **four-locus evolution-feed mechanism** connecting inter-Self events to home-perimeter Paper 2 evolution machinery — hand-off at FAI dissolution; per-mechanism feed structure with FAI-derived content reaching existing per-Self evolution mechanisms unchanged; layer-routing rule with three-case explicitness (DNA-layer content feeds DNA evolution; action-layer content feeds action-feedback evolution; instinct evolution takes no FAI input by architectural commitment, per Paper 2’s instinct/reasoning separation extended at the inter-Self perimeter); asymmetric ingestion sourced from per-perimeter governance. The fifth commits **configuration as substrate content with recursive applicability** — every dimension of the FAI mechanism is itself substrate content under all six Paper 1 commitments, with configuration of configuration substrate also substrate content and the recursion bottoming at human-authored governance authority per Paper 1 §3.3. The sixth offers the extension claim that **population-scale collective evolution** is the architectural object Paper 3’s five preceding claims compose into at population register when accumulated FAI events across many CKS-governed Selves operate under joint authority across population-level governance perimeters and the recursive applicability of configuration-as-substrate-content extended at population scope.

The unifying architectural move across all six claims is the extension of substrate-mediated coordination across the inter-Self perimeter under unified human governance, composed at population scope through accumulated FAI events under joint authority across population-level governance perimeters. Paper 1’s *outside-the-model vs. inside-the-model* spine extends through Paper 3 unchanged: inter-Self coordination, governance, conflict-handling, and evolution-feed live in the shared substrate outside any single LLM; instinct-layer content remains within each Self’s LLM and does not cross the inter-Self perimeter; and the architecture’s value at inter-Self and population scope comes from the same principled separation Papers 1 and 2 defend at

their respective scopes. Each claim either establishes the architectural primitive the spine requires, specifies the canonical operation over it, develops the dynamics the construction enables, or specifies governance properties that keep the construction safe under human authority across distinct perimeters. The resulting architecture answers the field-level architectural question §1.1 names: an LLM-era AI system that coordinates with other LLM-era AI systems under human governance at scale looks like CKS-governed Selves coordinating through a shared substrate that is itself first-class human-governed content carrying Paper 1’s six commitments within scope, with FAI as canonical operation over it, three-tier conflict handling within, an evolution-feed connecting events to home-perimeter Paper 2 evolution machinery, configuration recursively applicable as substrate content, all composing at population scope as collective evolution under joint authority across population-level governance perimeters.

The commitments are calibrated explicitly. The architecture commits to substrate-content commitment for the medium of inter-Self coordination, joint authority across distinct governance perimeters as architectural property of the technical substrate, conflict preservation as first-class state at the inter-Self perimeter, recursive applicability of configuration as substrate content, and population-scale collective evolution as composition of the preceding five commitments at population register. It does not commit to specific protocols, specific message schemas, specific security guarantees, specific deployment architectures, or specific scaling outcomes at population scope; what it commits to is the architectural pattern that research and engineering directions of those kinds would inherit from. The calibrated-humility voice anchor activated at §9.1 and threaded through §9, §10, and §11 holds at the trilogy’s closing as it has held throughout Claim 6’s territory. Population-scale CKS-governed Self deployments do not yet exist; the architectural commitment is to what the inheritance produces at population scope through composition, with specific population-scale dynamics — the precise trajectories of pattern propagation, the actual rates and shapes of specialization-preserving divergence, the empirical character of collective adaptation — awaiting empirical observation. Capability scaling delivers functional inter-Self coordination at scope honestly across foundation lab multi-agent systems, agent-protocol ecosystems, and federated-learning consortia; the differentiation Paper 3 commits to is at architectural-commitment register rather than at the functional-coordination claim. The architectural commitment is rendered confidently; the empirical-verification status of the population-scale extension is rendered with calibrated humility that does not collapse into general hedging on the architectural commitments themselves and does not conflate capability-scaling acknowledgment with architectural-commitment differentiation. The substrate-mediator-ladder-completion at the trilogy’s sixth rung is the trilogy’s anticipated landing rather than its empirically demonstrated outcome, calibrated through the vision-claim register Paper 2 §9 establishes.

The contribution is design-pattern writing in Papers 1 and 2’s genre. Paper 3 articulates a design pattern at the inter-Self perimeter and at population scope available to be adopted, varied, composed with adjacent patterns, or argued against, sitting at the intersection of six recognized lineages — multi-agent systems and inter-agent coordination; federated learning and cross-organizational ML; governance-as-code and policy-as-substrate work; multi-coordinator and joint-authority architectures; substrate-as-content and recursive-governance literature; trilogy continuity with Papers 1 and 2 — without being any of them. The trilogy is a design-pattern paper across three scopes: Paper 1 establishes the design pattern at cell scope with proof-of-concept grounding; Paper 2 extends the pattern to Self scope with the enterprise brain Self as architectural vision; Paper 3 extends the pattern to inter-Self scope and population scope with the calibrated-humility voice anchor at the extension-claim register. Each paper carries a core-theory contribution and an extension-claim contribution, with the parallel-extension-claim structure across the trilogy — Paper 1’s Claim 6, Paper 2’s Claim 6, Paper 3’s Claim 6 — making composition rather than fresh emergence theory architecturally honest at trilogy completion. The design pattern’s empirical validation through real-world cross-organizational deployment at scale is positioned as downstream work; what Paper 3 establishes is the architectural commitments and their composition at the inter-Self perimeter and at population scope. Future work — formal protocol specifications instantiating the FAI mechanism, security guarantees at the substrate-content register, message schemas for the substrate-mediator extension across organizational boundaries, deployment architectures for shared-substrate construction and dissolution dynamics, and empirical validation at scale — is downstream of the architectural pattern the trilogy specifies. Each direction is an extension theory the trilogy acknowledges; each would stand on its own claim set rather than on arguments the trilogy develops.

References

This references list is partial. It includes the academic and peer-reviewed sources most directly cited across Paper 3’s six claims, the contemporary industry-voice sources most directly engaged in closest-neighbor differentiation, and the primary architectural lineage anchors. Additional entries — including vendor documentation, standards-body publications, practitioner essays, conference workshop papers, and preprints requiring further verification — will appear in a revised version. Papers 1 (Li, 2026a) and 2 (Li, 2026b) are cited throughout Paper 3 as the substrate-cell and Self-scope foundations Paper 3 extends; the citation scheme uses “Paper 1 §X” and “Paper 2 §X” for section references with full citation only on first introduction in each major section. Paper 2’s References list functions as inheritance anchor for cross-paper lineage citations not surfaced in Paper 3 prose; entries below restate citations referenced in Paper 3 prose and the primary architectural lineage anchors specific to Paper 3’s six lineages.

Stage 9 reconciliation expanded the model-fusion-and-merging lineage anchors (Wortsman et al. 2022 model soups; Ilharco et al. 2023 task arithmetic; Yadav et al. 2023 TIES-Merging; Yu et al. 2024 DARE) to full canonical entries supporting Claim 2’s cluster (iii) closest-neighbor synthesis at the LLM-scale model-fusion register; resolved Yoon et al. 2021 FedWeIT to canonical ICML 2021 entry supporting Claim 4’s locus (b) closest-neighbor synthesis at the federated continual learning register; resolved Park et al. 2024 generative agents at thousand-person scale to canonical arXiv entry supporting §3.7 adjacent families and §9.4 Axis 3 emergence-theory contrast; resolved Kim et al. 2025 (arXiv:2512.08296) to canonical entry supporting §9.4 quote-ledger surface and the foundation-lab-multi-agent-systems-at-scale articulation; reconciled Packer et al. (MemGPT, 2023) to canonical arXiv entry; corrected the Fenoglio et al. 2024, Yang et al. 2025, and Rosendal & Oprescu 2025 entries from paraphrased to canonical titles; and inherited Star and Griesemer 1989 plus Bai et al. 2022 from Papers 1 and 2 References as canonical entries. Sources requiring further verification — vendor documentation surfaces, standards-body publication metadata, practitioner essays, and preprints with metadata not available at Stage 9 publication-readiness window — defer to a revised version per the partial-references convention; the architectural defense at §3.7’s four-family closest-neighbor structure and §9.4’s three-axis conjunction holds through Family-organization commitments and inheritance from architecturally engaged peer-reviewed and standards-body lineages, not through single-source load-bearing on deferred entries.

Anthropic. (2024). *Model Context Protocol (MCP) specification*. <https://modelcontextprotocol.io> (retrieved May 2026). (Standardized contextual exchange under cross-vendor stewardship; cited at §1.1, §3.1, §4.3, §9.4.)

Atlan. (2026). *Agent Memory Architectures: 22 Patterns for Production-Grade Agent Memory*. Atlan practitioner publication. <https://atlan.com> (retrieved March 2026). (Quote closed at §4.2 per cumulative one-quote-per-source ledger; cited substantively at §4.3 Pattern 4 actor-aware edges, §6.3 preserve-tier neighbors. The Pattern 19 orchestrator/aggregator and Pattern 21 arbiter agent reformulations applied per master synthesis §7(d) vocabulary discipline. Retained as canonical entry per cross-paper one-quote-per-source ledger transparency requirement, with the partial-references convention otherwise applying to practitioner literature.)

Bai, Y., Kadavath, S., Kundu, S., et al. (2022). Constitutional AI: Harmlessness from AI Feedback. arXiv:2212.08073 [cs.CL]. <https://arxiv.org/abs/2212.08073> (51 authors; first three named, “et al.” abbreviation per citation discipline for very-large-author preprints. Substrate-as-content lineage anchor at training-time and objective-shaping register; cited at §3.5, §8.3. Canonical entry inherited from Paper 2 References.)

Fenoglio, D., Dominici, G., Barbiero, P., Tonda, A., Gjoreski, M., & Langheinrich, M. (2024). Federated Behavioural Planes: Explaining the Evolution of Client Behaviour in Federated Learning. *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*. arXiv:2405.15632 [cs.LG]. <https://arxiv.org/abs/2405.15632> (Cited at §3.2, §3.7 Family 4 supporting density, §9.3 Family 4, §9.4.)

Ilharco, G., Ribeiro, M. T., Wortsman, M., Schmidt, L., Hajishirzi, H., & Farhadi, A. (2023). Editing Models with Task Arithmetic. *Proceedings of the Eleventh International Conference on Learning Representations (ICLR 2023)*. arXiv:2212.04089 [cs.LG]. <https://arxiv.org/abs/2212.04089> (LLM-scale model-fusion lineage

anchor cited at §5.3 cluster (iii); task vectors as additive and subtractive editing primitive.)

Kim, Y., Gu, K., Park, C., Park, C., Schmidgall, S., Heydari, A. A., Yan, Y., Zhang, Z., Zhuang, Y., Malhotra, M., Liang, P. P., Park, H. W., Yang, Y., Xu, X., Du, Y., Patel, S., Althoff, T., McDuff, D., & Liu, X. (2025). Towards a Science of Scaling Agent Systems. arXiv:2512.08296 [cs.AI]. <https://arxiv.org/abs/2512.08296> (Quote closed at §9.4 per cumulative one-quote-per-source ledger; cited at §3.1, §9.4, §11.8. Quantitative scaling principles for multi-agent coordination across 180 configurations in v1, expanded to 260 configurations in subsequent revisions; foundation-lab multi-agent-systems-at-scale articulation.)

Li, W. (2026a). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* [Paper 1 of the CKS theory series]. (Cited as “Paper 1” throughout Paper 3.)

Li, W. (2026b). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance* [Paper 2 of the CKS theory series]. (Cited as “Paper 2” throughout Paper 3. Paper 2’s References list functions as inheritance anchor for cross-paper lineage citations not surfaced in Paper 3 prose, including: Booch et al. 2021 SOFAI; Bottino et al. 2026 OIDA; cognitive-architectures lineage Anderson 1993, Laird et al. 1987, Langley & Choi 2006, Sun et al. 2001; organizational-design lineage Argyris & Schön 1978, Galbraith 1973, Mintzberg 1979, Senge 1990; Alexander 1979, 2002 architectural-pattern lineage; Davenport & Prusak 1998, Nonaka & Takeuchi 1995, Wenger 1998 KM canon; Engin 2025 HAIG; Hartwell et al. 1999, Jablonka & Lamb 2014, Kirschner & Gerhart 2005, Laland et al. 2015, Maynard Smith & Szathmáry 1995, Vanchurin et al. 2022, West-Eberhard 2005 evolutionary-biology lineage; Hodgson 2017, Inmon 2005, Kimball & Ross 2013, Skelton & Pais 2019, Steimann 2000 substrate-sharing and structural lineage; Kahneman 2011 fast-and-slow; Kerr et al. 1972 apoptosis; Lehman & Stanley 2011 novelty search; Mosqueira-Rey et al. 2023, Wu et al. 2022 HITL; Orlikowski 1992 organizational-technology duality; Quirke et al. 2025 Beyond Monoliths; Seck 2026 Cognitive Core; Ziabari et al. 2025 spectrum-of-reasoning. Paper 3 inherits these references via Paper 2 without restatement except where specifically engaged in Paper 3 prose.)

McMahan, B., Moore, E., Ramage, D., Hampson, S., & y Arcas, B. A. (2017). Communication-efficient learning of deep networks from decentralized data. *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS 2017)*, PMLR 54:1273–1282. <https://proceedings.mlr.press/v54/mcmahan17a.html> (FedAvg foundational paper; FL aggregation lineage anchor cited at §3.2, §5.3, §6.3, §7.3.)

Packer, C., Fang, V., Patil, S. G., Lin, K., Wooders, S., & Gonzalez, J. E. (2023). MemGPT: Towards LLMs as Operating Systems. arXiv:2310.08560 [cs.AI]. <https://arxiv.org/abs/2310.08560> (Tiered memory architecture cited at §3.1, §4.3; the Letta agent framework derives from this design pattern.)

Park, J. S., Zou, C. Q., Shaw, A., Hill, B. M., Cai, C., Morris, M. R., Willer, R., Liang, P., & Bernstein, M. S. (2024). Generative Agent Simulations of 1,000 People. arXiv:2411.10109 [cs.HC]. <https://arxiv.org/abs/2411.10109> (Complex-adaptive-systems and emergence-theoretical lineage cited at §3.7 adjacent families, §9.3 adjacent families, §9.4 Axis 3.)

Rosendal, D., & Oprescu, A. (2025). A Taxonomy and Resolution Strategy for Client-Level Disagreements in Federated Learning. *Proceedings of the 2025 IEEE International Conference on Big Data*. (Strongest articulation at disagreement-as-substrate register at federated scope; cited at §3.2, §6.3 resolve-via-orchestration-tier neighbors. Exclusion policies as first-class configuration with isolated model-update tracks for mutually excluding clients.)

Star, S. L., & Griesemer, J. R. (1989). Institutional ecology, ‘translations’ and boundary objects: Amateurs and professionals in Berkeley’s Museum of Vertebrate Zoology, 1907–39. *Social Studies of Science*, 19(3), 387–420. <https://doi.org/10.1177/030631289019003001> (Boundary-objects lineage anchor cited at §4.5 obj 3. Canonical entry inherited from Paper 1 References.)

Wortsman, M., Ilharco, G., Gadre, S. Y., Roelofs, R., Gontijo-Lopes, R., Morcos, A. S., Namkoong, H., Farhadi, A., Carmon, Y., Kornblith, S., & Schmidt, L. (2022). Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. *Proceedings of the 39th Interna-*

tional Conference on Machine Learning (ICML 2022), PMLR 162:23965–23998. arXiv:2203.05482 [cs.LG]. <https://arxiv.org/abs/2203.05482> (LLM-scale model-fusion lineage anchor cited at §5.3 cluster (iii); uniform-soup and greedy-soup recipes for averaging fine-tuned model weights.)

Yadav, P., Tam, D., Choshen, L., Raffel, C., & Bansal, M. (2023). TIES-Merging: Resolving Interference When Merging Models. *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*. arXiv:2306.01708 [cs.LG]. <https://arxiv.org/abs/2306.01708> (LLM-scale model-fusion lineage anchor cited at §5.3 cluster (iii); Trim, Elect Sign, and Merge as three-step interference-resolution method on task vectors.)

Yang, Y., Chai, H., Shao, S., Song, Y., Qi, S., Rui, R., & Zhang, W. (2025). AgentNet: Decentralized Evolutionary Coordination for LLM-based Multi-Agent Systems. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*. arXiv:2504.00587 [cs.MA]. <https://arxiv.org/abs/2504.00587> (Decentralized cross-organizational frameworks lineage anchor cited at §1.1, §3.1, §4.3; DAG-structured retrieval-augmented coordination across LLM-based agents.)

Yoon, J., Jeong, W., Lee, G., Yang, E., & Hwang, S. J. (2021). Federated Continual Learning with Weighted Inter-client Transfer. *Proceedings of the 38th International Conference on Machine Learning (ICML 2021)*, PMLR 139:12073–12086. arXiv:2003.03196 [cs.LG]. <https://arxiv.org/abs/2003.03196> (FedWeIT; federated continual learning lineage anchor cited at §7.3 locus (b); weight-decomposition into global federated parameters and sparse task-specific parameters with weighted inter-client transfer.)

Yu, L., Yu, B., Yu, H., Huang, F., & Li, Y. (2024). Language Models are Super Mario: Absorbing Abilities from Homologous Models as a Free Lunch. *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*. arXiv:2311.03099 [cs.CL]. <https://arxiv.org/abs/2311.03099> (DARE: Drop And REscale; LLM-scale model-fusion preprocessing lineage anchor cited at §5.3 cluster (iii); randomly sets delta parameters to zero with rescaling factor $1/(1-p)$ to mitigate interference during merging.)

Industry-voice, standards-body, vendor-documentation, and practitioner-literature sources cited across the claims include: AgenticAI Foundation institutional governance; Adimulam et al.’s 2026 orchestration-of-multi-agent-systems articulation; Anthropic Claude multi-agent system framings; Apache Airflow, Temporal.io, and Prefect multi-orchestrator workflow patterns; ANP (Agent Network Protocol) and IBM ACP at protocol-spec register; the AgentAuditor 2026 work and adjacent agent-evaluation literature; Aragon, Compound, MakerDAO, Arbitrum DAO constitutional architectures and the MakerDAO Endgame and Optimism Collective bicameral structures supplying recursive-governance instances; AWS Cedar at policy-engine register; AWS Deterministic Governance Engine literature; CNCF (Cloud Native Computing Foundation) Automated Governance Maturity Model; CrewAI hierarchical-with-manager-agent configurations; Conftest and Checkov policy-as-code documentation; Cognitive Core (Seck), Governance-First Architecture, and Governing Multi-Agent AI Systems enterprise blueprints supplying intra-organizational substrate-content commitment instances; CockroachDB and YugabyteDB multi-leader replication documentation; Couchbase Mobile conflict-resolution documentation as preserve-tier MVCC neighbor; cross-border privacy recourse frameworks at legal-layer register; data trusts and cooperatives at organizational-structure register; data mesh implementations with first-class data-product metadata; data-sharing agreements and joint venture dispute mechanisms supplying authority structures from outside the technical substrate; the European Commission Gaia-X federation services and Data Usage Agreements specifications; FATE, FLARE, and OpenFL platform documentation as population-scope FL consortium platforms cited across §1.1, §3.2, §3.7 Family 4, §9.3, §9.4; the Fauscette April 2026 Arion Research piece on cross-organizational agents at practitioner-literature register; Google Cloud Agent2Agent (A2A) protocol specification and Gemini Enterprise Agent Platform documentation; HADA and SettleFL crypto-economic coordinator at cross-perimeter joint-authority technical-mechanism register; HashiCorp Sentinel, Pulumi CrossGuard, and CloudFormation Guard at policy-as-code/IaC register; HGAFL multi-coordinator FL architecture; Holland-style complex-adaptive-systems apparatus, Indian PMEAC’s CAS framework, and Molt Dynamics as complex-adaptive-systems framing instances; IDS (International Data Spaces Association) IDS-RAM Reference Architecture Model with Provenance Storage and Collection Points; IETF and W3C drafts at agent-protocol-spec register; ISO/IEC 42001; JAMS AI Disputes Clauses and Rules at legal-layer dispute-resolution register; the Kothari & Iyer 2025 closest-neighbor work; Kubernetes Custom Resource Definitions (CRDs) at substrate-content register within single-cluster

scope; LangChain LangGraph documentation for typed shared state, per-key reducers, durable execution with checkpointing as substrate-pattern-register multi-agent framework primitive cited at §3.1, §4.3, §5.3, §6.3, §7.3; Microsoft AutoGen multi-agent framework documentation; multi-tenant SaaS configuration services as cross-tenant-substrate register; the Nguyen 2026 CAIA SSRN preprint at register-mismatch positioning territory; NIST AI Risk Management Framework; NIST AI Agent Standards Initiative; OpenAI AgentKit and Agents SDK documentation; OpenID Foundation AI Identity Management Community Group at standards-body register; Open Policy Agent (OPA) and Rego policy language; Casbin policy framework; the Raza et al. and TechRxiv survey institutional-AI work; the Strwythura ontology pipeline; the Vector Institute “Transparency in Agentic AI” articulation; W3C Solid Project as substrate-content register reference; the Ye and Tan 2026 Agent Contracts work at resource-budget substrate-content register; the post-microservices retrospective with the Amazon Prime Video monolith-rebuild case as its popular crystallization (inherited from Paper 2 References); the choreography-vs-orchestration distinction in BPMN at process-modeling-language register; Terraform/Sentinel at IaC policy-as-code register; threshold MPC structures and TEE attestation at cryptographic-and-hardware register; Okta agent-orchestration with authority metadata; the 2026 conflict-of-laws RAG example at preserve-tier neighbor register; the broader 2024–2026 multi-agent-systems-at-scale landscape including swarm and graph-topology architectures. Sectoral and regulatory sources, vendor enterprise agent-deployment platform documentation, additional preprints requiring verification, and practitioner essays referenced in closest-neighbor differentiation are not enumerated here; they will appear with full citations in a revised version.